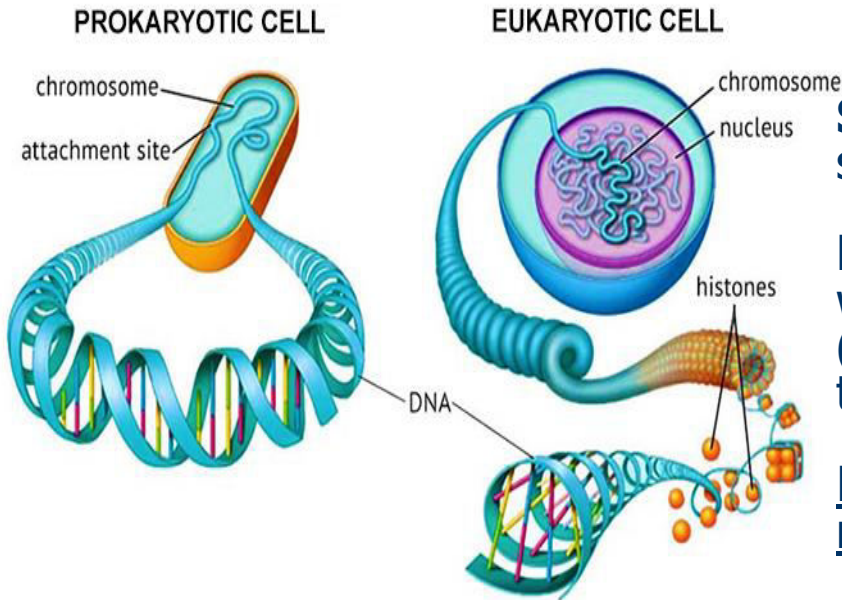




Chapter 2.1 DNA Structure and Replication

DNA

- Contains genetic code
- Nucleus of eukaryotic cells
- Cytoplasm of prokaryotic cells



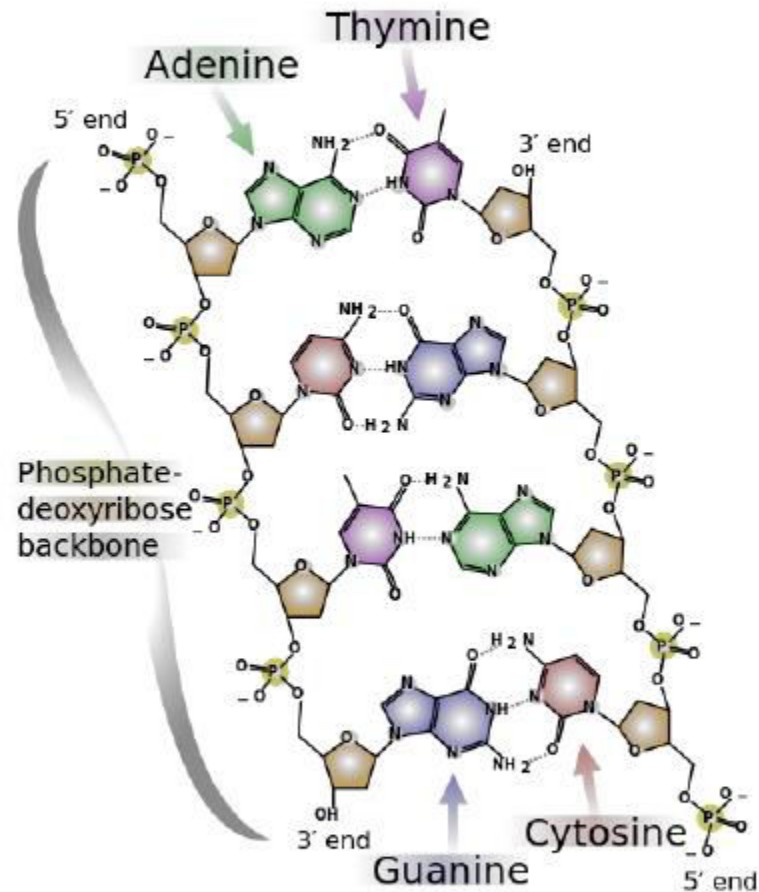
Storage of genetic information stored as DNA

Eukaryotic → DNA is associated with various types of proteins (nucleoproteins) **Histone** present in the nucleus

Prokaryotic → Does not have a nuclear membrane and Histone

DNA Structure

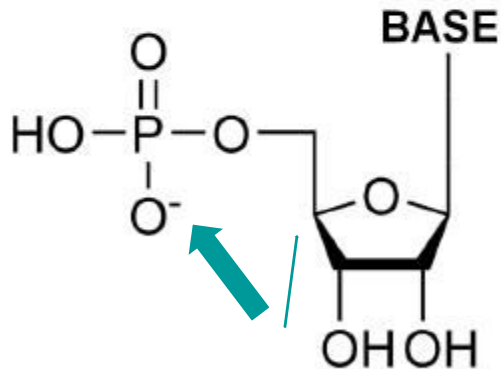
- Sugar (ribose) backbone
- Nitrogenous base
- Phosphate bonds



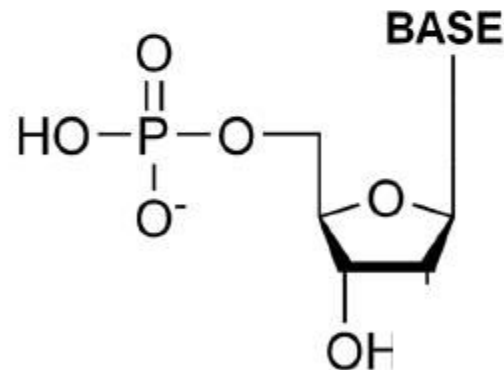
Wikipedia/PublicDomain

Nucleotides

- DNA: Polymer
- Nucleotide: Monomer
 - Pentose sugar
 - Nitrogenous base
 - Phosphate group



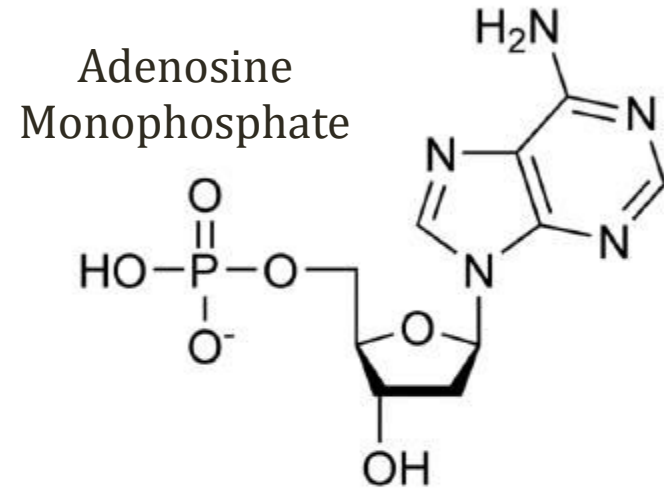
Ribonucleotide



Deoxyribonucleotide

Nucleoside Vs. Nucleotide

- Nucleotide
 - Nitrogenous base
 - Sugar
 - Phosphate group
- Nucleoside
 - Base and sugar
 - No phosphate group



Nitrogenous Bases

Purines



Adenine

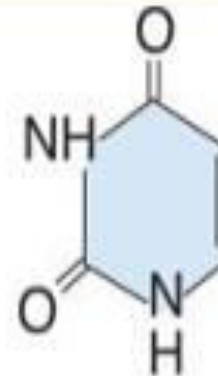


Guanine

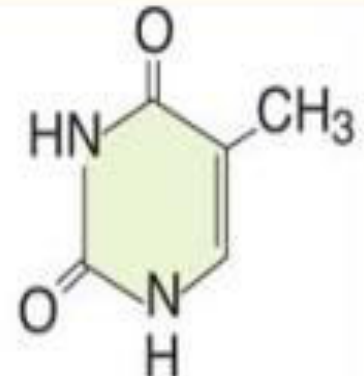
Pyrimidines



Cytosine



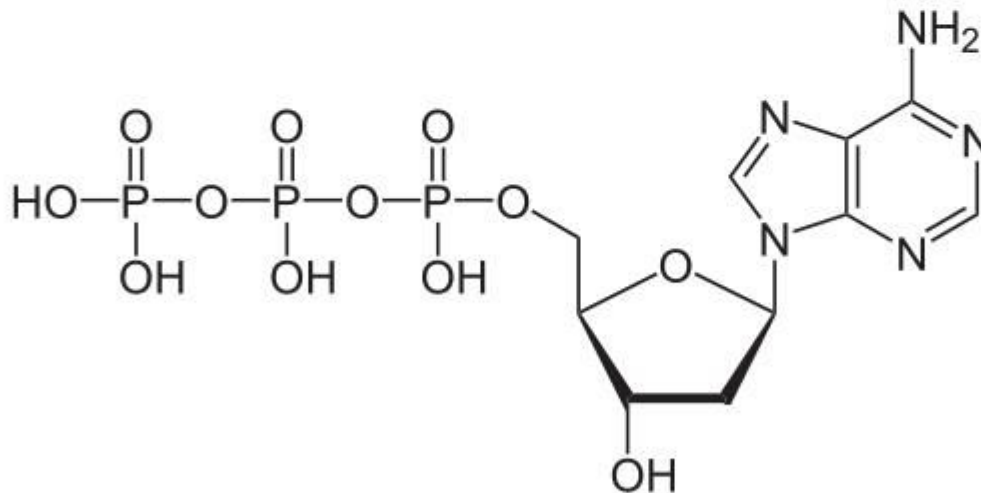
Uracil



Thymine

Nucleotides

- Synthesized as monophosphates
- Converted to the **triphosphate form**
- **Added to DNA**



Deoxyadenosine Triphosphate

Structure of DNA

- DNA is a **double-stranded** molecule (two strands)
- The two strands **wind** around each other forming a **double helix**
- **Each strand** of DNA molecule is a **polydeoxyribonucleotide**:
(i.e. many monodeoxyribonucleotides)
acid **b**ase (A,G,C,T) + **s**ugar (deoxyribose) + **p**hosphoric
- **Monodeoxyribonucleotides**: linked by 3' - 5' phosphodiester bonds (PDE) to form polydeoxyribonucleotide
- In **eukaryotic cells** (with a nucleus): DNA is associated with **various types of proteins (nucleoproteins)** present in the **nucleus**
- In **prokaryotes**, the protein-DNA complex is present in the **nucleoid**

3' - 5' phosphodiester bonds (PDE)

- **PDE** bonds join the **5'-hydroxyl** group of the sugar of one nucleotide to the **3'-hydroxyl** group of the sugar of an adjacent nucleotide through a **phosphate** group –to give a long un-branched chain of nucleotides.

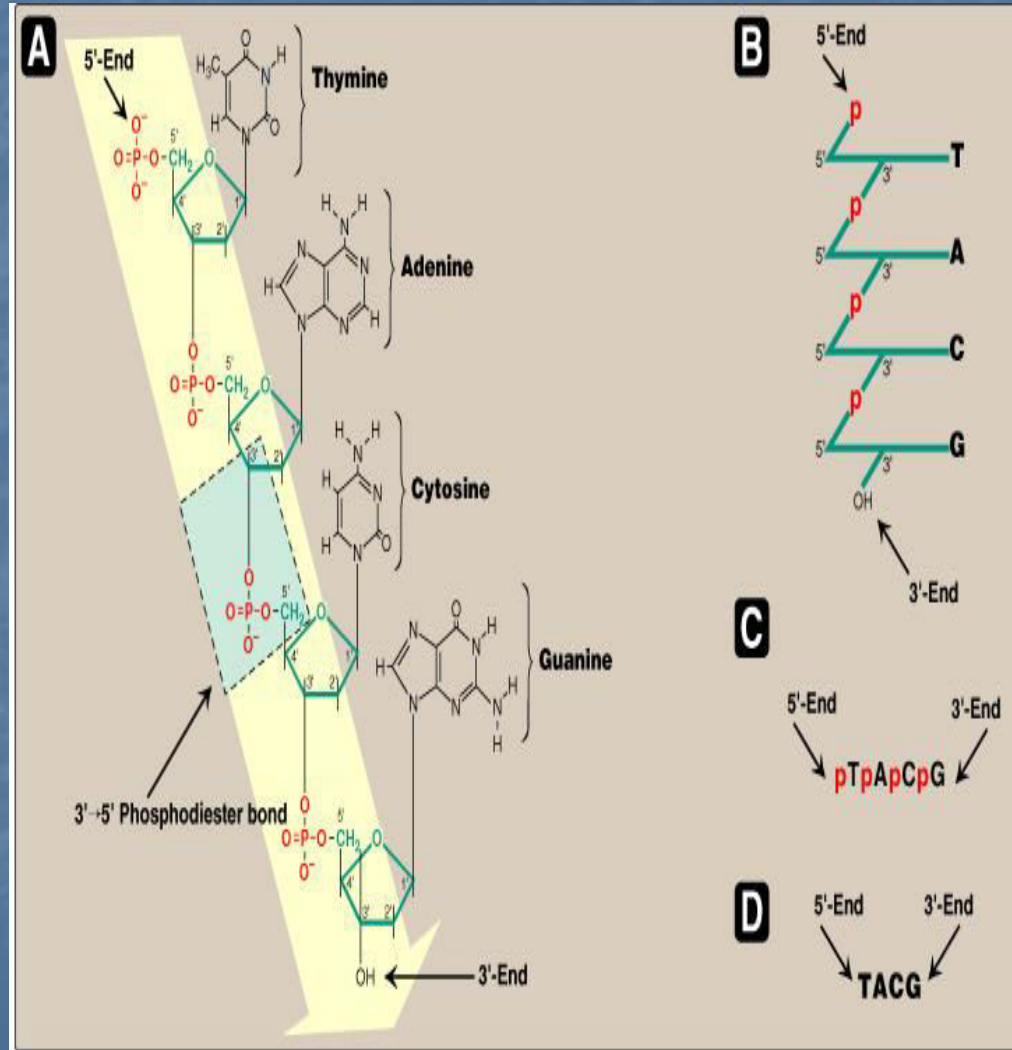
- The resulted long un-branched chain has a **polarity**,
5'-end the end with free phosphate
3'-end the end with free hydroxyl group
both ends are **not** attached to other nucleotides

- The bases along the backbone are written in sequence from **5'-end** of the chain to the **3'-end**

- Phosphodiester bonds can be **cleaved** by
1- chemicals
2- hydrolyzed enzymatically by **nucleases**.
(deoxynucleases: for **DNA** , ribonucleases: for **RNA**)

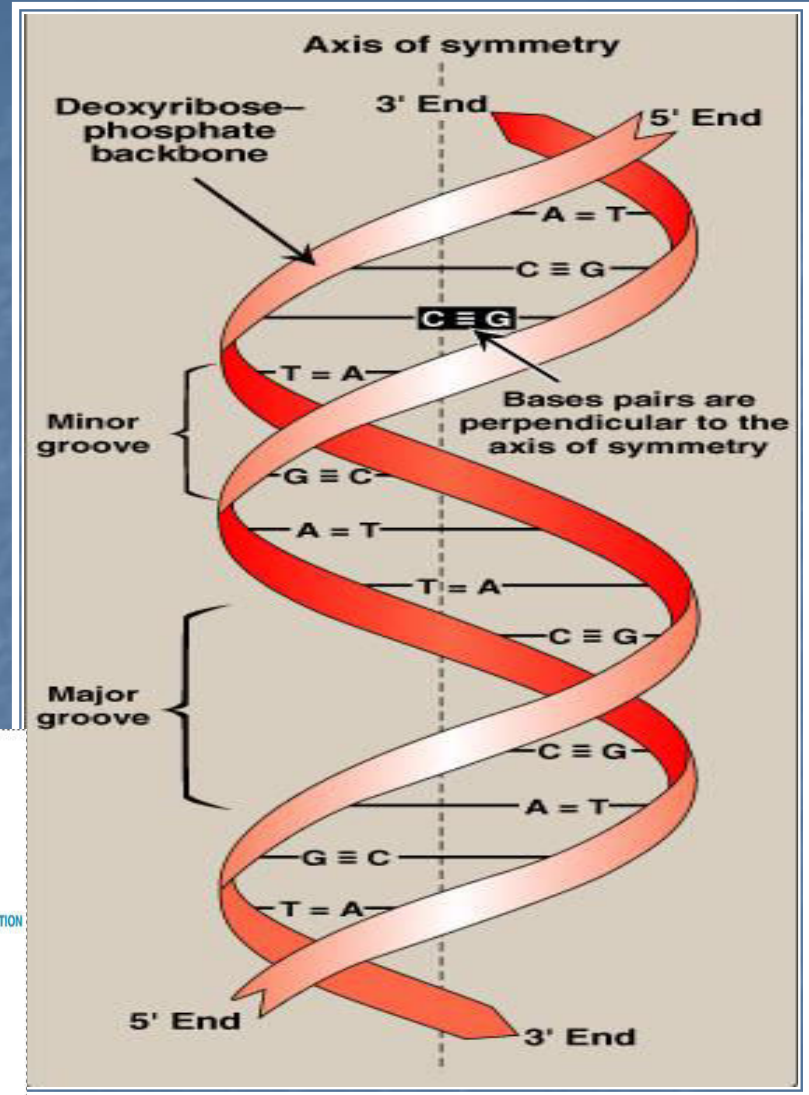
endonucleases: cleave the chain at positions in the interior of the chain

exonucleases: cleave the chain by removing individual nucleotides from one of the two ends



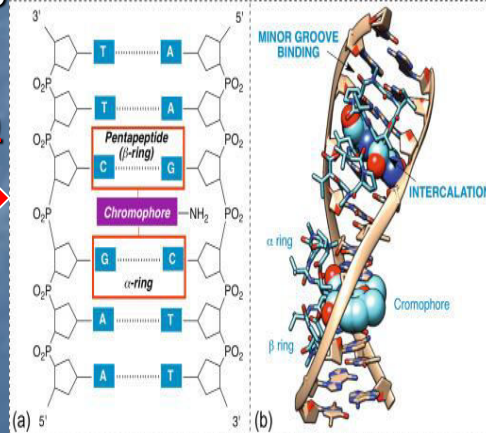
Double helix

- In the double helix, the two chains are coiled around a common axis called the axis of symmetry
- The chains are paired in an antiparallel manner
the 5'-end of one strand is paired with the 3'-end of the other strand
- The sugar-phosphate backbone of each chain is on the outside of the molecule (hydrophilic), whereas the bases are stacked inside (hydrophobic)
- Major groove & minor groove:** provide access for the binding of regulatory proteins for their recognition sequences along the DNA chain.



Dactinomycin (actinomycin D) (anticancer drug):

Anti-tumor activity is associated with DNA intercalation. It binds place in the DNA structure where topoisomerase binds with DNA (minor groove)



- intercalate into the minor groove of the DNA double helix - interfere with replication & transcription

Base Pairing

- DNA

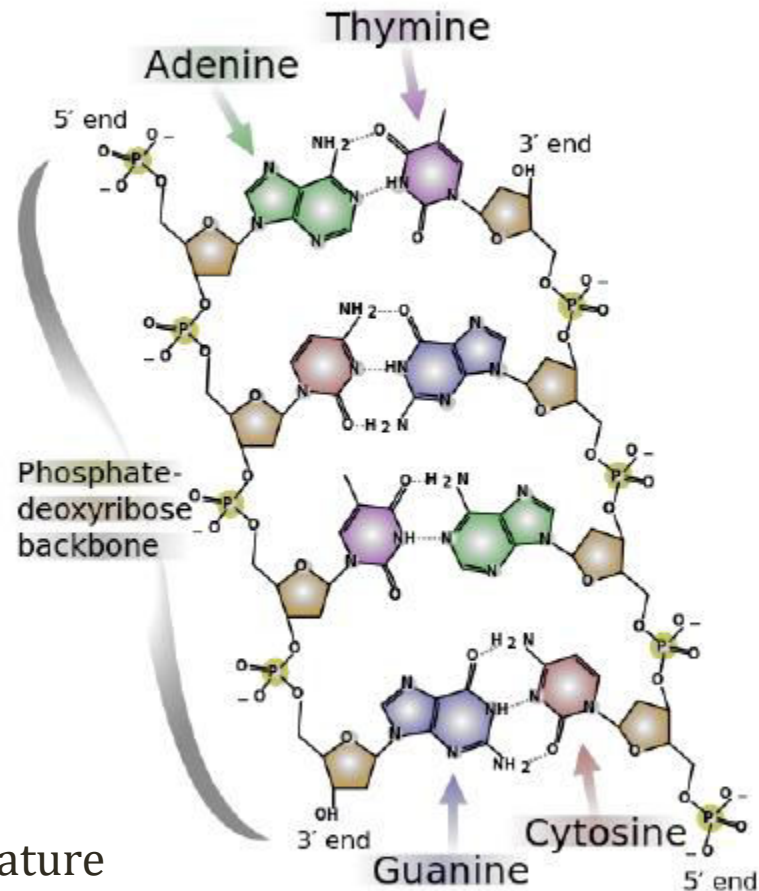
- Adenine-Thymine
- Guanine-Cytosine



- RNA

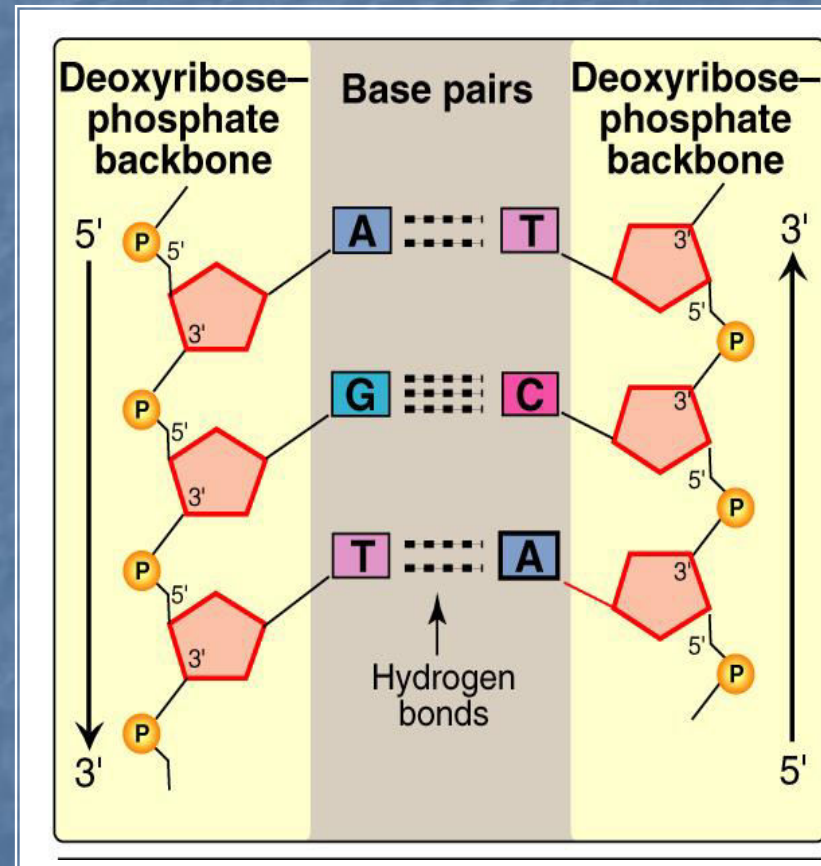
- Adenine-Uracil
- Guanine-Cytosine

More C-G bonds = $\uparrow$ Melting temperature



Base pairing

- The **bases** of one strand of DNA are **paired** with the bases of the second strand.
 - Adenine is paired with **Thymine**
 - Guanine is paired with **Cytosine**
- The base pairs are **perpendicular to the axis** of the helix
- Accordingly, one polynucleotide chain of the DNA double helix is always the complement of the other
- **Chargaff's Rules**: in any sample amount of double-stranded DNA, the amount of adenine equals the amount of thymine & the amount of guanine equals the amount of cytosine...
 - the total amount of purines equals the total amount of pyrimidines
- The base pairs are held together with **hydrogen bonds**
 - two between **A** & **T**
 - three between **G** & **C**



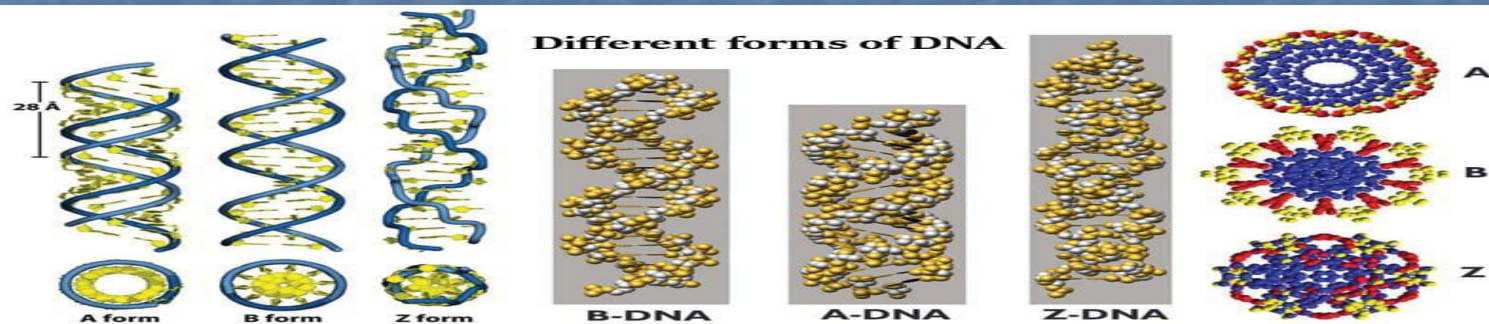
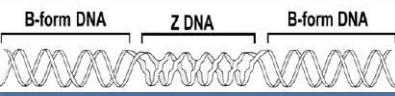
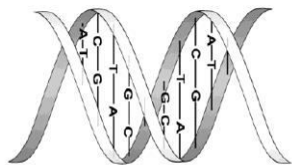
Separation of the two DNA strands in the double helix

- The two strands of DNA separate when hydrogen bonds between the paired bases are disrupted
- Disruption occurs in the laboratory when
 - 1- *pH* of the DNA solution is altered so that the nucleotide base ionizes
 - 2- Melting temperature (T_m): the temperature at which one-half of the the helical structure is lost.
 - The loss of helical structure is called **denaturation**
 - **In 1 & 2 the phosphodiester bond is not broken .**
 - Denaturation can be monitored by measuring DNA absorbance at 260 nm.
 - ✓ **single stranded DNA has a higher absorbance at 260nm than a double – stranded DNA**
- ❖ DNA that contains high concentrations of **A & T** (two hydrogen bonds) denatures at lower temperature than DNA rich in **G & C** (three hydrogen bonds)
- ❖ Complementary DNA – strand can reform of double helix called **(renaturation or reannealing)**

Structural forms of the double helix

A -form	B - form	Z-form
Right handed helix & is produced by dehydrating the B – form .	Right handed helix	Left handed helix
Eleven base pair per turn .	Ten residues per turn (360)	Twelve base pair per turn .
Tilted 20 a way from the perpendicular .	Bases are perpendicular	-----
The conformation found in DNA , RNA hybrids or RNA,RNA double stranded (not single) region is very close to the A- form .	Chromosomal DNA is consist primarily from thisB- form .	In DNA that have a sequence of alternating purine & pyrimidine . e.g: poly CG .

Canonical B-form DNA



Transition between these helical forms of DNA may play an important role in regulating Gene expression .

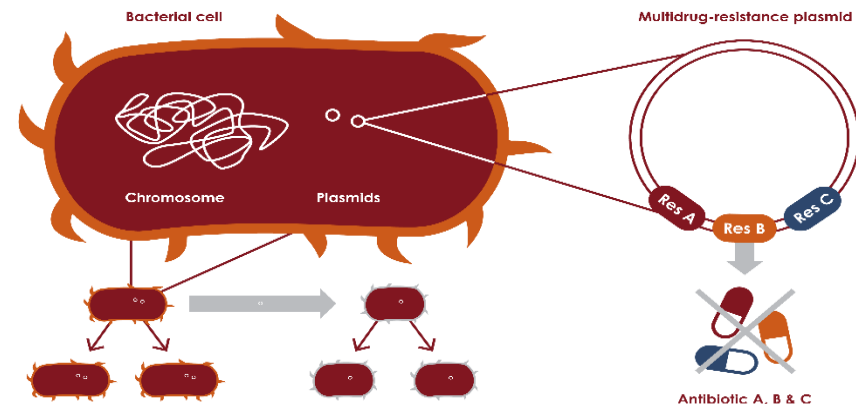
Circular DNA molecules

Eukaryotes	Prokaryotes
-One long linear molecule of double stranded DNA which bound to protein to Form chromatin. - closed circular DNA in mitochondria -----	- single , double – stranded , supercoiled , circular chromosome which bound to histone – like protein & RNA to form a nucleoid . Plasmid : a) small b) circular c) extrachromosomal DNA

❑ plasmid function :

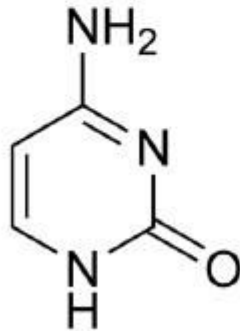
- carries genetic information .
- undergoes replication that may or may not be synchorized to chromosomal division .
- antibiotic resistance . 
- facilitate transfer of genetic information .
- vectors in recombinant DNA technology . 

Multidrug-resistance plasmids



DNA Methylation

- **Methyl group** added to **cytosine**
 - Occurs in segments with CG patterns (“CG islands”)
 - Both strands
- **Inactivates transcription** (“epigenetics”)
- Human DNA: ~70% methylated
- Unmethylated CG stimulates immune response



Cytosine



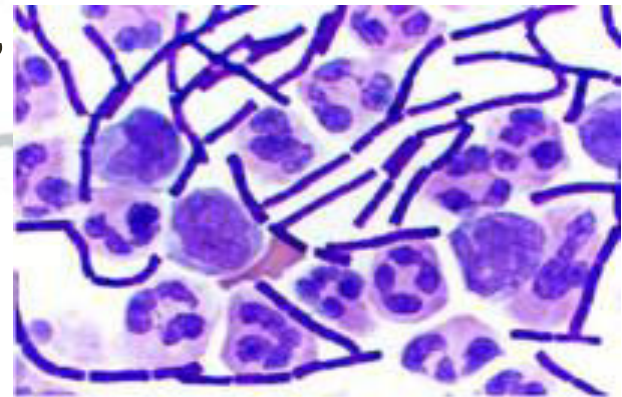
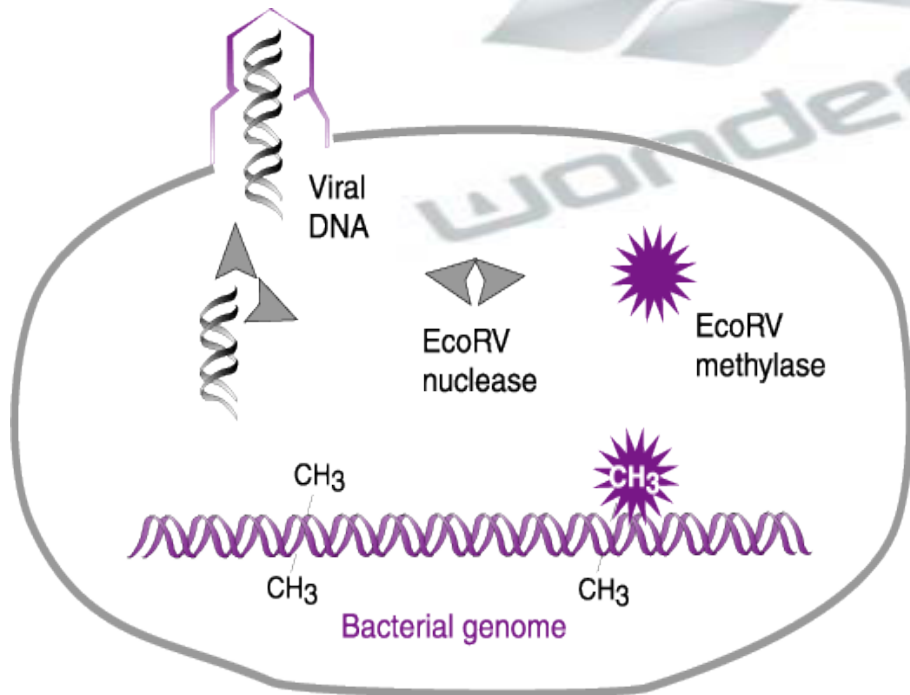
5-methyl cytosine

Epigenetics → stable phenotypic changes (known as **marks**) are **reversible by using the enzyme “DNA methyltransferase” enzyme**

Methylation is important for **regulating gene expression** by inactivating the replication (by methylation of the cytosine)

Bacterial DNA Methylation

- Bacteria methylate **cytosine and adenine**
- **Methylation protects bacteria from viruses (phages)**
- Non-methylated DNA destroyed by **endonucleases**
- “Restriction-modification systems”



The defense mechanism is termed the restriction-modification system (R-M system), and it consists of **two components: DNA methyltransferase and restriction endonuclease.**

Do not destroy the DNA of the same organism

Organization of the eukaryotic DNA

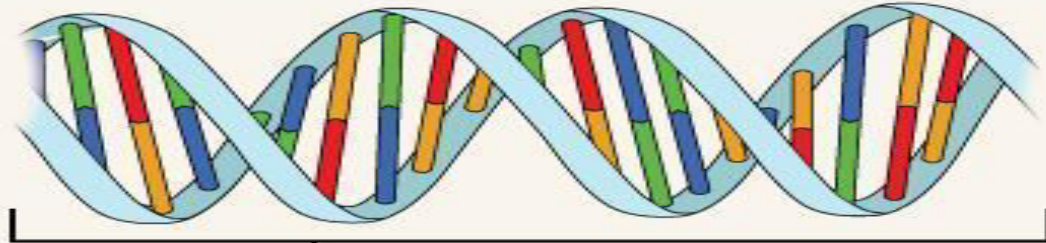
- Typical human DNA contains 46 chromosomes
- Total DNA is ~ one meter long
- It is important to effectively packaging the such large amount of genetic material into a volume of the size of a cell nucleus for effective replication & gene expression.
- To fulfill so, DNA interacts with large number of proteins
- Eukaryotic DNA is associated with basic proteins called **histones**.
- These serve to order the DNA into basic structural units called nucleosomes (resemble beads on a string)
- Nucleosomes are further arranged into more complex structures that organize and condense the long DNA molecules into chromosomes

Chromatin

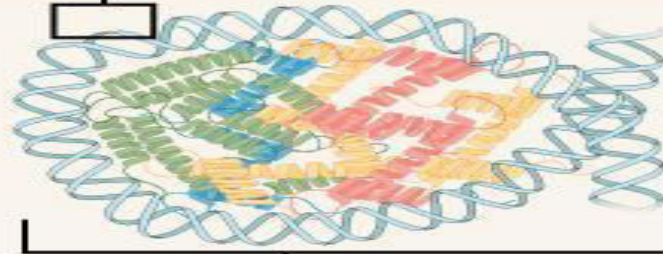
- Found in **the nucleus of eukaryotic cells**
- DNA plus proteins = **chromatin**
- Chromatin condenses into **chromosomes**

Organization of Eukaryotic Chromosomes

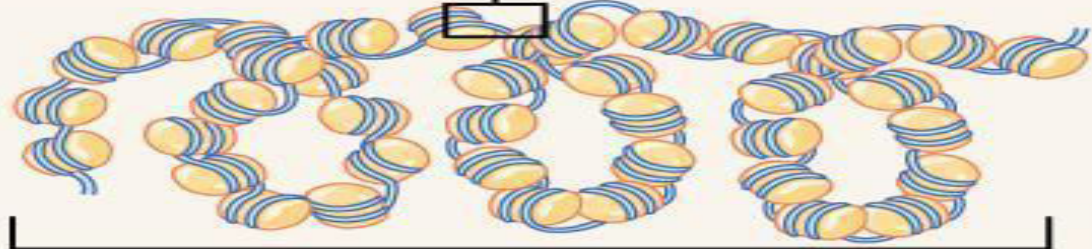
DNA double helix



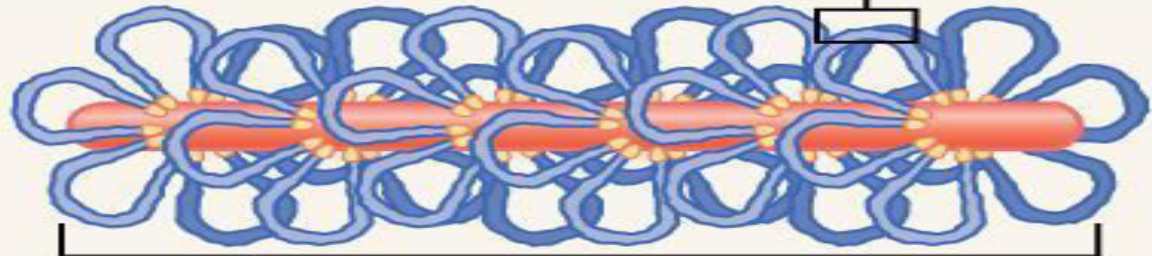
DNA wrapped around histone



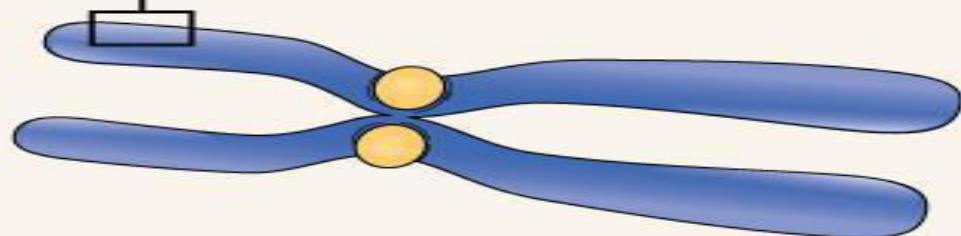
Nucleosomes coiled into a chromatin fiber



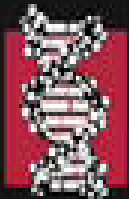
Further condensation of chromatin



Duplicated chromosome



DNA organization in Eukaryotic cell



www.dnalc.org

Histones and the formation of nucleosomes

- 5 classes of histones H1, H2A, H2B, H3 & H4
- Histones are positively charged at physiologic pH (high contents of lysine & arginine)
Arginine and **lysine** are **+ve ly** charged (capable to form **bond** with -ve ly charged DNA)
- They can form ionic bonds with negatively charged DNA
(in addition, Mg^{2+} can neutralize -Ve charged DNA phosphate groups)

1- Formation of the nucleosome

- Two molecules each of H2A, H2B, H3 and H4 form the structural core of the **individual nucleosome** (bead).
- Around this core, a segment of DNA double helix is wound **twice** forming a negatively super twisted helix.
- Neighboring nucleosomes are joined by **linker DNA** ~ 50 base pairs long.
- **H1** binds to the **linker DNA** chain between the nucleosome beads.
- **H1** is the most tissue-specific and species-specific of the histone.

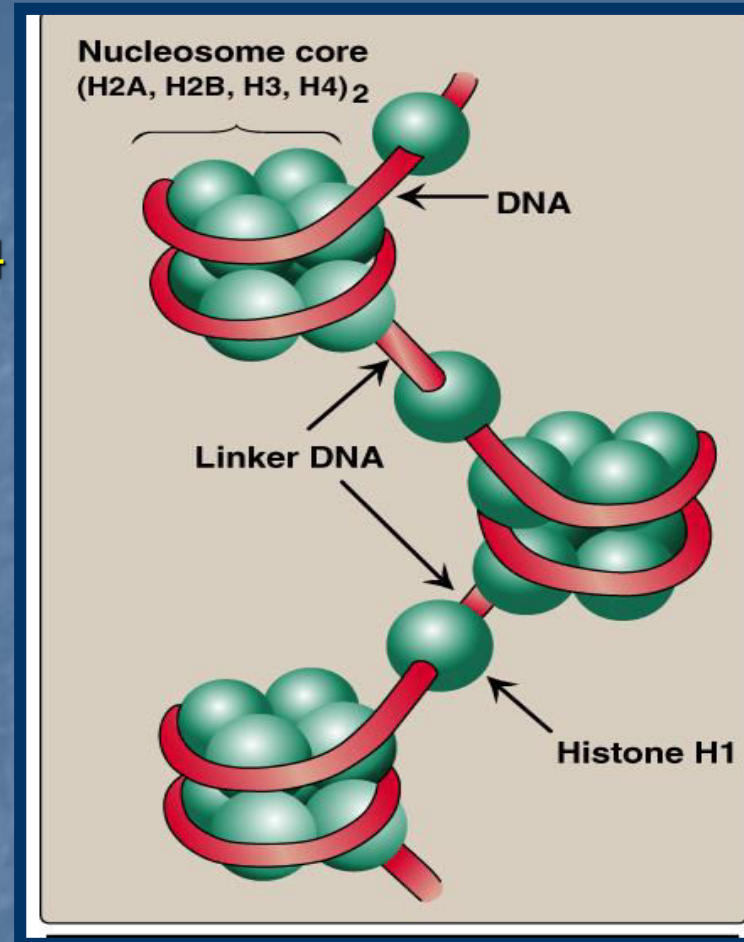
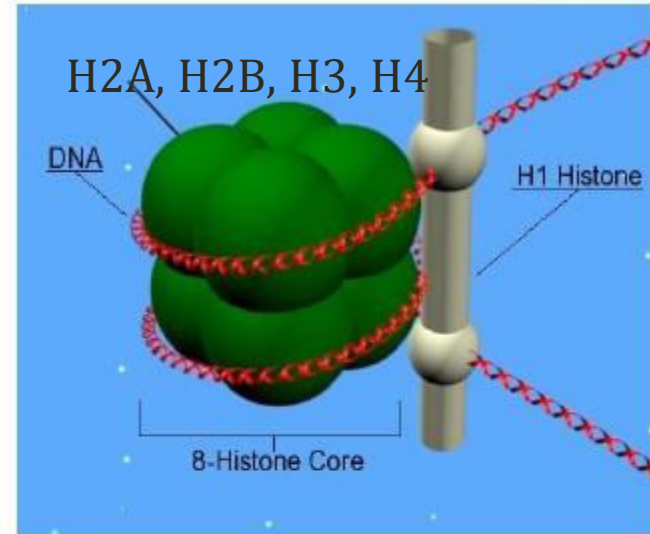


Fig. structure of nucleosomes

Histones

- Peptides
 - H1, H2A, H2B, H3, H4
- Contain **basic** amino acids
 - High content of **lysine, arginine**
 - **Positively** charged
 - Binds **negatively** charged phosphate backbone
- **H1 distinct from others**
 - Not in nucleosome core
 - Larger, more basic
 - Ties beads on string together



2- Higher levels of organization

- Nucleosomes can be packed more tightly to form polynucleosomes (nucleofilament **or** 30-nm fiber)
- The fiber is organized into **loops** that are anchored to **nuclear scaffold proteins**
- Additional levels of organization lead to the final **chromosomal** structure

Nuclear Scaffold Protein → **localizes the chromosome to specific areas of the cell (nucleus)...**

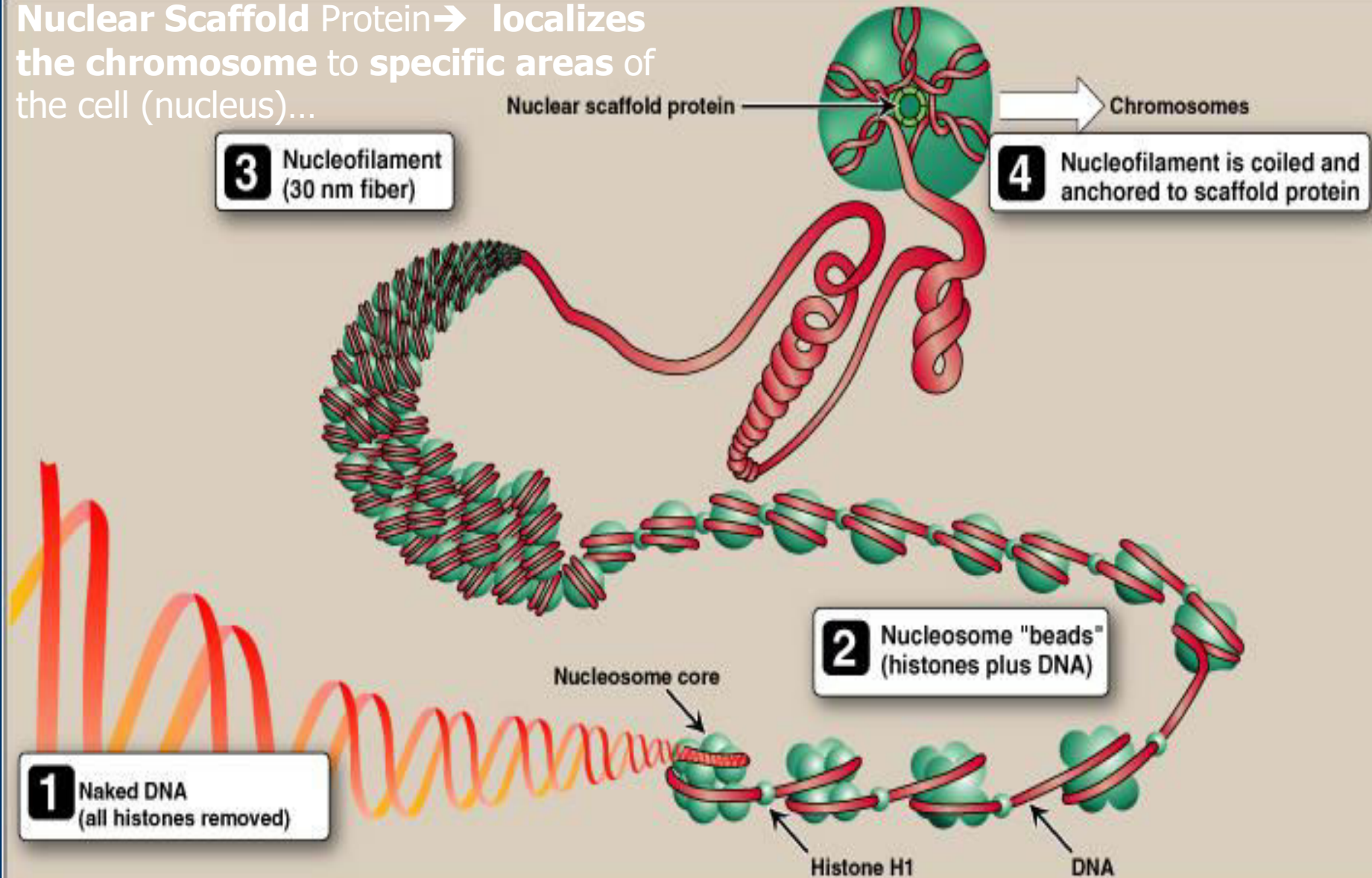
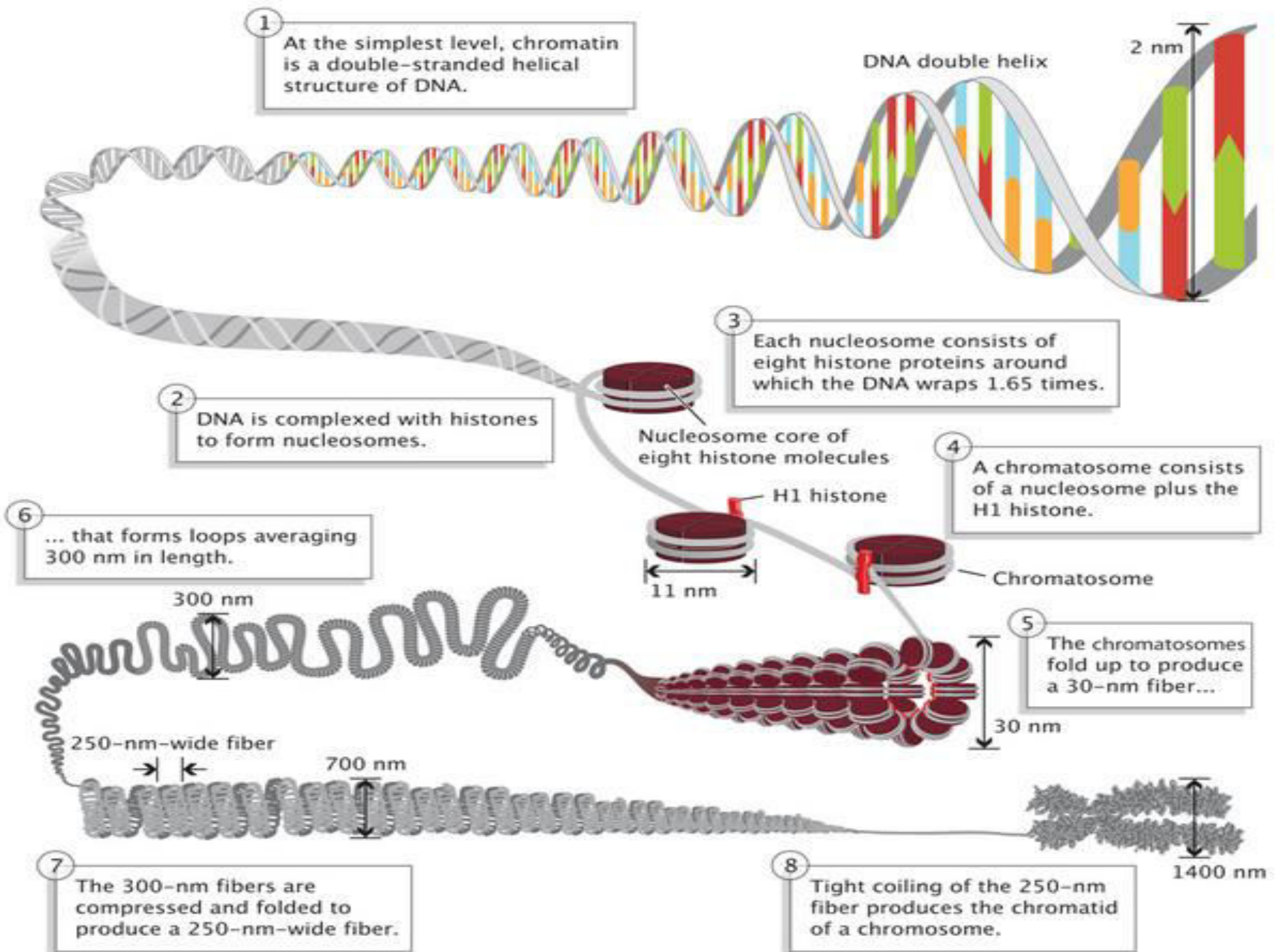


Fig. structural organization of eukaryotic DNA



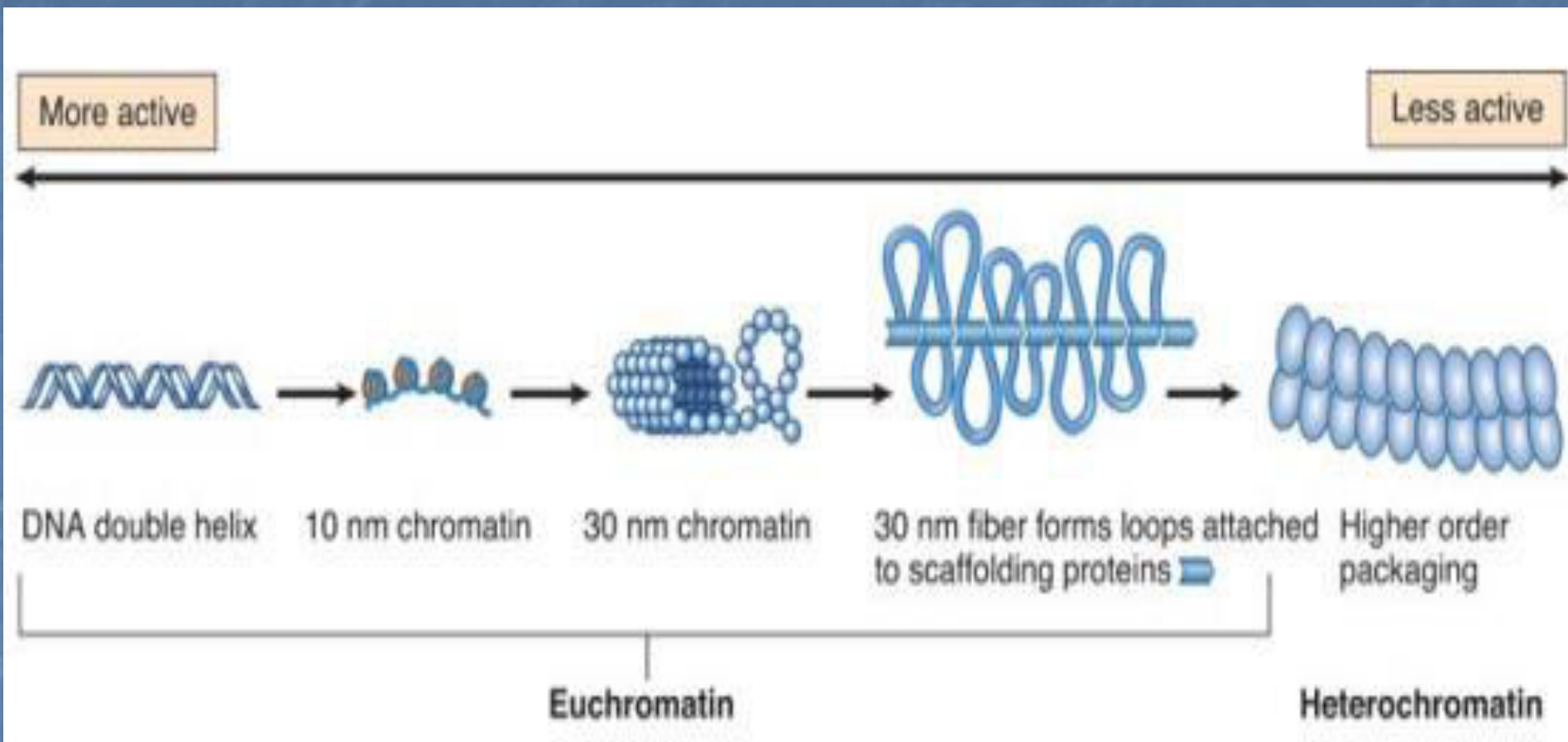
Chromatin Types

- **Heterochromatin**

- Condensed
- Gene sequences **not transcribed** (varies by cell)
- **Significant DNA methylation**

- **Euchromatin**

- Less condensed
- Transcription
- **Significant histone acetylation**

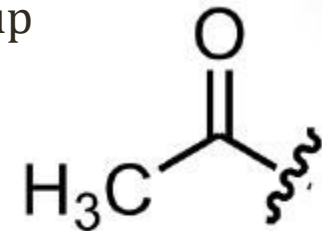


Euchromatin is the **lightly packed** form of **chromatin**, whereas **heterochromatin** refers to the **condensed form (higher-order packaging)**.

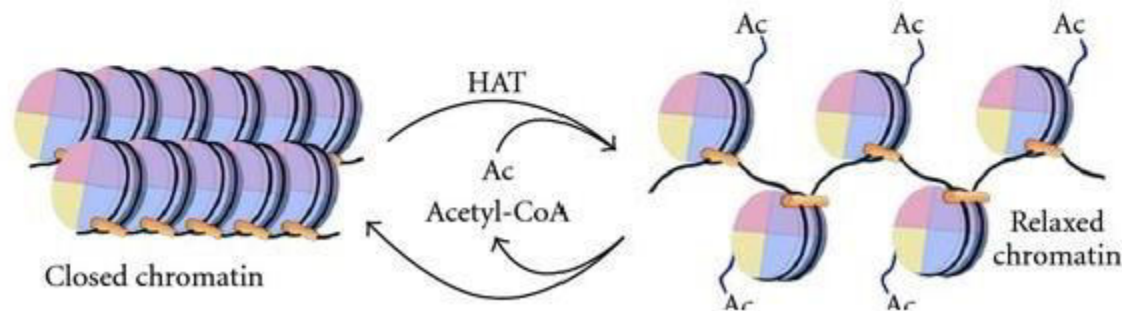
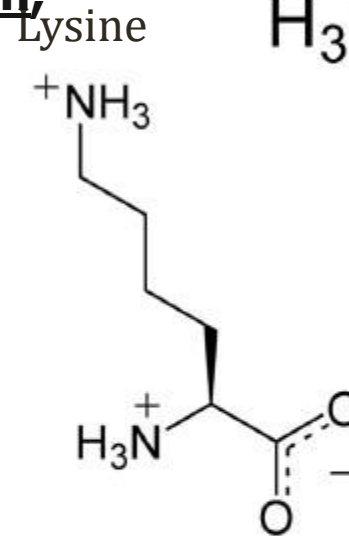
Histone Acetylation

Epigenetics → reversible regulation of gene expression, including **acetylation** and **methylation**,

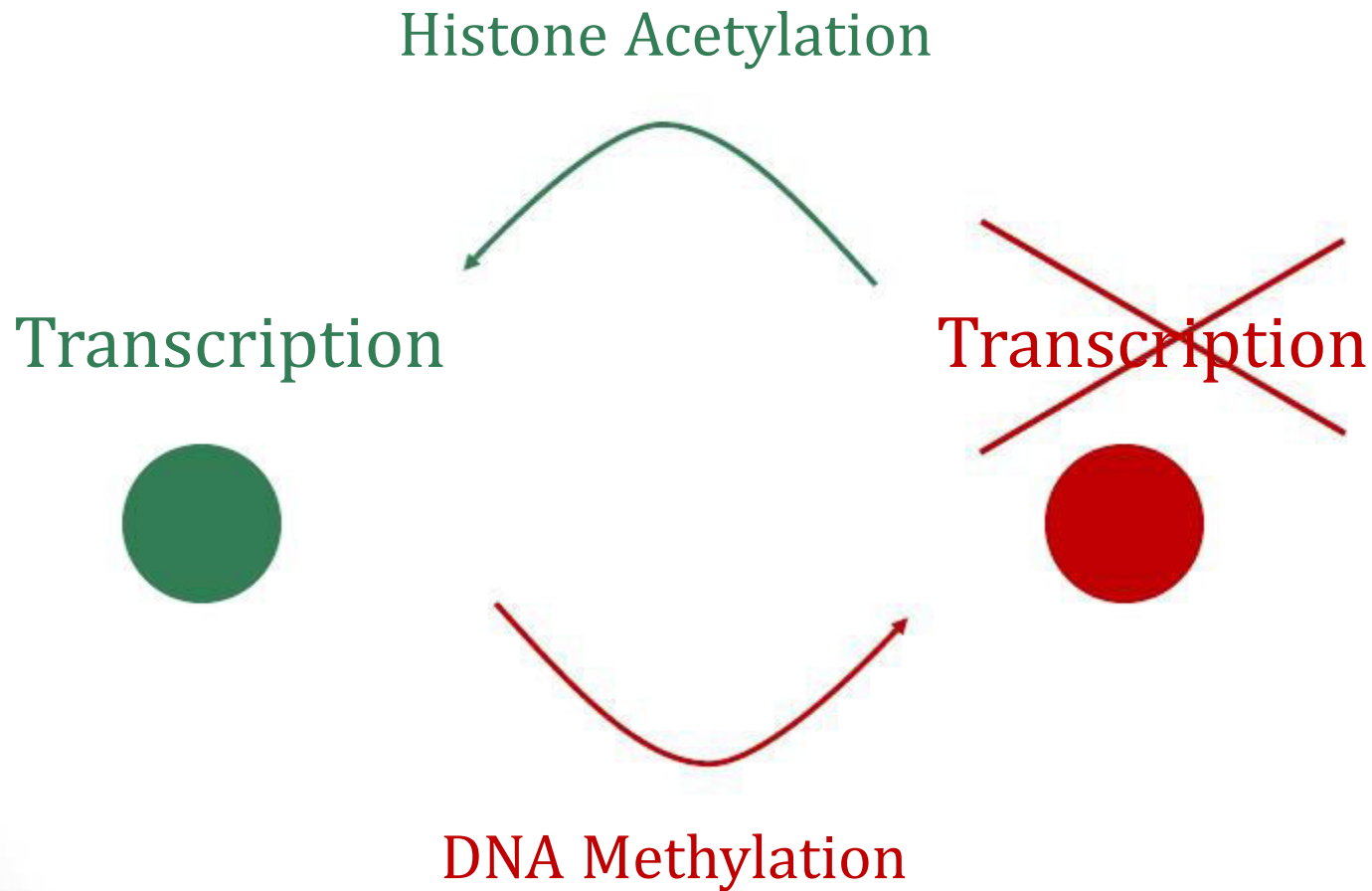
Acetyl
Group



- **Acetylation**
 - Acetyl group added to **lysine**
 - **Relaxes chromatin for transcription**
- Deacetylation
 - Reverse effect



Epigenetics



Histone deacetylase inhibitors

HDACs

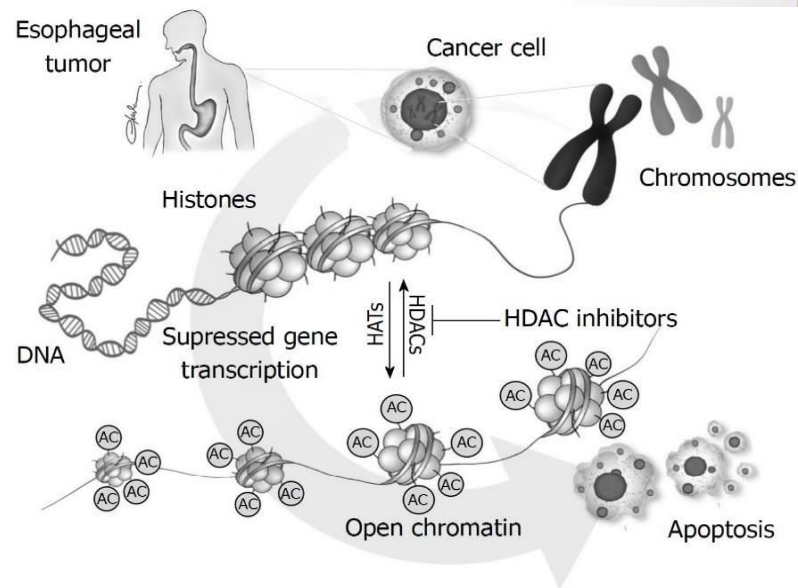
- Potential therapeutic effects

- Anti-cancer

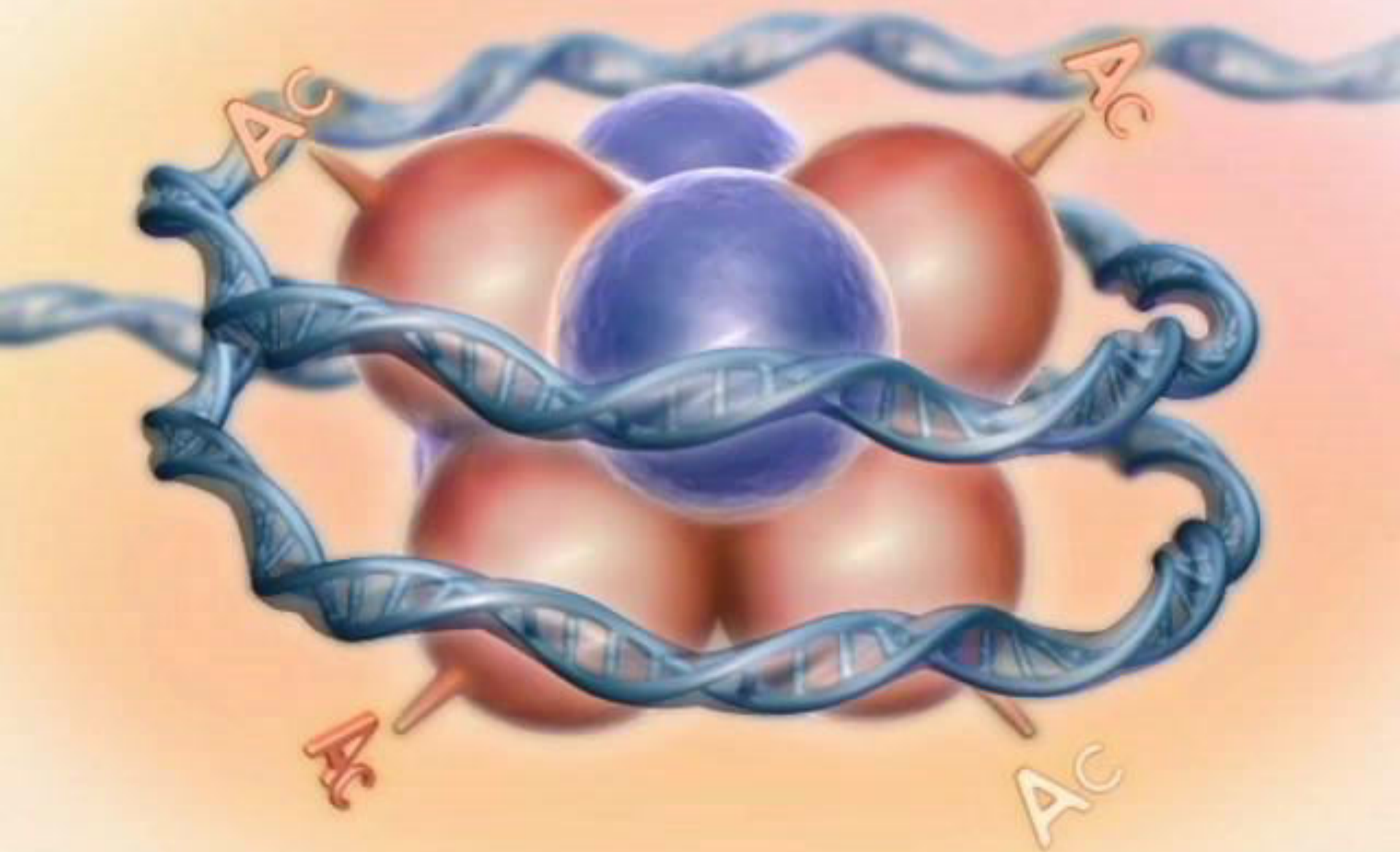
The cancer cell has abnormal epigenetics

Increased expression of HDACs in some tumors

- Huntington's disease
 - Movement disorder
 - Abnormal huntingtin protein
 - Gain of function mutation(mutant protein)
 - Possible mechanism: histone deacetylation → **gene silencing**
 - **This leads to neuronal cell death in the striatum**



Dokmanovic et al. **Histone deacetylase inhibitors: overview and perspectives**
Mol Cancer Res. 2007 Oct;5(10):981-9.



Fate of nucleosomes during DNA replication :

- highly structure chromatin must be relaxed .
- nucleosome displaced ,dissociation of the nucleosome core from the DNA is incomplete .
- All parental histones associated with (only one) of the parental DNA strand . (remember only one) (very important)
- Synthesis of new histones occurs simultaneously with DNA replication
And nucleosome containing the newly synthesized histones associate with
Only one of the new daughter helices .
- The parental histones octamer are **conserved** (NOT SEMICONSERVE)

* **Prokaryotic DNA synthesis (REPLICATION) :**

* **Semiconservative Replication :**

- When the two strands of the DNA double helix are separated, each can serve as a template for the replication of a new complementary strand.
- Each of the individual parental strands remains intact in one of the two new Duplexes . (T) . (important)
- (i.e. one of the parental strands is conserved in each of the two new duplexes) .

A. Separation of the two complementary DNA strands:

- **Before the replication , separation of the two parental strands must occur first why ?**

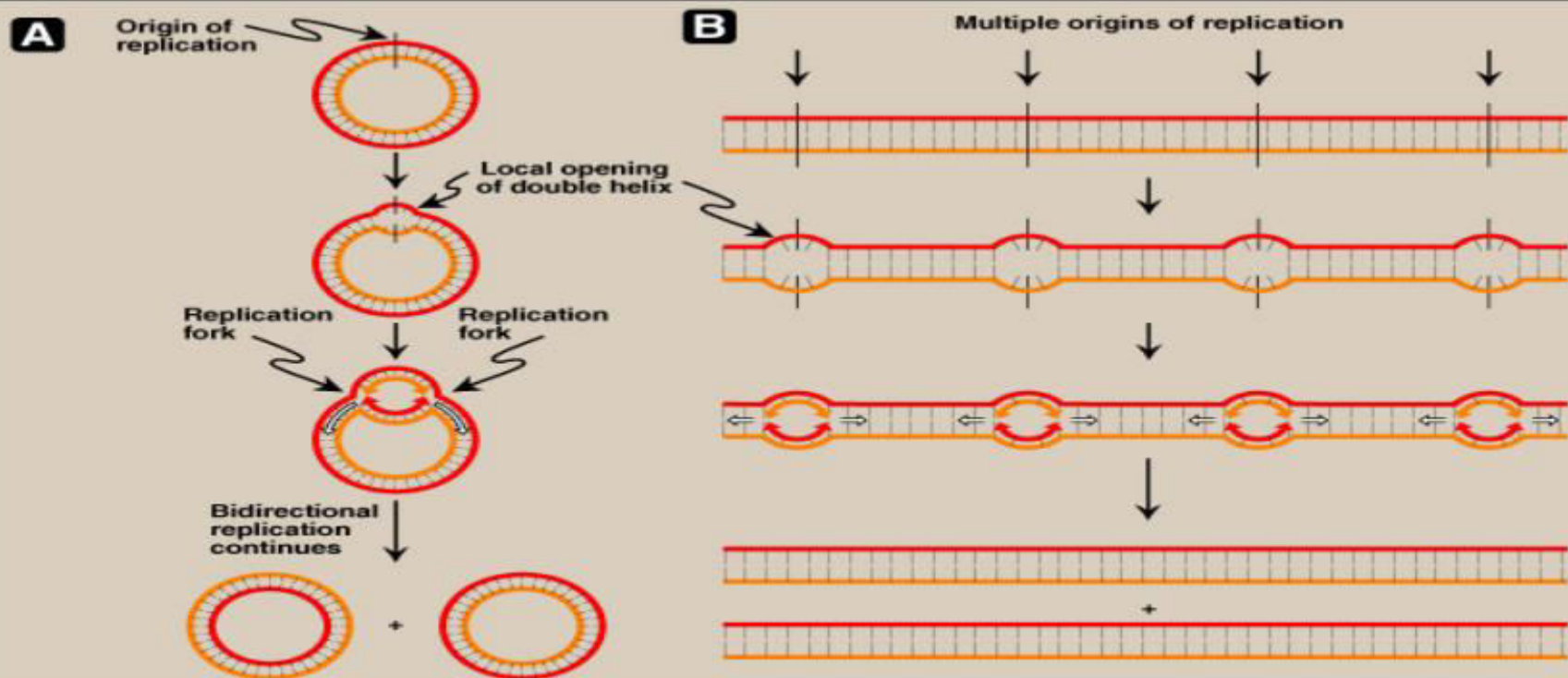
B. Origin of Replication:

Prokaryotes

DNA replication begins at **single**, unique nucleotide sequences (this site is called **origin of replication**)

Eukaryotes

-In eukaryotes, replication begins at **multiple** sites along the DNA helix this is referred to as **short consensus sequence** (exclusively composed of **AT base pairs**)
- **multiple origins of replication**.



Replication of DNA: origins and replication forks. A. Small prokaryotic circular DNA. B. Very long eukaryotic DNA.

C. Formation of the replication fork :

- As the two strands unwind & separate they form a V where active synthesis occurs , this region is called replication fork
- replication fork moves along the DNA molecule as synthesis occurs in both direction (bidirectional)

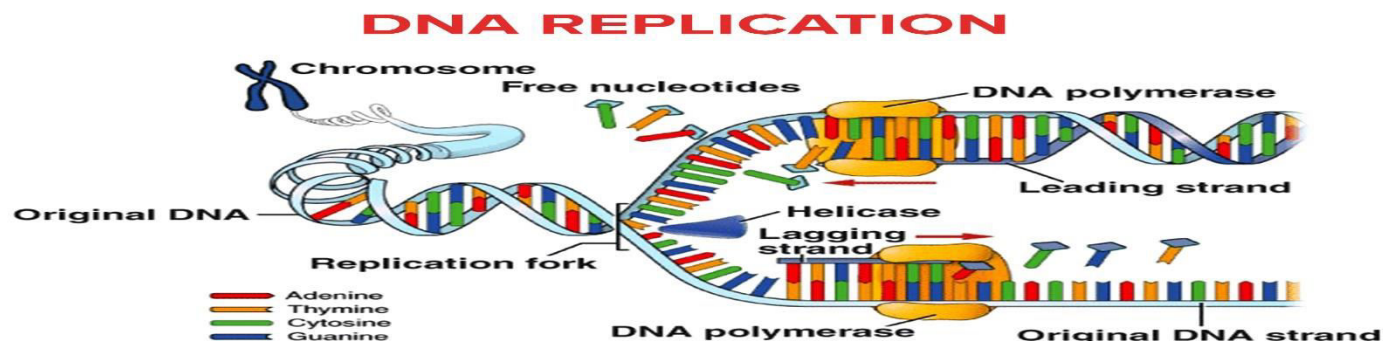
1. proteins required for DNA strand separation :

✓ called **prepriming** complex & include :

DNA protein

Single-stranded DNA-
binding(SSB) proteins

DNA helicases



DNA protein

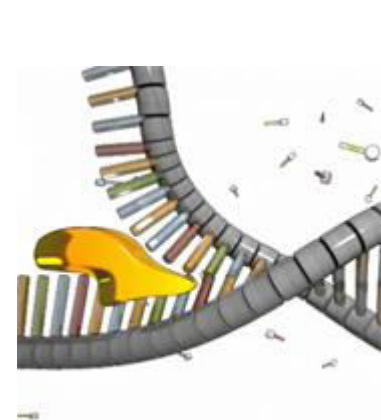
- 20 to 50 monomers bind to specific nucleotide sequences at origin of replication (which rich in AT base)
- ATP is required
- Cause double stranded DNA to melt (separate) & , forming localized regions of single stranded DNA .

Single-stranded DNA-binding(SSB) proteins

- Also called helix-destabilizing proteins
- Bind only to single stranded DNA
- Bind cooperatively (binding of one molecule make it easier for binding of additional molecules)
- Are not enzyme (but shift the equilibrium between single & double stranded DNA in the direction of single stranded form)
- Functions of SSB proteins:
 - I. Keep the two strands of DNA separated in the area of replication origin (prevent reformation of double-helix
 - II. Protect the DNA from nucleases (that cleave single stranded DNA)

DNA helicases

- Bind to single stranded DNA near the replication fork , then move to neighboring double stranded region , & unwinding the strands
- ATP is required
(**be careful DNA helicase Bind to single strand**)



* Solving the problem of supercoils☺ :

☐ **Positive** supercoils (supertwist) in the region of DNA **ahead** (be careful not before) of the replication fork interfere with further unwinding of double helix .

DNA topoisomerases

are responsible for removing these supercoils

☐ Negative supercoils :

contain fewer turns of helix than relaxed DNA.

☐ Positive supercoils :

contain more turns of helix than relaxed DNA.

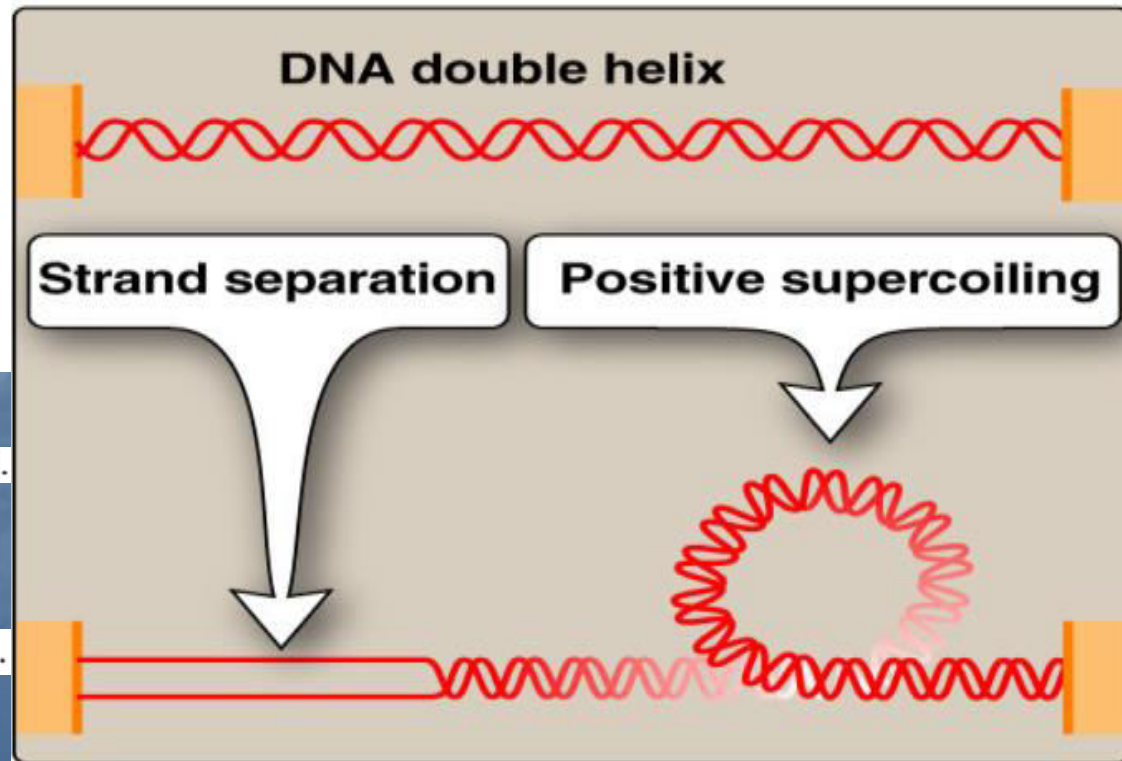


Figure 29.11

Positive supercoiling resulting from DNA strand separation.

DNA topoisomerases I

- Reversibly cut a single strand of double helix . (MCQ)(NOT double)
- (by nuclease then ligase) (هام جدا)
- * nuclease (strand – cutting)
- * ligase (strand resealing)

- They do not require ATP (but store the energy from cleaving of phosphodiester bond)

- Reliving (relaxing) of:
 - ✓ Negative supercoils in **prokaryotic** DNA (e.g. in *E. coli*)
 - ✓ Negative & positive supercoils in **eukaryotic** DNA

- Transient "nick" in one DNA Strand (NOT Both strand)
- the intact DNA –strand is passed through the break before resealed .

DNA topoisomerases II

- cuts double strands
- causes a second stretch of the DNA double helix to pass through the break & finally , reseals the break .

USE ATP

- Relax both **positive & negative supercoils**
- Required for separation of interlocked molecules of DNA following a chromosomal replication (**in prokaryotes & eukaryotes**)

- DNA gyrase : (لم يخلو منه اختيار)
- a) A type II topoisomerase
- b)found in *E.coli*
- c) Introduce negative supercoils
- d)into relaxed circular DNA
- e)(to facilitates future replication of DNA)
- f) Use ATP .
- Antimicrobials against bacteria DNA gyrase), quinolones (as ciprofloxacin)
- Anticancer agents : etoposide (target human topoisomerase II)

DNA Gyrase→ It relaxes positive supercoils and **reduces topological** strain in an **ATP-dependent** manner

Ciprofloxacin→ Antibiotics inhibit **DNA gyrase** function and **convert them into toxic enzymes** that produce permanent double-stranded breaks in the bacterial chromosome

Etoposide is used to treat a certain type of **lung cancer** (by forming a complex with **TI 2**, causing **ds DNA break**)

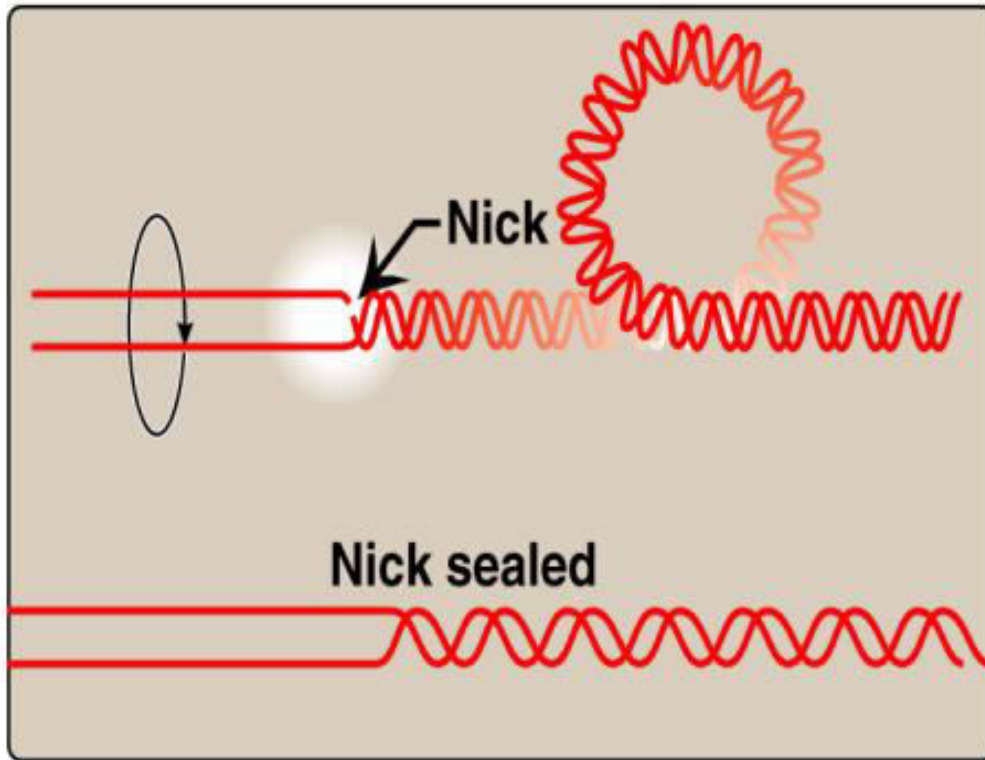


Figure 29.12
Action of type I DNA topoisomerases.

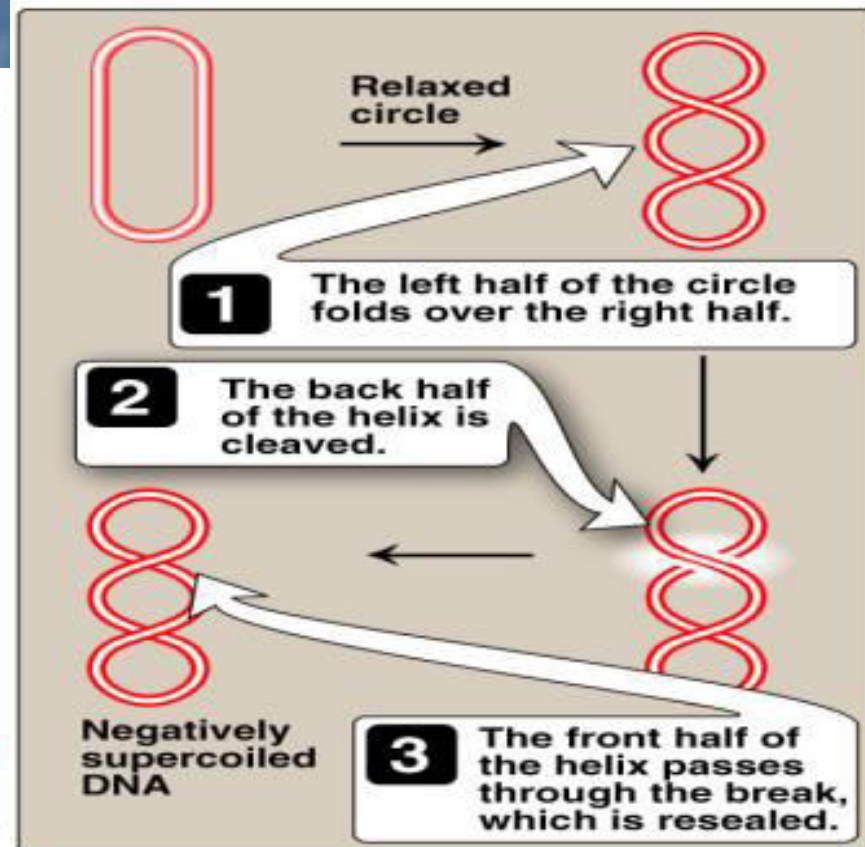


Figure 29.13
Action of type II DNA topoisomerase.

Topoisomerase 1 and 2



Topoisomerase solves

A KNOTTY PROBLEM

D.Direction of DNA replication:

- DNA polymerases (that responsible for of DNA template are):
 - ✓ Only read parental nucleotide sequences in $3' \rightarrow 5'$ direction
 - ✓ They synthesized the new DNA in $5' \rightarrow 3'$ antiparallel direction

■ So, beginning with one prenatal double helix,
the two newly synthesized DNA must grow in opposite directions:

1- One in $5' - 3'$ direction towards the replication fork

2- One in $5' - 3'$ direction away from the replication fork.

1. Leading strand:

- copied in $5' \rightarrow 3'$ direction , toward the replication fork
- synthesized almost continuously

2. Lagging strand:

- copied in $5' \rightarrow 3'$ direction , away from the replication fork
- synthesized discontinuously with small fragments of DNA near the replication fork , is called okazaki fragments
- note : okazaki fragments are eventually joined to become a single continuous strand

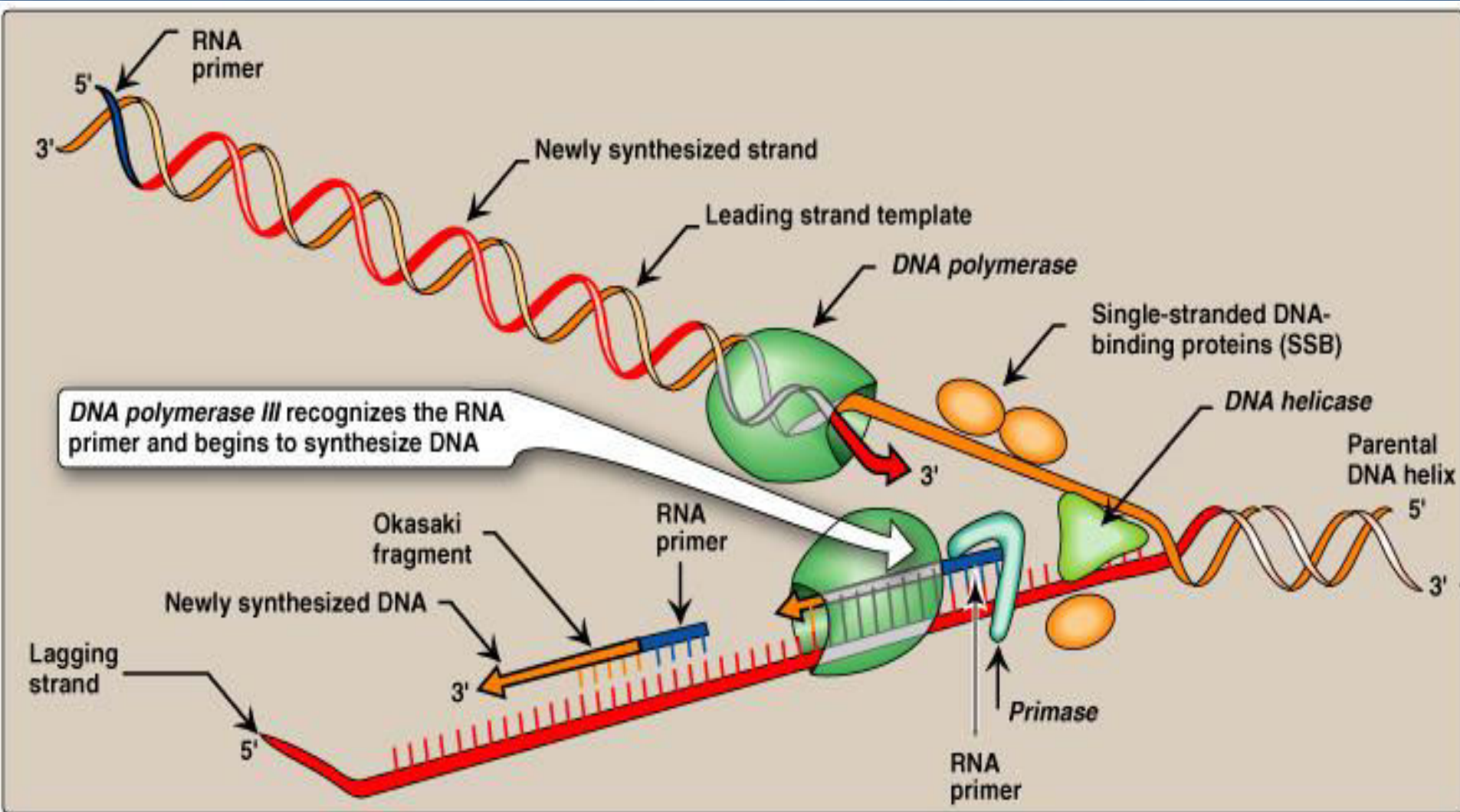


Figure 29.16
Elongation of the leading and lagging strands.

Primase makes each **RNA primer** (10 Ntds length)

Polymerase III are prokaryotic enzyme

D)- RNA Primer Synthesis : (we will mention a lot of details)

- DNA polymerase **can not** initiate synthesis of a complementary strand of DNA on a totally single stranded template Rather, they require an **RNA primer** .

■ RNA primer :

- a short , double stranded (**Not single**) fragment of (RNA paired to the single stranded DNA template) . (very important)
- has a free –OH group on the 3`-end of the RNA that serves as an acceptor of the first nucleotide by action of DNA polymerase .

■ This note always come :

* in de novo DNA synthesis , the free 3 – OH is provided by(**short stretch**) of :

- a) RNA (T)
- b) DNA (F)

the answer is **short stretch of RNA** .

1) primase :

- It is a specific RNA polymerase .

Q , primase is DNA polymerase . (F) (RNA polymerase) .

- it is DNA dependent (T) – its RNA dependent (F) (very important)
- It synthesizes short (not long) (MCQ) fragments of RNA (~ 10 nucleotides), complementary & antiparallel to DNA template.
- RNA primers hybrid with DNA template (U in RNA pairs with A in DNA) (Hybrid complex)

* In the replication fork

- In the **leading strand**: only one RNA primer is synthesized.
- In the **lagging strand**, many RNA primers are constantly synthesized along the strand .

■ RNA primase building block :

- a) 5 -ribonucleoside triphosphate . (NTP) (be careful not deoxyribo)
(**ATP** , **GTP** , **CTP** , **UTP**) (**NO TTP**)
- b) pyrophosphate is released so the reaction is **irreversible** .

2) *Primosome* :

- (Primase + Prepriming complex)
- So primosome consist of 4 componenet :
 - a) primase .
 - b) DnaA protein .
 - c) Single – stranded DNA – binding (SSB) protein .
 - d) Helicases .
- This complex initiates Okazaki fragment formation by moving along the template for the lagging strand in **the 5'-3direction**
- periodically recognizing specific sequences of nucleotides that direct it to create an RNA primer that is synthesized in the 5'-3' direction.

▣ revision :

- a) 3 – end OH is the first acceptor nucleotide by the action of DNA polymerase .
- b) okazaki fragment in **the 5'-3direction** .
- c) RNA primer is synthesized in the 5'-3' direction.

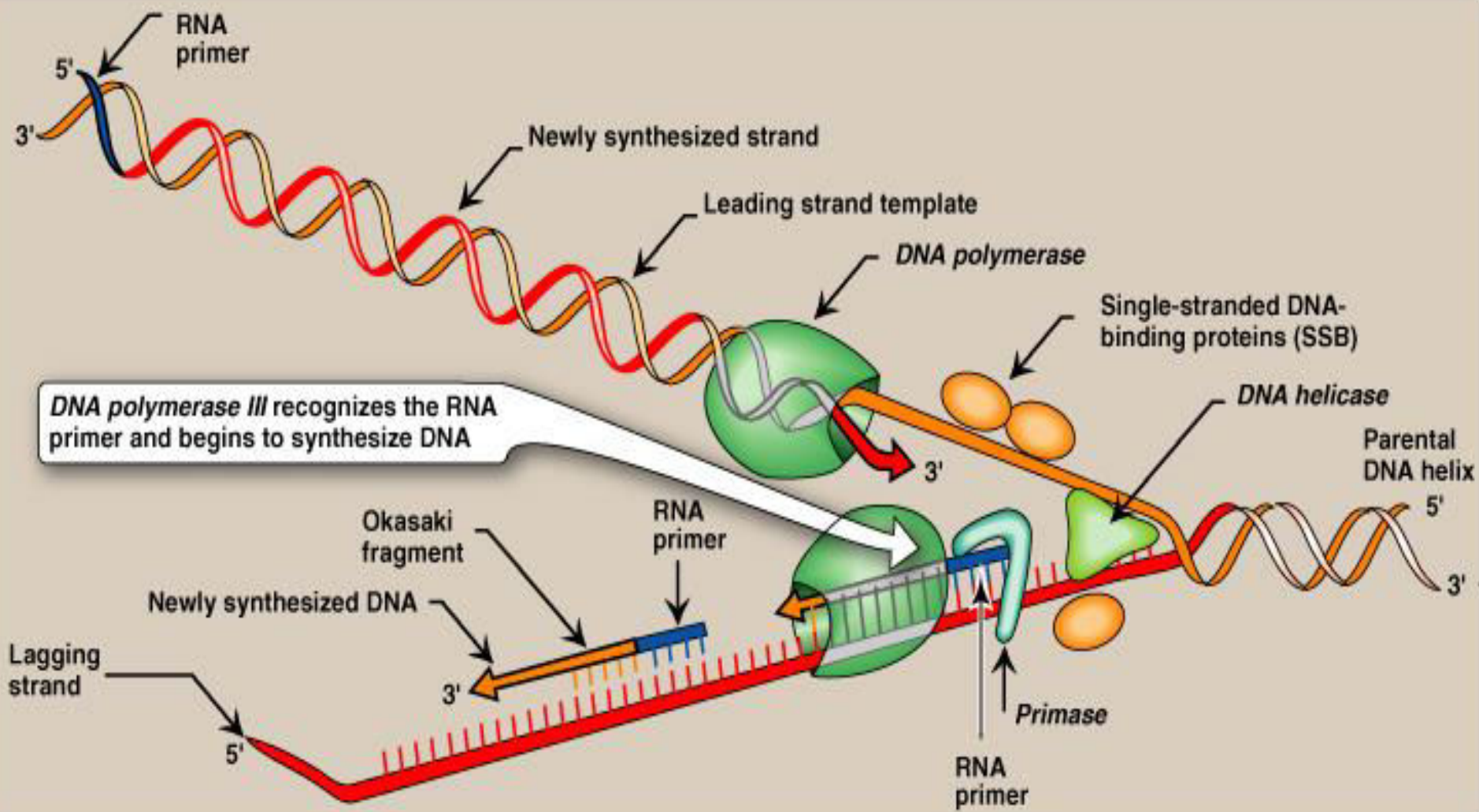


Figure 29.16
Elongation of the leading and lagging strands.

E) Chain elongation :

- DNA polymerase add deoxyribonucleotide one at a time .
_(not like RNA primase which add ribonucleotide)

* DNA polymerase III :

- elongates a new DNA strand by adding deoxyribonucleotides dNTP (dATP, dGTP, dCTP, TTP)
(NOT UTP) to the 3'-end of the growing chain (one at a time) starting from the RNA primer.
- The sequence of the nucleotides that are added is dictated by the base sequence of the template strand.
- The new strand grows in the 5'- 3' direction, antiparallel to the parental strand.
- DNA polymerase **III** is a highly processive enzyme .
{ processive enzyme means : remains bound to the template strand }

☐ Energy required:

- Pyrophosphate (Ppi) is released when each new nucleotide is added to the growing chain
- Further hydrolysis of Ppi to two Pi
- Accordingly, a total of two high-energy bonds are used to drive the addition of each dNTP

☐ **Rivision :**

Location of work: Leading & lagging DNA strands starting from the RNA primer(s).

Enzyme : DNA polymerase III

Materials: dNTPs (dATP, dGTP, dCTP, dTTP)

Direction of elongation: 5'-3' direction antiparallel to parental strand

Energy required: two high-energy bonds

example: $dATP \Rightarrow dAMP + Ppi$ ($Pi + Pi$)

2) Proofreading of newly synthesized DNA:

- Misreading of the template sequence could result in mutations. (may be lethal) .

=> very important

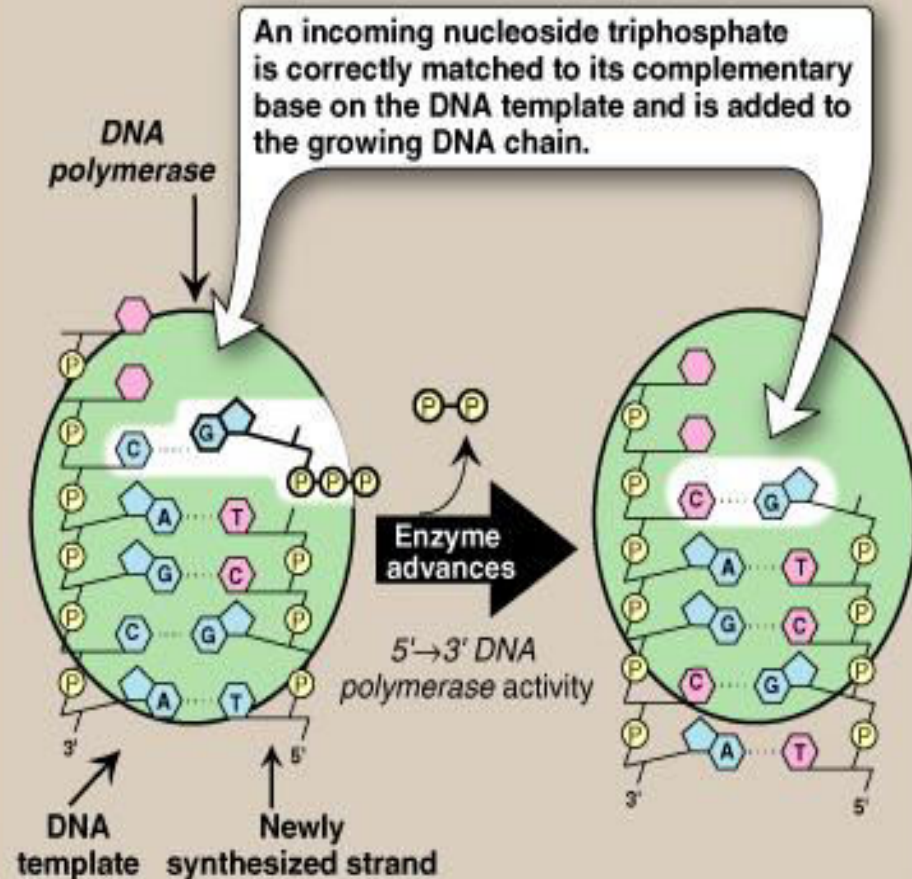
- To ensure replication correctness of sequence, DNA polymerase III has a proofreading activity (3'-5' exonuclease activity)
(3' to 5' be careful)

■ Steps:

- 1- DNA polymerase III checks for the correctness of matching of added nucleotide to its complementary base on the template.
- 2- If a wrong nucleotide is added, 3'-5' exonuclease activity edits the mistake

■ Why 3'-5' exonuclease does not degrade correctly paired nucleotide sequence?

A POLYMERASE FUNCTION



B PROOFREADING FUNCTION

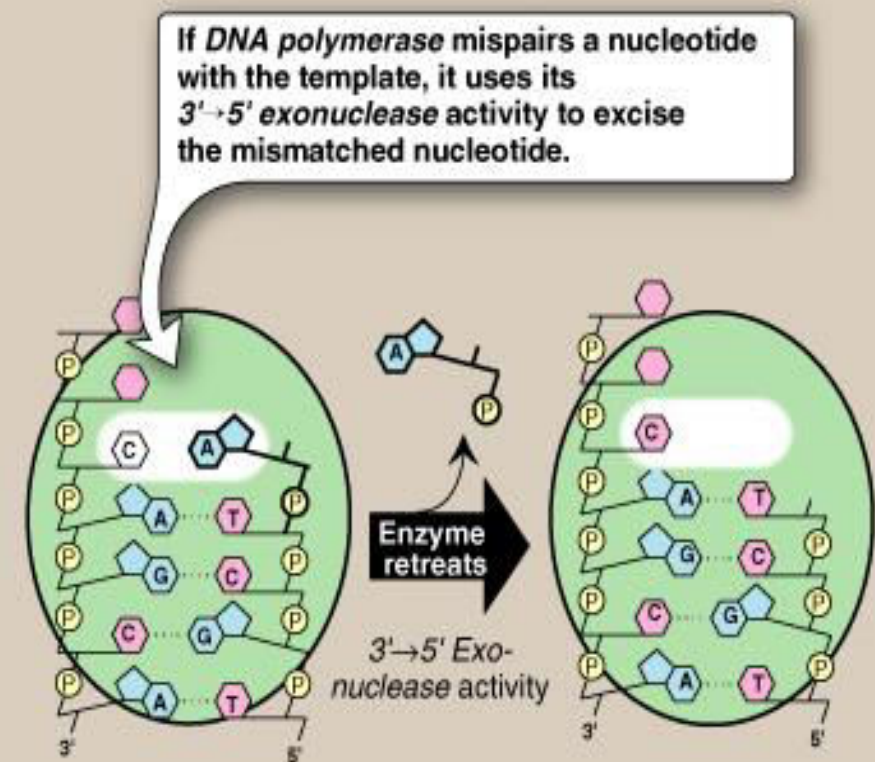


Figure 29.17

3'→5'-Exonuclease activity enables *DNA polymerase III* to "proofread" the newly synthesized DNA strand.

Conclusion : (the excision must be done in the reverse Direction from that of the synthesis)

F . Excision of RNA primers and their replacement by DNA :

- On the lagging strand :

(DNA polymerase III continues to synthesize DNA until it is blocked by an RNA primer.)

■ At that stage,

- 1- RNA primer is removed (excised) by DNA polymerase I
(by 5'-3' **exonuclease activity**) (be careful not endonuclease)
SO (the DNA polymerase I locates the space " nick " between the 3-end
Of the DNA newly synthesized by DNA polymerase III & the 5 – end
Of the adjacent RNA primer .
- 2- Then, the gap is filled by DNA polymerase I
(by 5'-3' **polymerase activity**)
- 3- DNA polymerase I proofreads the filled gap DNA
(by 3'- 5' **exonuclease activity**)

■ *The final result :*

* This removal / synthesis / proofreading continues (**one nucleotide at time**) , until
The RNA is totally degraded & the gap is filled with DNA .

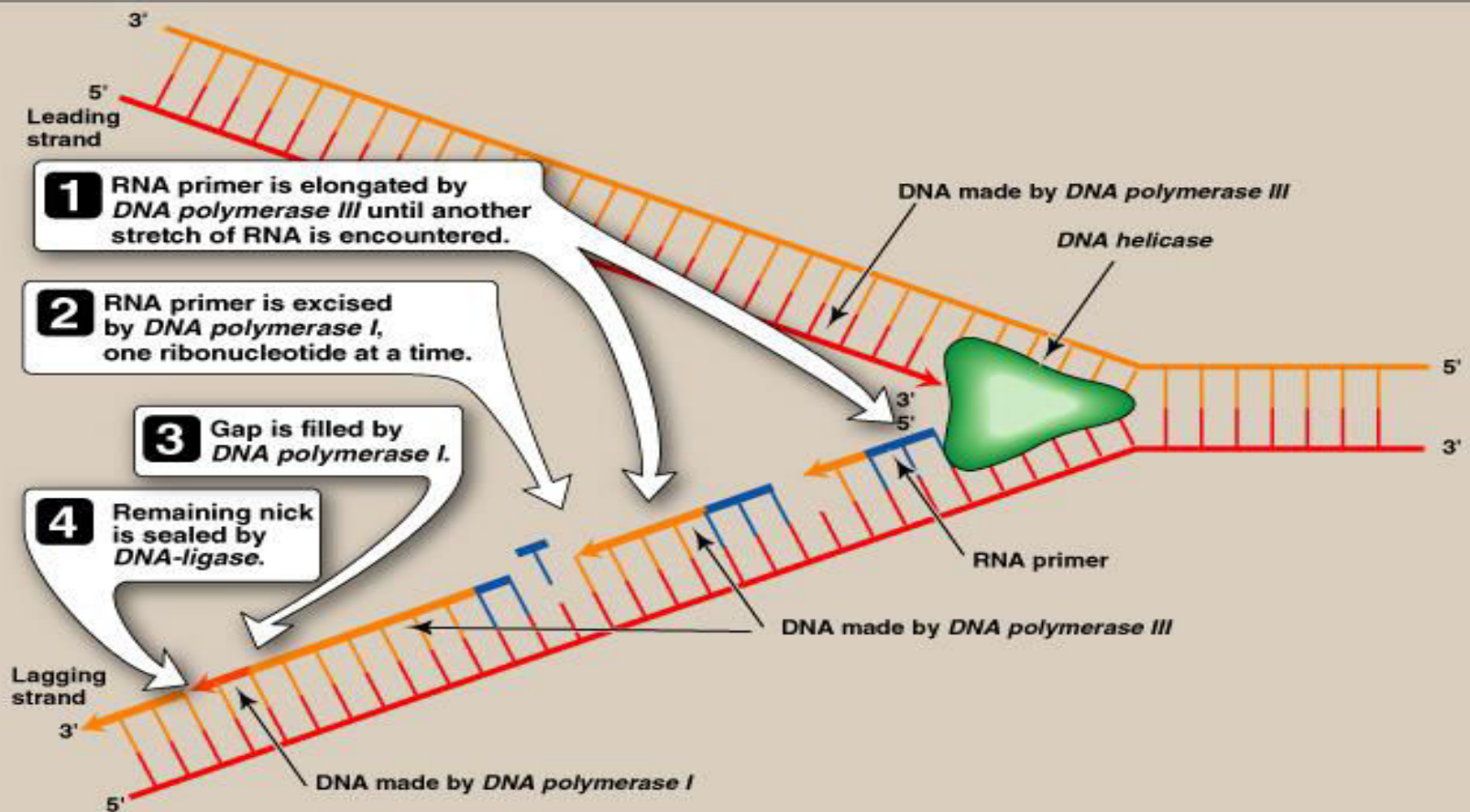


Figure 29.19

Removal of RNA primer and filling of the resulting "gaps" by *DNA polymerase I*.

DNA pol 1 → has a 5' to 3' Exonuclease and polymerase activity, at the RNA primer site, and proofreading the framed NTds

→ for proofreading 3' to 5' exonuclease activity ,,,, for primers 5' to 3' Exonuclease

→ remove 1 Ribo NTds at a time (Exonuclease) and Replace it with Deoxy RIBO Ntds

DNA Polymerase III

5`-3`polymerase activity
synthesis of new DNA
strand

3`-5` exonuclease activity
proof reading of new
strand

Function

Synthesis ---- Proofread

DNA Polymerase I

5`-3`polymerase activity
synthesis (filling) of gap of
removed RNA primer

Step 2

3`-5` exonuclease activity
proof reading of the DNA
filling the gap of removed
RNA primer



Step 3

5`-3` exonuclease activity
removal (excision) of RNA
primers. Step 1

Function

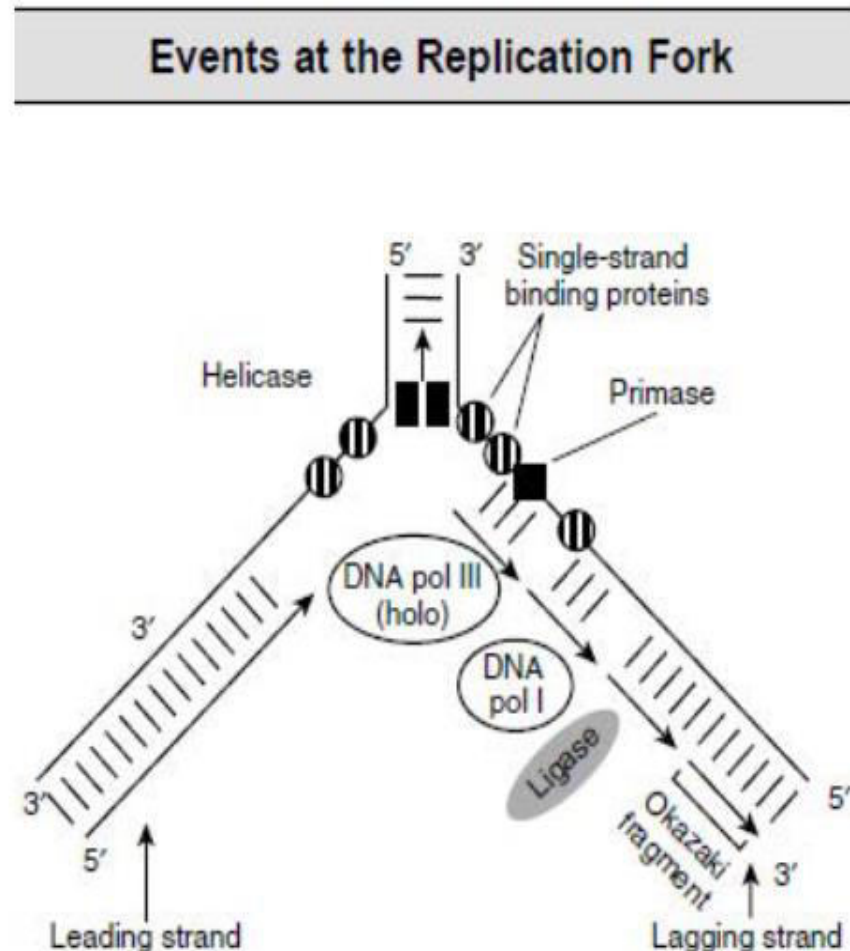
Removal ----synthesis -----
Proofread

* DNA ligase :

■ The final phosphodiester linkage between the 5'-phosphate group on the DNA chain synthesized by DNA polymerase III and the 3'-OH group on the chain made by DNA polymerase I is catalyzed by DNA Ligase.

■ This process requires energy derived from the cleavage of ATP TO AMP + P_{Pi}

Step in Replication	Prokaryotic Cells
Origin of replication (ori)	One ori site per chromosome
Unwinding of DNA double helix	Helicase
Stabilization of unwound template strands	Single-stranded DNA-binding protein (SSB)
Synthesis of RNA primers	Primase
Synthesis of DNA Leading strand Lagging strand (Okazaki fragments)	DNA polymerase III DNA polymerase III
Removal of RNA primers	DNA polymerase I (5'→3' exonuclease)
Replacement of RNA with DNA	DNA polymerase I
Joining of Okazaki fragments	DNA ligase
Removal of positive supercoils ahead of advancing replication forks	DNA topoisomerase II (DNA gyrase)
Synthesis of telomeres	Not required



DNA replication process



Chromosome formation and replication process

1. DNA Wrapping

* Eukaryotic DNA Replication :

- The process of eukaryotic DNA replication closely follows that of prokaryotic DNA synthesis.

Eukaryotes	prokaryotes
<u>Multiple origins of replication</u>	Single origin of replication
RNA primers are removed by <u>RNase H</u>	RNA primers are removed by DNA polymerase 1
Single – stranded DNA binding protein & ATP dependent DNA helicase	Single – stranded DNA binding protein & ATP dependent DNA helicase (so it's the same)

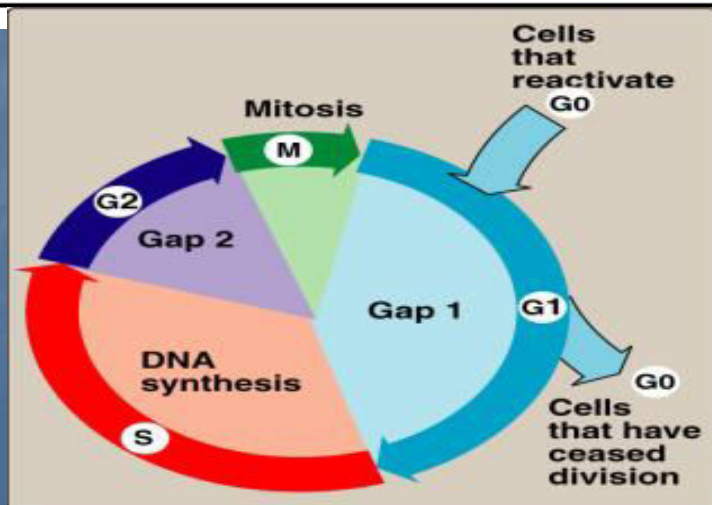


Figure 29.21
The eukaryotic cell cycle.

* Eukaryotic Cell Cycle :

- Cell cycle: the events surrounded DNA replication and cell division (mitosis)
- G1 phase (Gap1):the period before replication
- S (synthesis) phase: DNA replication occur
- G2 phase (Gap2):the period after replication and before mitosis
- M phase (mitosis):cell division occur
- G0 phase :
 - cell that reactive and enter the cycle (through Gap1)
 - cell ceased division exit the cycle (through Gap1) e.g. mature neurons
- (some cells leave the G0 phase and reenter G1pase to resume division)

*Eukaryotic DNA polymerase:

- At least 5 classes have been identified. Categorize on the basis of MW, cellular location, sensitivity to inhibitors, and templates or substrates on which they act.
- They are designated by Greek letters (α alpha, β beta, δ delta, γ gamma, ϵ epsilon)
- (Note that all polymerase have specific function + proofreading except α and β)

■ *** Pol α** : (multisubunit enzyme)

- one subunit has(1) **primase activity**:

which **initiate** (only) synthesis on leading strand (one primer) and on lagging strands (beginning of each Okazaki fragment)

- (2) **polymerase activity**:

Which extend RNA primer by add short (not long) DNA pieces from 5' $\rightarrow$ 3'

■ **pol δ** :

-**complete (NOT initiate)**
DNA synthesis on leading strand and elongate each Okazaki fragment

-proofread using 3' $\rightarrow$ 5' exonuclease

***pol β and pol ϵ**

- For repair .

-Pol ϵ also to proofread

POLYM- ERASE	FUNCTION	PROOF- READING
<i>Pol</i> α	● Contains primase ● Initiates DNA synthesis	—
<i>Pol</i> β	● Repair	—
<i>Pol</i> γ	● Replicates mitochondrial DNA	+
<i>Pol</i> δ	● Elongates leading strands and Okazai fragments	+
<i>Pol</i> ϵ	● Repair	+

Figure 29.22
Activities of eukaryotic DNA polymerases (pols).

*pol γ

Mitochondria DNA(which is circular)

-circular DNA occur in :

Mitochondria, plants, and prokaryotic

Reverse Transcriptase :

☐ A retrovirus as HIV, carries the genome in the form of single-stranded RNA molecule.

☐ Following infection of a host cell, the viral enzyme, **reverse transcriptase**, uses **the RNA as a template for the synthesis of viral DNA**

☐ Then, DNA becomes integrated into host chromosomes.

☐ Reverse transcriptase moves along the RNA template in the 3'-5' direction,

☐ synthesizing new DNA in the 5'-3' direction

No proofreading in reverse transcriptase high mutation rate of such viruses

☐ As an attempt to prevent HIV infection from progressing to AIDS, patients are treated by inhibitors of reverse transcriptase (nucleosides or non-nucleosides inhib.), in addition to protease inhibitors (which targets another HIV maturation enzyme) , thus produce a mixture " cocktail " that requires the virus to develop multiple resistance in order to continue to replicate .

* Inhibitors of DNA synthesis by nucleoside analogs :

* DNA chain growth can be blocked by the incorporation of certain nucleoside analogs that have been modified in the sugar portion of the Nucleoside (as the 3'-end is not present for DNA chain elongation)

■ NUCLEOSIDE ANALOGS

1- Removal of (OH) from 3'-carbon of deoxyribose

{ 2', 3' dideoxyinosine (didanosine) }

2- Conversion of deoxyribose to another sugar (arabinose)

* slow the division of rapidly growing cell & viruses

Cytosine arabinoside (cytarabine, araC) : cancer therapy

Adenine arabinoside (vidarabine, araA) : antiviral agent

3- Chemically modification of the sugar moiety

Zidovudine (AZT)

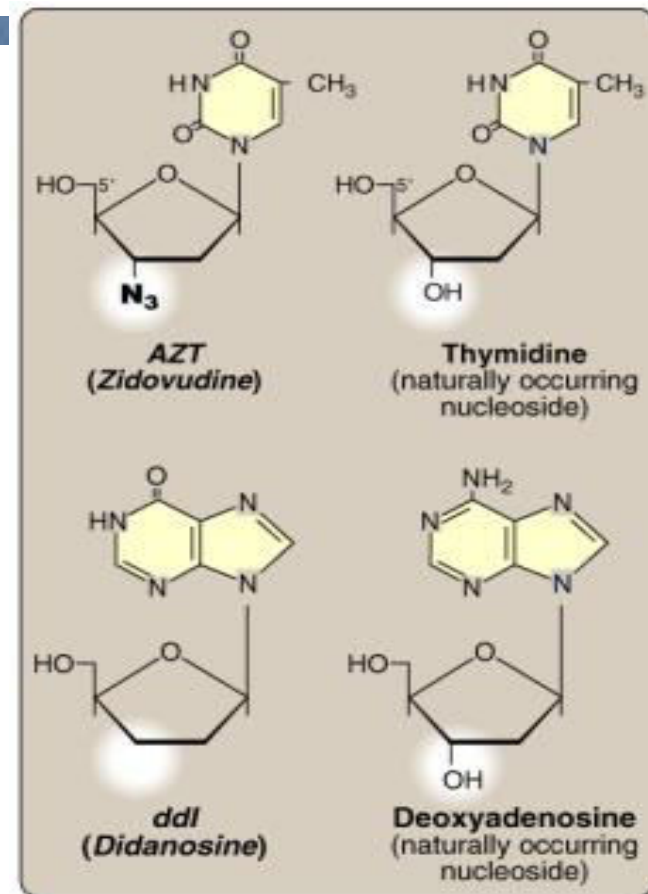


Figure 29.24
Examples of nucleoside analogs that lack a 3'-hydroxyl group.

Drugs→ Formed by the **modification** of **carbon in the ribose sugar** with **no or modified OH** group(most (**ART**) **HIV drugs** synthesized)

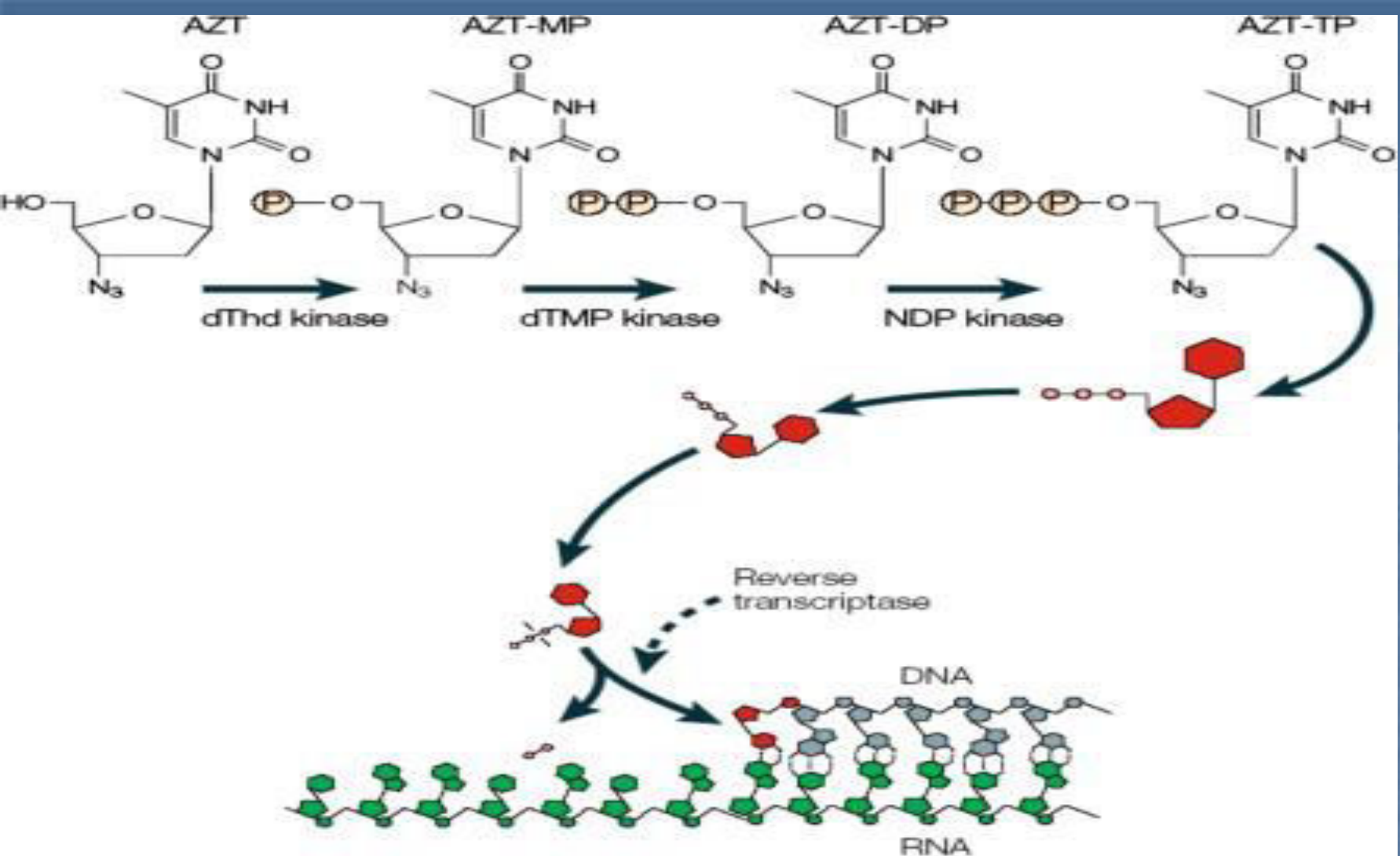
Didanosine is used **to treat (HIV) infection**.

Didanosine is in a **class** of medications called **nucleoside reverse transcriptase inhibitors** (NRTIs) act as “**Chain terminators**” (**competitive**) because of there is **no free OH** to continue the elongation process

Didanosine is phosphorylated to active metabolites (NSd tri P) that **compete for incorporation into viral DNA**.

Zidovudine→ contains a **3'-azide group** rather than a 3'-hydroxyl group.

does not cure HIV, it may **decrease** your **chance of developing** (**AIDS**) and **HIV-related illnesses (Cancer)**



Zidovudine, a structural analog of thymidine, is a prodrug that must be phosphorylated to its "active 5'-triphosphate metabolite", zidovudine triphosphate (ZDV-TP).

It inhibits the activity of HIV-1 reverse transcriptase (RT) via "DNA chain termination" after incorporation of the nucleotide analog (zidovudine monophosphate).

- These sugars are generally supplied as nucleosides which are then to nucleotides by cellular salvage enzyme converted
 - These nucleotides blocks DNA replication by preventing further chain elongation
 - So, these compounds slow the division of rapidly growing cells (cancer) and viruses.
- Arabinose (modified sugar)** → for Viruses (A arabinose) and cancer (C arabinose) treatment

HIV Replication



Reverse transcriptase inhibitor mechanism of action

Reverse Transcriptase Enzyme Action

Fragment from the RCSB PDB video *The Structural View of HIV Therapy*

Telomerase

- In **eukaryotic cells**

After removal of the RNA primer from the extreme 5`-end of the lagging strand, there is no way to fill the remaining gap with DNA.

- To **solve this problem** & to **protect the ends of the chromosomes** from attack by nucleases



noncoding sequences of DNA complexed with proteins are found at these ends, called telomeres.

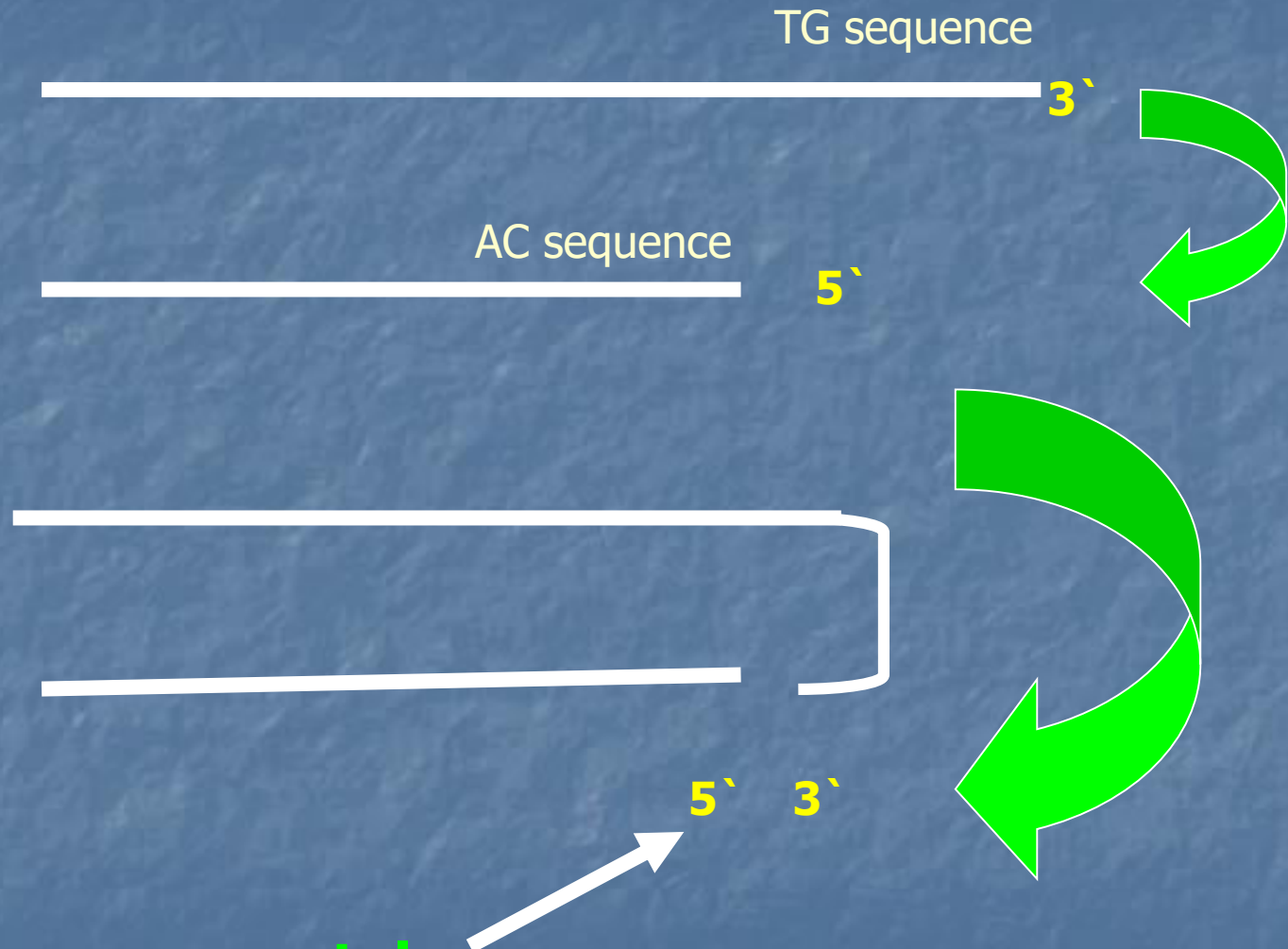
Telomeres

- **DNA** of telomeres consists of repetitive sequence of T`s & G`s, base paired to a complementary chain of A`s and C`s
T x G y : where x and y are in the range of 1 : 4
- The **TG strand** is longer than its complement leaving a region of single stranded DNA at the 3`-end of the double helix (few hundred nucleotides long).
- This single stranded region folds back on itself, forming a structure that is stabilized by **protein** to protect the ends of the chromosomes
- In **aging cells**, the ends of their chromosomes get slightly shorter with each cell division until the telomeres are gone and DNA essential for cell function is degraded.
This phenomenon is related to cellular aging & death.

DNA of the telomere



Single stranded TG region folds back on itself



telomere
formed by DNA & protein protects the ends of the chromosome

In **aging cells, the ends of their chromosomes get slightly shorter with each cell division until the telomeres are gone and DNA essential for cell function is degraded.**

This phenomenon is related to cellular aging & death.

So, what is a telomerase ?

- Cells that *do not age* (as germ-line cells and cancer cells) contain an enzyme called **telomerase** responsible for replacing these lost ends of telomeres.
- **Telomerase** is a special kind of reverse transcriptase that carries its own RNA molecule of about 150 nucleotides long.
This RNA contains copies of **A/C** sequence that is complementary to **T/G** repeat sequence

Steps of telomere elongation

- 1-The telomerase RNA serves as template for extending **T/G** DNA strand (longer strand)
- 2-The 3'-end of the RNA serves as a primer for DNA polymerase to extend the **A/C** DNA strand (shorter strand)
- 3-Once the next repeat sequence is complete, telomerase RNA is translocated to the newly synthesized end of the DNA and the process is repeated.

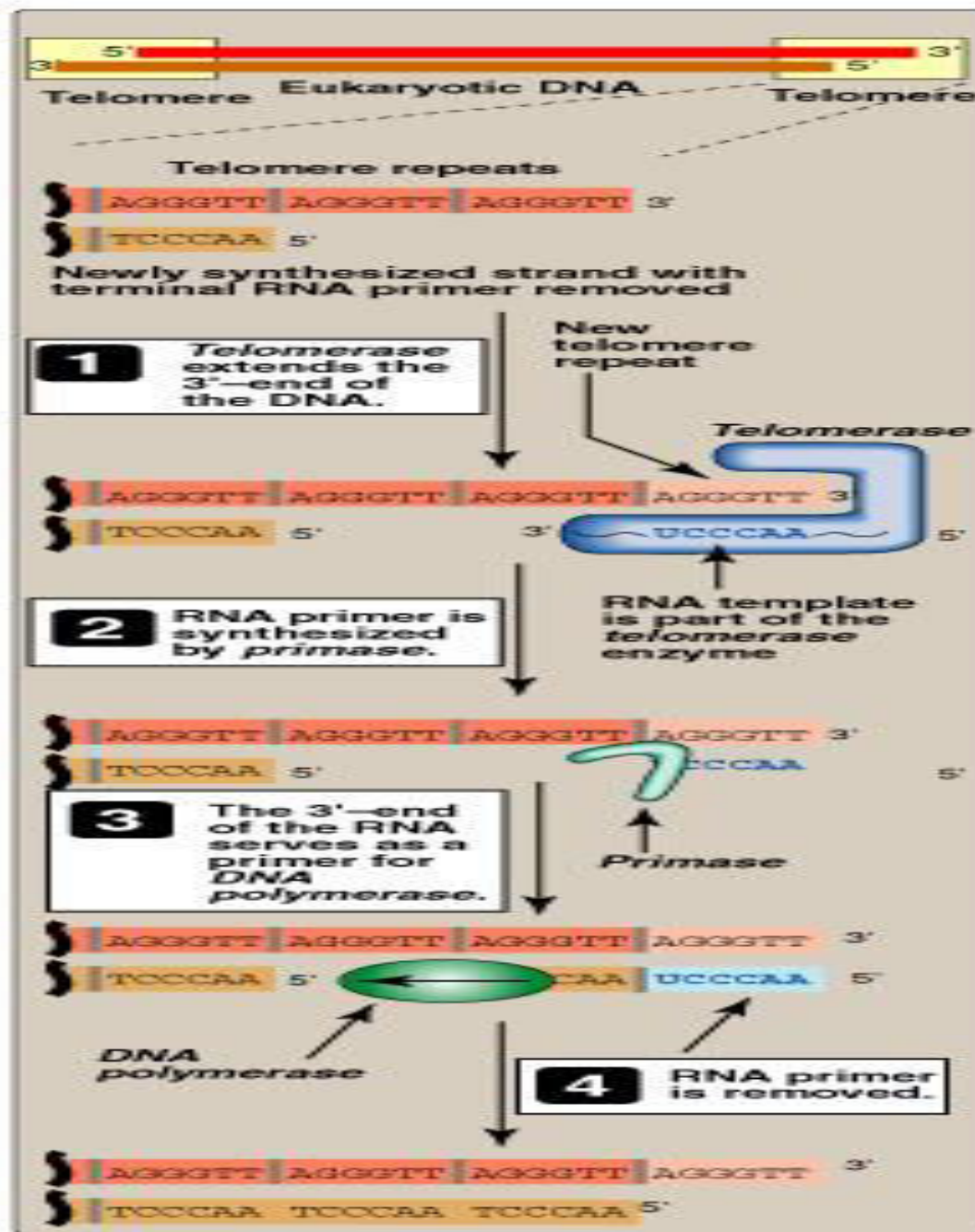


Figure 29.23
Mechanism of action of *telomerase*.

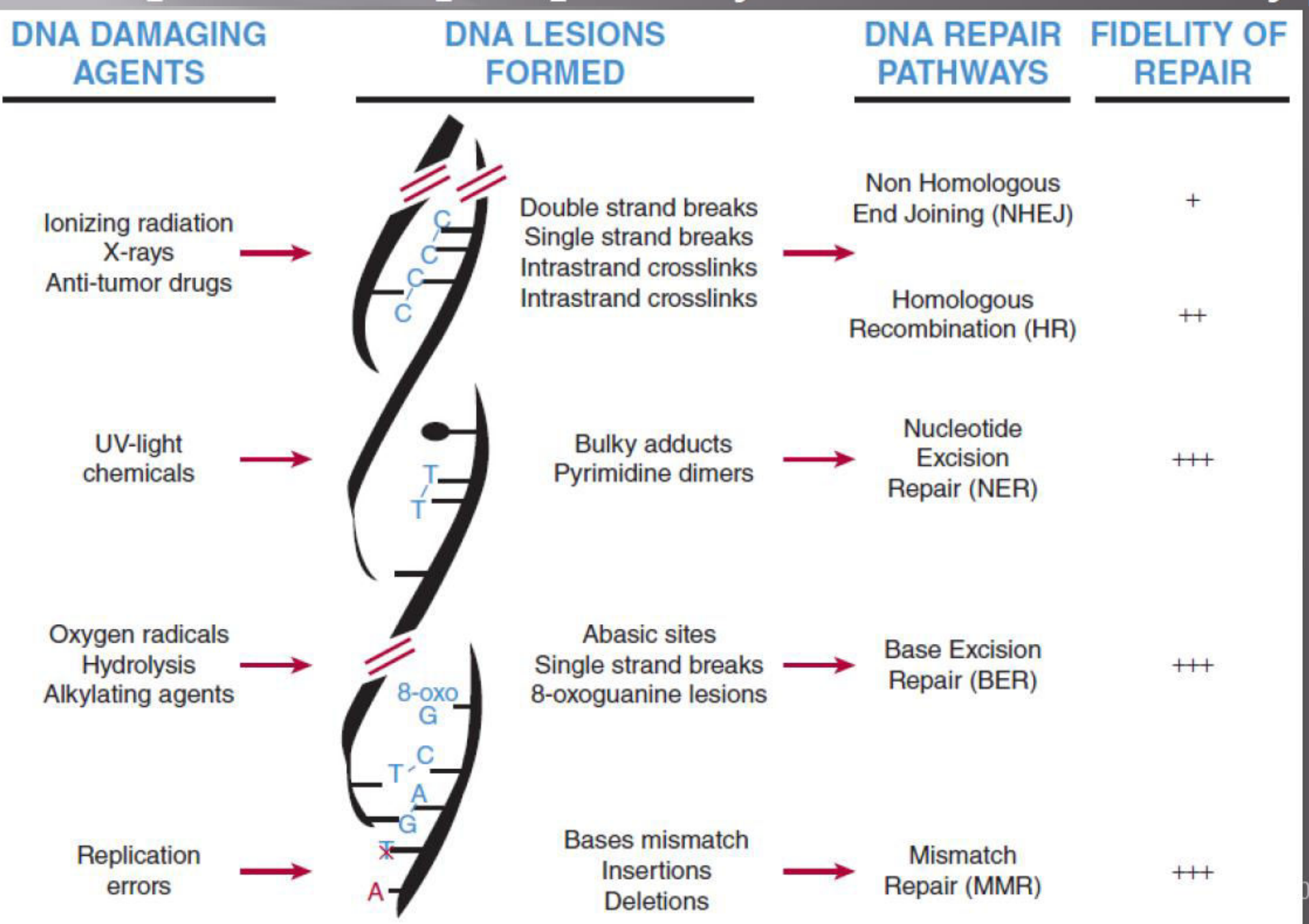
Functionally similar proteins

Prokaryote	Eukaryote	function
Hexameric Dna β complex	Hexameric minichromosomal maintenance (MCM) complex	Unwinds the DNA
SSB	Replication protein A	Prevent reannealing
DnaG	Pol α	Primase synthesizes RNA primer
Pol III	Pol ϵ (leading) and Pol δ (Lagging)	Processive, leading and lagging strand synthesis
Pol I	Rnase H and FEN1 (flap of endonuclease)	RNA primers removed
Pol II	β	DNA repair
Pol I		Gap filling following DNA replication, repair, and recombination
	γ	Mitochondrial DNA synthesis
β subunit of Pol III	PCNA (Proliferating cell nuclear antigen)	Sliding clamp for high procesivity

Inhibitors of DNA Replication

Antibacterial agents	Action
Ciprofloxacin	Bacterial DNA gyrase
Nalidixic acid	Bacterial DNA gyrase
Anticancer agents	
Adriamycin	Human topoisomerase
Etoposide	Human topoisomerase
6-mercaptopurine	Human DNA polymerase
5-fluoro uraci	Thymidylate synthase
Cytosine arabinoside (Cytarabine)	Nucleoside analog
Anti viral agent	
adenine arabinoside (Vidarabine)	Nucleoside analog

Multiple DNA repair pathways of variable accuracy



Summary

- ▣ Each strand of the double helix serves as a **template (semiconservative replication)**.
 - ▣ occurs in the **S phase of the cell cycle and begins at the origin of replication**.
 - ▣ strands are separated locally, forming **replication forks**.
 - ▣ Replication of dsDNA is **bidirectional**
 - ▣ **Helicase unwinds** the double helix.
 - ▣ As the two strands of the double helix are separated, **supercoils are produced**
- DNA topoisomerases Types I and II** remove supercoils

- ▣ The primer for de novo DNA synthesis is a short stretch of RNA synthesized by **primase**
- ▣ **DNA polymerases synthesize new DNA strands only in the 5'→3' direction.**
- ▣ one of the newly synthesized stretches of nucleotide chains must grow in the 5'→3' direction toward the replication fork (**leading strand**)
- ▣ In E. coli DNA chain elongation is catalyzed by **DNA polymerase III**
- ▣ The enzyme “**proofreads**” the newly synthesized DNA with its 3'→5' **exonuclease activity**.
- ▣ RNA primers are removed by **DNA polymerase I, using its 5'→3' exonuclease activity**.
- ▣ The final phosphodi-ester linkage is catalyzed by **DNA ligases**
- ▣ at least five **eukaryotic DNA polymerase**
- ▣ **Nucleoside analogs containing modified sugars can be used to block DNA chain growth.**



Thank You

3.2 RNA Structures and Transcription

Tamrat Nida (M.Sc)
Dep't of Med.
Biochemistry,
AAU, CHS, SM

RNA STRUCTURE & SYNTHESIS

OVERVIEW :

- ✓ Deoxyribonucleotides (DNA) are the genetic master plan of an organism
- ✓ DNA serves as template ... thru RNA(working copies)
(this process is called **transcription**)

✓ **AND then bigining of thr synthesis of :**

- a) messenger RNAs are translated into sequence of amino acids ,
- b) ribosomal RNA ,
- c) transfer RNA .

☑ SO the RNA are three types

- the transcription is highly selective

✓ This selectivity is the reason of biochemical differentiation

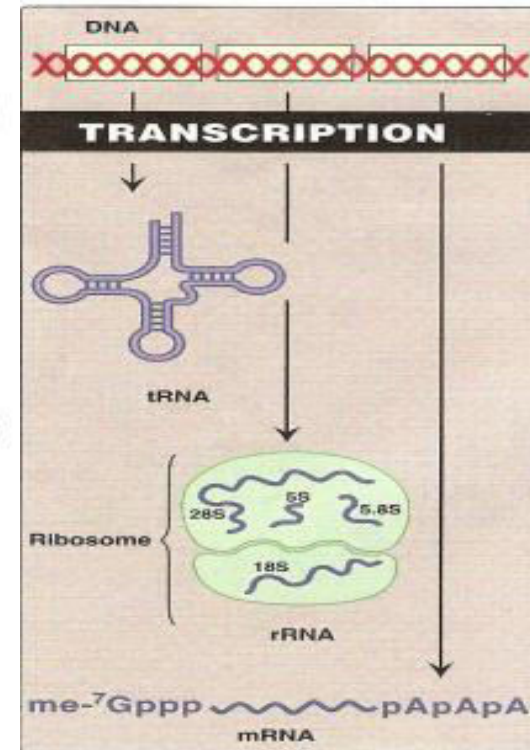


Figure 30.1
Expression of genetic information

■ highly selective

- **Explanation** : many transcripts , are made of some region of DNA . In other region , Few or no transcripts are made .
- **Reasons** :
 1. signals sequence in DNA instruct RNA polymerase where & How to Start & stop transcription .
 2. regulatory proteins .
- **ultimate result** : Biochemical differentiation .

the RNA transcript; undergo various modifications (terminal additions, base modification , trimming, internal segmental removal)

then by **splicing** convert the inactive primary transcripts into a functional molecule .

Q- splicing can make the transcript into functional form 

STRUCTURE OF RNA :

- 3 types (participate in protein synthesis):
 1. ribosomal RNA(rRNA)
 2. transfer RNA(tRNA)
 3. messenger RNA(mRNA)

- ▣ differ from each other in size, function, and structural modification .
- ▣ Like DNA they are Unbranched polymeric molecules composed of mononucleotide joined together by phosphodiester bonds .

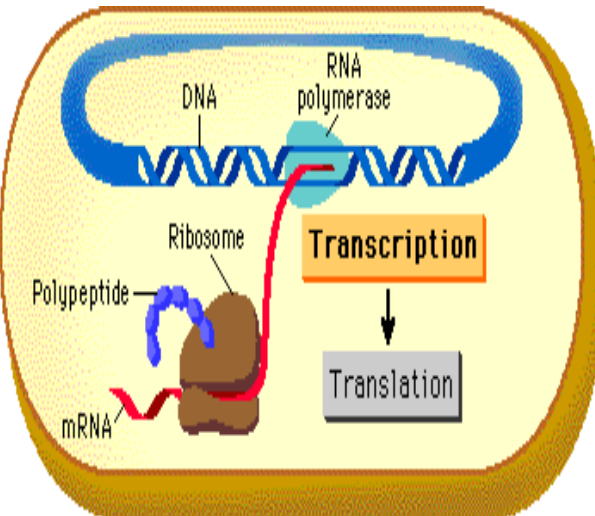
DNA	RNA
bigger	smaller
deoxyribose	ribose
thymine	uracil
double	Single strand (capable to folding into complex structure)
More stable	Less stable

▣ NOTE :

- in eukaryotes small RNA molecules found in the nucleus (snRNAs)

PROKARYOTIC VERSUS EUKARYOTIC TRANSCRIPTION

Prokaryotic Transcription



Prokaryotic transcription occurs in the cell cytoplasm.

Here, transcription and translation happen simultaneously.

mRNA is transcribed directly from template DNA molecule.

RNA polymerase consists of five subunits.

Holoenzyme (RNA polymerase + sigma factor) recognizes and binds directly to the promoter.

Eukaryotic transcription occurs in the cell nucleus.

Transcription and translation differ in space and time.

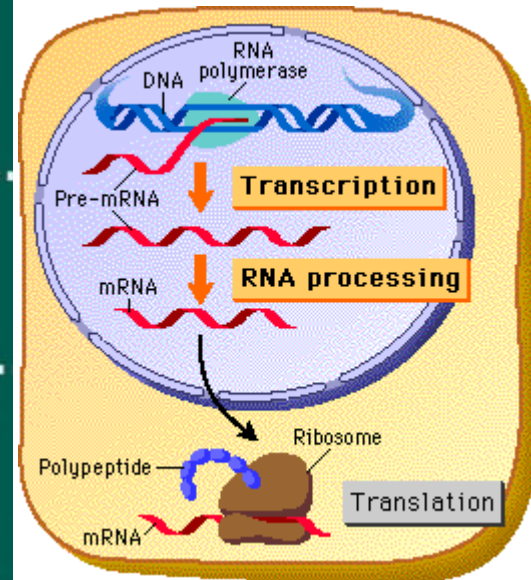
Initially a pre-mRNA molecule is formed and then processed to yield a mature mRNA.

RNA polymerase consists of 10-17 subunits.

Promoter recognition cannot be carried out by RNA polymerase alone.

Pediaa.com

Eukaryotic Transcription



A. Ribosomal RNA :

- rRNA make up 80% of total RNA in the cell (most)
- Found in ribosomes (sites for protein synthesis) in association with several proteins (not histones)

70S- In prokaryotes there are three rRNA species : (23S , 16S , 5S)

80S- In eukaryotes there are four rRNA species : (28S , 18S , 5.8S , 5S)

Q- 5S are found only in eukaryotes

*Note: S=Svedberg unit which is related to MW & shape of the compound . ()

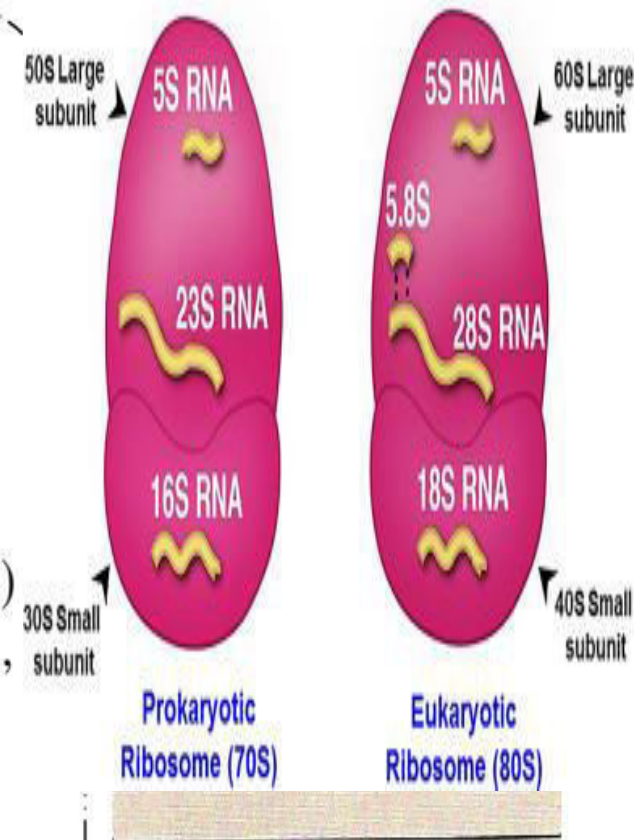
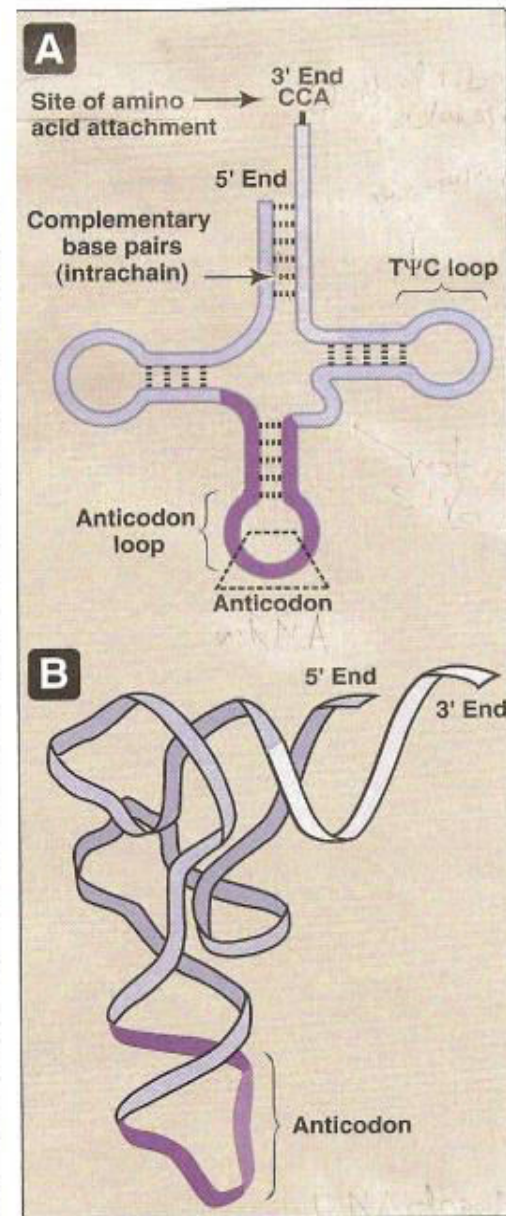


Figure 30.2

Prokaryotic and eukaryotic rRNAs.

B. Transfer RNA :

- tRNA make up 15% of total RNA in the cell .
- is the smallest RNA molecules (4S) (always come and he said smallest in amount but the true is smallest in size) , & have (74-95) nucleotides residues .
 - at least one specific tRNA molecule for one amino acid → so, at least there are 20 species of tRNA .
- tRNA molecules contain :
 - a)unusual bases (e.g. pseudouracil TΨ C loop { not uracil })
 - b)extensive intrachain base pairing (not interchain) (
- each tRNA serve as adaptor molecule () that carries its specific amino acid that attached by covalent bond to 3'-end (CCA) to the site of protein synthesis (NOT from nucleus to cytoplasm) . there it recognizes the genetic code word on an mRNA , which specifies the addition of its amino acid to the growing peptide chain

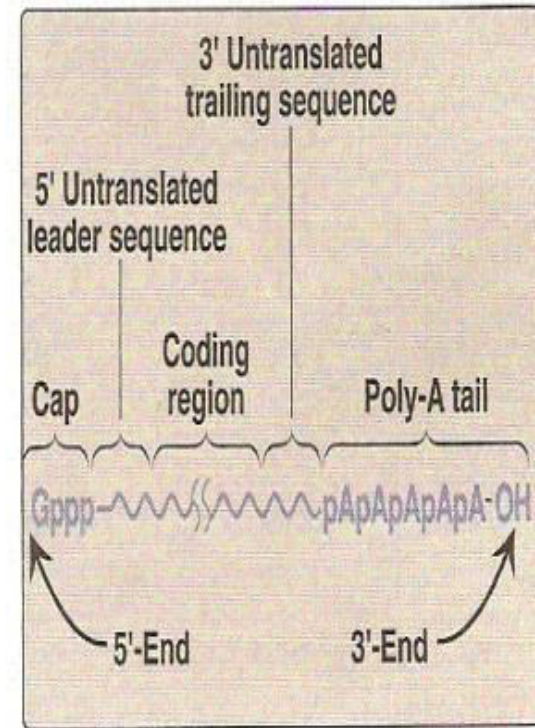


Figure

A. Characteristic tRNA structure.
B. Folded tRNA structure found in cells.

C. Messenger RNA :

- mRNA comprises only about 5% of RNA in the cell .(smallest in amount not size)
- Most heterogenous & base sequence type (500 to 6000 nucleotide).
- carries information from nuclear DNA to the cytosol where it used as template for protein synthesis . (not like tRNA) .
- Eukaryotic mRNA (NOT prokaryotes) contains :(always come in exams)
 - a. long sequence of Adenine (poly-A tail) on the 3 -end of RNA (only one P)
 - b. one molecule of 7-methyguanosine "cap" on the 5-end attatched " backward " (5' $\rightarrow$ 5') (here we have triphosphate linkage)



Figure

Structure of eukaryotic messenger RNA.

TRANSCRIPTION OF PROKARYOTIC GENES :

- eukaryotes & prokaryotes differ in;

1) RNA polymerase .

2) signals that control transcription

3) varieties of modification that RNA transcripts .

A - Properties of prokaryotic RNA polymerase :

- In bacteria, one species of RNA polymerase synthesizes all of the RNA **except** for the short RNA primers which synthesize by specialized enzyme, primase.

(RNA primer needed for DNA replication)

- RNA polymerase – multisubunit enzyme that is:

- 1- recognize a nucleotide sequence (the promoter region) at the beginning of the length of DNA that is to be transcribed.
- 2- next , makes a complementary RNA copy of DNA template strand.
- 3- then , recognize the end of DNA sequence to be transcribed (the termination region).

- RNA is synthesized from its **5'-end to its 3'-end antiparallel** to its DNA template strand.
- the template is copied as in DNA synthesis :**

DNA		RNA
G	specifies	C
C	specifies	G
T	specifies	A
A	specifies	U

But

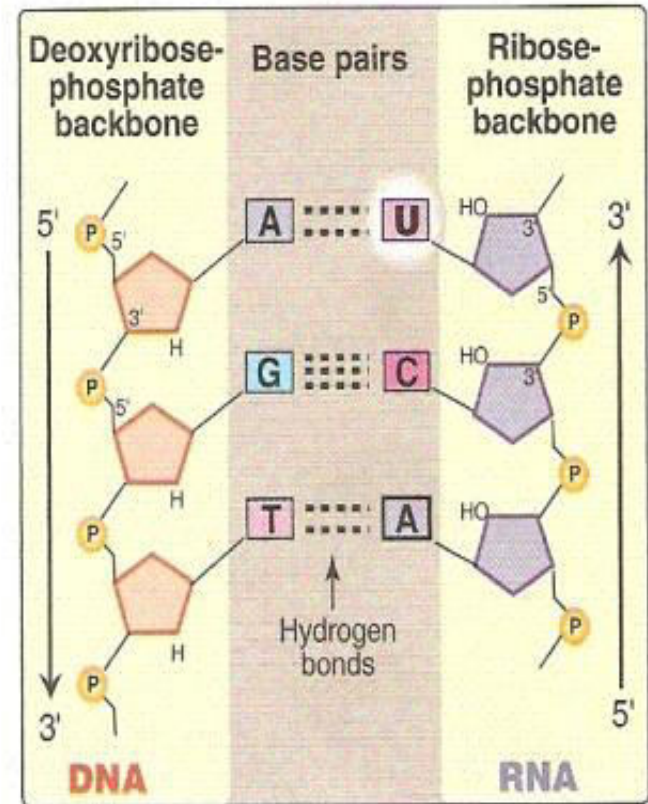
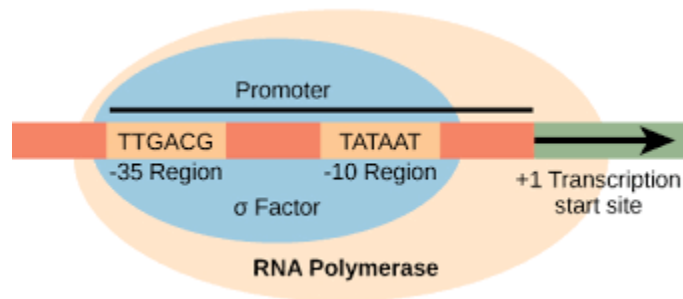


Figure 30.5

Antiparallel, complementary base pairs between DNA and RNA.

- **transcription unit:** extend from the promoter to termination region
- **primary transcript:** product of the process of transcription by RNA polymerase.

Q- Promoter >>> initiation of transcription.



* transcription by RNA polymerase involves a core enzyme and several auxillary proteins:

1-core enzyme :

- Four enzyme's peptide subunits, 2α , 1β , and $1\beta'$.
- responsible for $5' \rightarrow 3'$ RNA polymerase activity
- lacks specificity, that is, it cannot recognize the promoter region on DNA template

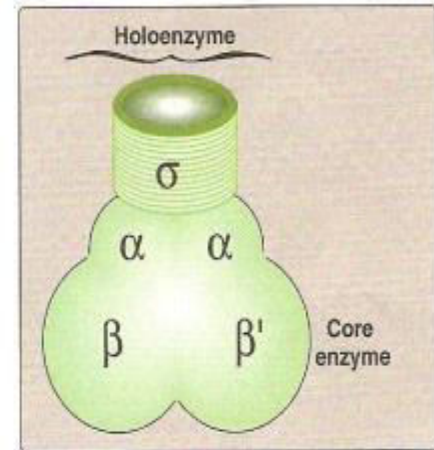


Figure
Prokaryotic RNA polymerase.

2-holoenzyme :

- holoenzyme = σ (sigma) subunit + core enzyme
- **σ subunit (sigma factor) function :**
- ✓ Enable RNA polymerase to recognize the promoter region on the DNA

Note : (Different σ factor recognize different groups of genes)

Q- Concerning prokaryotic transcription :

>Sigma (σ) factor enables RNA polymerase to recognize the promoter region.

Q- Holoenzyme (except)

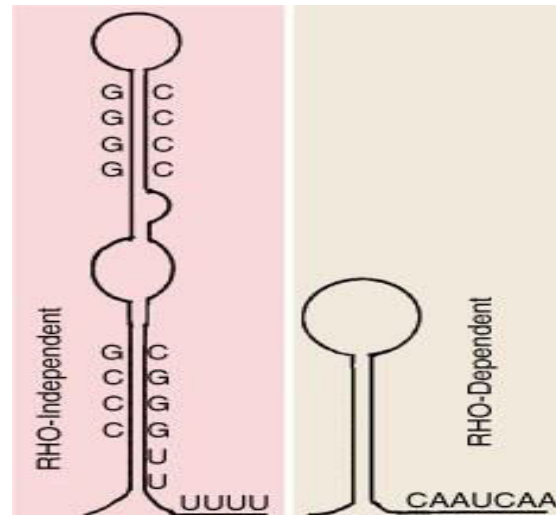
> Its Rho-factor recognizes the termination signal.

3-termination factor :

- Some regions on DNA that signal the termination of transcription are recognize by RNA polymerase itself

- others are recognized by specific termination factors e.g. rho (ρ) factor of E.coli .

Transcription termination in prokaryotes can be rho-independent (intrinsic terminators exist in the RNA polymerase) G-C rich sequence causes a pause in the movement of the RNA polymerase and dissociation from DNA.



rho-dependent, i.e., the RNA polymerase requires the cofactor rho for the termination of transcription. rho recognizes 50 to 90 bases before the “hairpin facilitates termination.”

B . Steps in RNA synthesis :

- Three phases :

1) initiation . 2) elongation . 3) termination .

■ within DNA molecule : both strand serve as template for RNA .

■ **within a specific stretch of double helix : only one strand serve as template .**

1) Initiation :

- binding of RNA polymerase holoenzyme to the promoter region .
- consensus sequence of the promoter region are highly conserved . { so , different promoters contain very similar or identical Sequences }

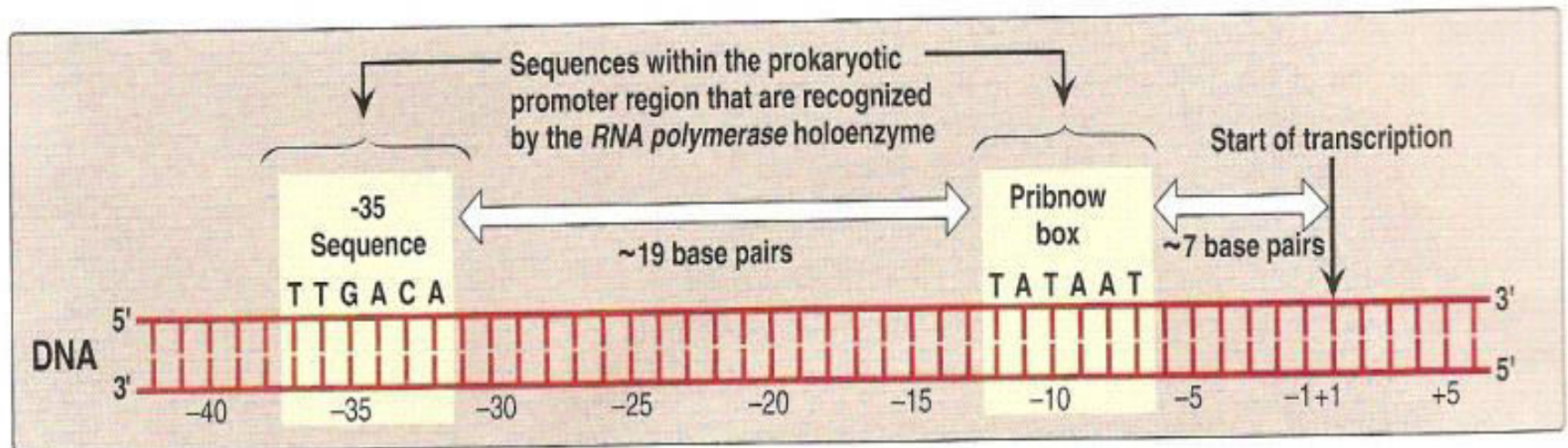


Figure 30.7

Structure of the prokaryotic promoter region.

- Those that are recognized by prokaryotic RNA polymerase σ factor :

a) Pribnow box :

- stretch of six nucleotides (5 - TATAAT- 3) centered about eight to ten nucleotide (in the figure he said 7 nucleotide) to the left of the Transcription start site that code for the initial base of the mRNA
- ✓ The regulatory sequence are in 3 to 5 in template strand & 5 to 3 in nontemplate strand .

■ **NOTE :** (Regarding the transcription site):

- (up stream) (prior) assigned a negative number . e.g : pribnow box -9 .
 - at & after (downstream) assigned a positive number . e.g : the first base at the transcription site is +1 .
- i.e : (there is no base designated "0")

B) - 35 sequence :

- **A consensus sequence (5- TTGACA – 3)** , which is centered about 35 bases To the **left** of the transcription start site .

■ Note :

- ✓ A mutation in either Pribnow box or the – 35 sequence can affect the transcription of the gene controlled by mutant promoter .

2) Elongation:

- Once the promoter region has been recognized by the holoenzyme, *RNA polymerase* begins to synthesize a transcript of the DNA sequence (usually beginning with purine), & the σ subunit is released.
- Unlike *DNA polymerase*, *RNA polymerase*: .(always come in exams)
 - a. does(not) require a primer
 - b. has(no) known endonuclease or exonuclease activity. It, therefore, has(no) ability to repair mistakes in the RNA, as does *DNA polymerase* during DNA synthesis.(always come in exams)
- *RNA polymerase* uses **ribonucleoside triphosphats**, & releases pyrophosphate (Tor F) each time a nucleotide is added to the growing chain.

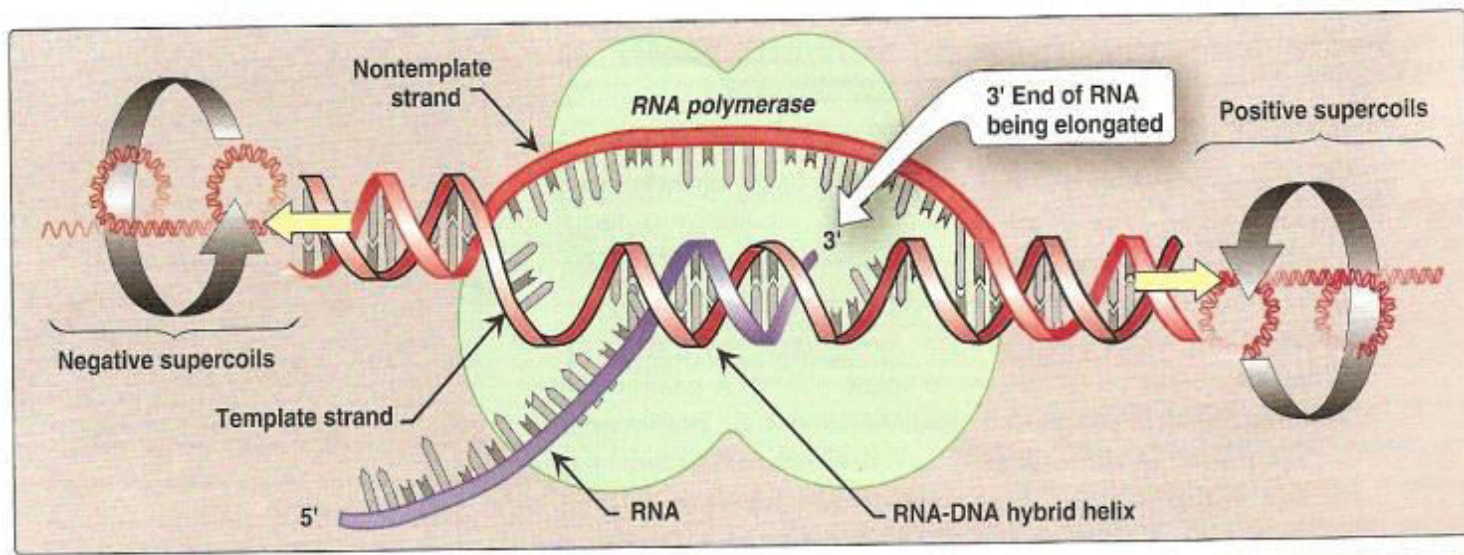


Figure 1
Local unwinding of DNA caused by *RNA polymerase*.

- **Note :**
- As in DNA synthesis, two high-energy bonds are thus used for the addition of each nucleotide.
 - The binding of the enzyme to the DNA template result in local unwinding of the DNA helix.
 - This process can generate supercoils that can be relaxed by *DNA topoisomerases I & II*.

Q- Regarding to Prokaryotic transcription
>>> Supercoils that relaxed by DNA topoisomerase I & II.

3) Termination:

- The process of elongation of the RNA chain continues until a termination signal is reached.
- An additional protein, ρ (**rho**) **factor**, may be required for the release of the RNA product (ρ - **dependent termination**). Alternatively, the tetrameric *RNA polymerase* can, in some instance, recognize termination regions on the DNA template (ρ - **independent termination**).

a. Rho-dependent termination:

- ✓ protein not enzyme (
- ✓ have some enzymatic activity
- ✓ requires the participation of an additional protein, ρ **factor**. (
- ρ **factor** binds to a C-rich region near the 3'-end of the newly synthesized RNA, & migrates along behind the *RNA polymerase* in the 5'→3' direction until the termination site is reached.

■ Note :

- Rho factor has ATP- dependent RNA-DNA helicase activity (very important).that : hydolyzes ATP,& uses the energy to unwind the 3'-end of the transcript from the template.
 - ✓ This facilitates the movement of the protein along the RNA/DNA duplex.
- At the termination site, ρ factor displace the DNA template strand, facilitating the dissociation of the RNA molecule.

So, what you should know it : ☺

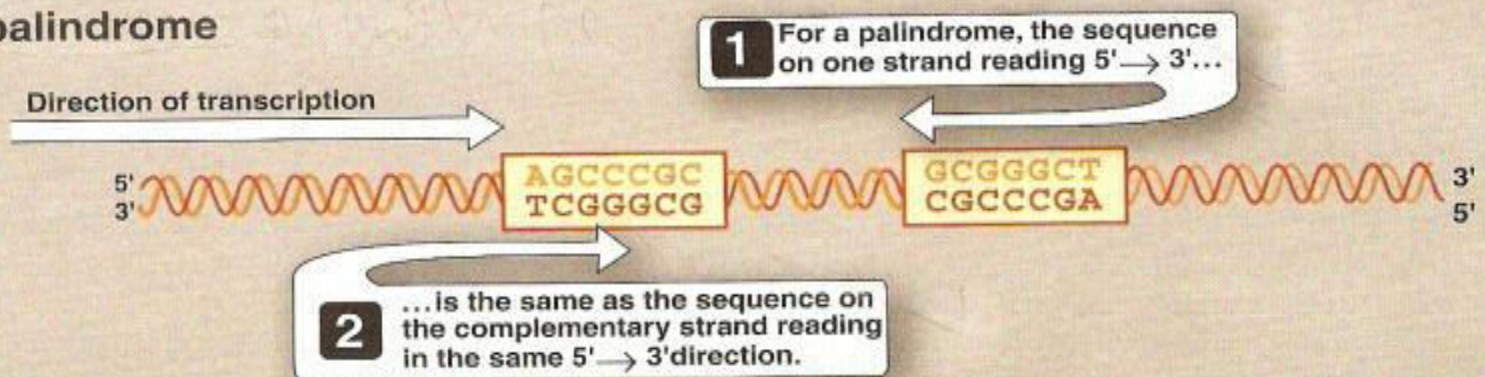
1) protein not enzyme .

2) bind to a C-rich region near the 3'-end .

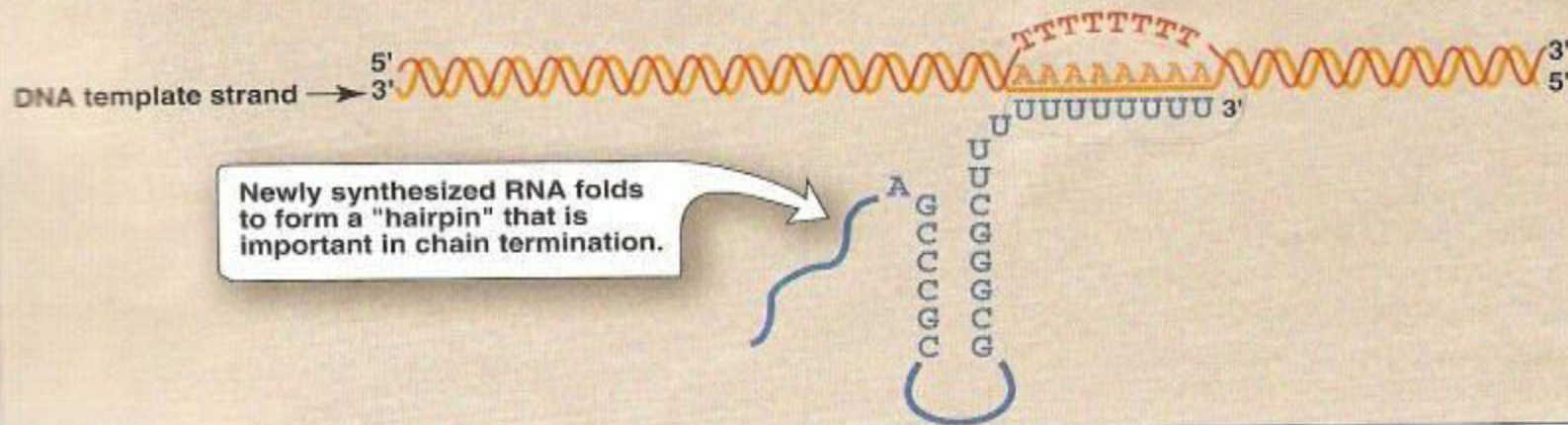
3) Rho factor has ATP- dependent RNA-DNA helicase activity

b. Rho- independent termination :

A DNA palindrome



B Role of palindromes in the termination of RNA synthesis



Figure

Rho-independent termination of transcription. A. An example of a palindrome in double-stranded DNA. B. A transcribed DNA palindrome codes for RNA that can form a hairpin turn.

* requires that the newly synthesized RNA have two important structural features:

First,

- the RNA transcript must be able to form a stable hairpin turn.

Q- Regarding to termination Prokaryotic transcription

>>> RNA form a stable hairpin turn.

■ when we ask what is the function of a stable hairpin turn ?

Answer : it slows down the progress of *RNA polymerase* & causes it to pause temporarily.

■ Note : A palindrome is a region of a double-stranded DNA in which each of the two strand contain stretches that have the same nucleotide sequence when read in the same (for example, 5'→ 3') direction

■ Note : as a result of the presence of a palindrome

The hairpin turn of the RNA is complementary to a region of the DNA template near the termination region exhibits two-fold symmetry.

Secondary structure of RNA is stabilizes by a lot of G & C in the base of the stem Of hairpin .

Second ,

beyond the hairpin turn,

- the RNA transcript contains a string of U's. The bonding of U's to the corresponding DNA template A's is weak. This facilitates the separation of the newly synthesized RNA from its DNA template, as the double helix “zips up” behind the *RNA polymerase*

■ Rivision : (Rho- independent termination) ☹->☺

** staple hairpin turn ** two-fold symmetry

** *Secondary structure of RNA is stabilizes by a lot of G & C in the base of the stem Of hairpin

** beyond the hairpin turn,

- the RNA transcript contains a string of U's. The bonding of U's to the corresponding DNA template A's is weak

4) Action of antibiotics:

- some antibiotics prevent bacterial cell growth by inhibiting RNA synthesis.

■ Note : Dactinomycin:

- known to biochemists as actinomycin D .

- It bind to the DNA template & interferes with the movement of the RNA polymerase along the DNA.

- was the first antibiotics to find therapeutic application in tumor chemotherapy (MCQ)

■ Note : rifampin : (MCQ)

رکز علیه کثیر دائماً یجی ورکز انه ما یأثر
eukaryotes

transcription by binding to the β subunit of prokaryotic RNA polymerase (Not eukaryotes) (MCQ) (always come), thus interfering with the formation of the first phosphodiester bond .

- Is useful in the treatment of tuberculosis

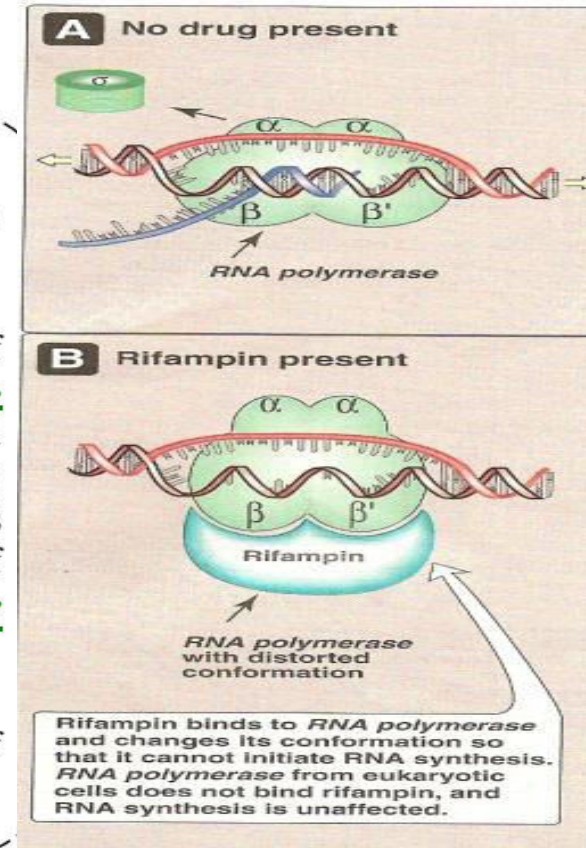


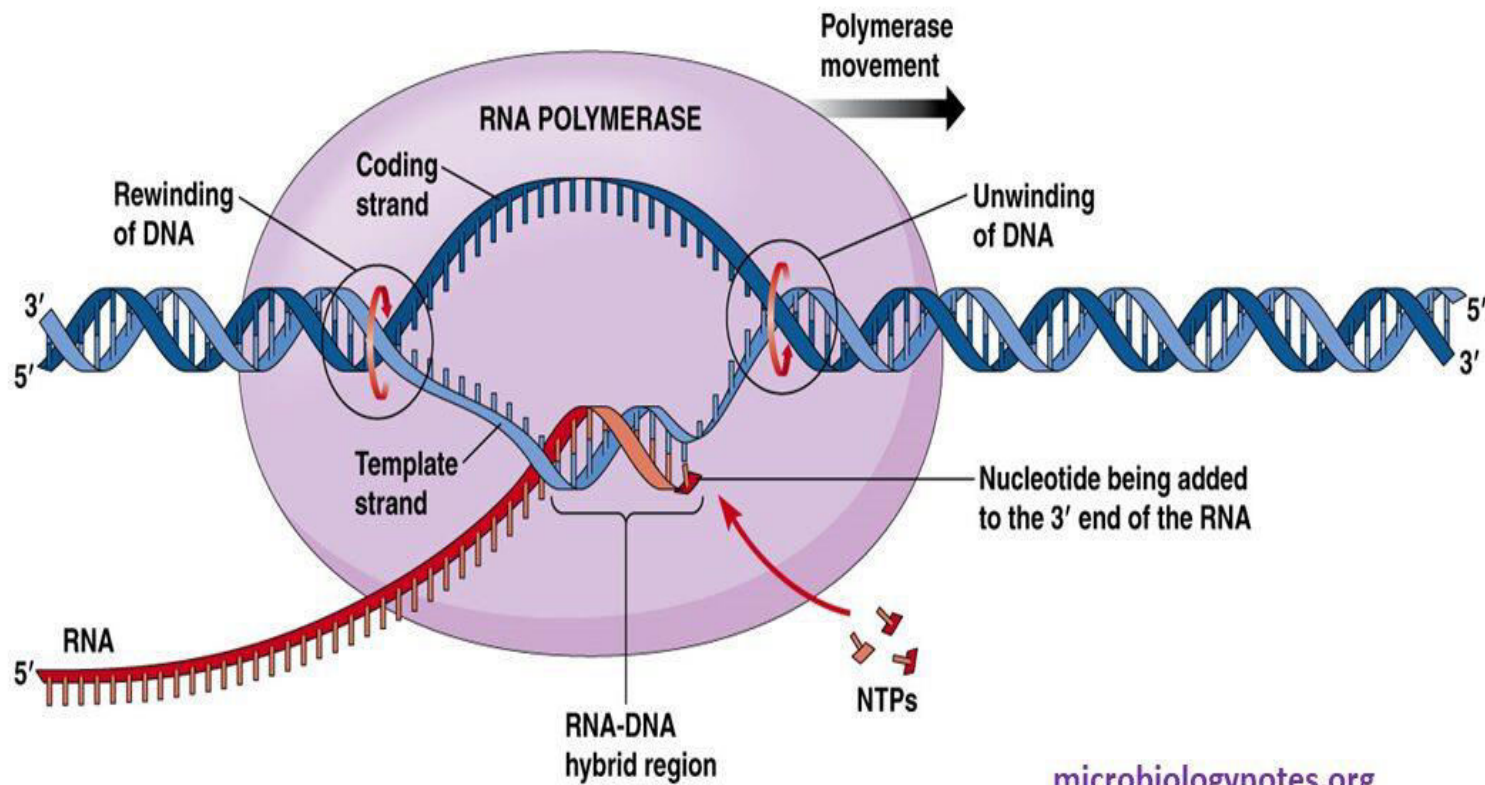
Figure 30.10
Inactivation of RNA polymerase by rifampin.

Actinomycin D → Intercalation agent (minor groove) and is an apoptosis inducer (cytotoxic drug) for a cancer cell (Anti-cancer), by inhibiting RNA synthesis

Rifampin → Antibacterial (Tuberculosis), inhibits bacterial DNA-dependent RNA polymerase (stops the initiation process)

Selective inhibition of bacterial RNA polymerase → B/C It does not affect the eukaryotic cell transcription process

Transcription in prokaryotes: Initiation, Elongation and Termination



TRANSCRIPTION OF EUKARYOTIC GENES :

- transcription is more complicated in eukaryotes than prokaryotes .

■ **Note :** you know that RNA polymerase bind to promoter region and initiate Transcription . (with it , several transcriptin factors bind either to Promoter region or some distance from it)

■ transcriptin factors function : it is protein that determines what genes are to be transcribed .

For all these to happen we should have double helix DNA that assume a loose Conformation and dissociate temporarily from the nucleosome core.

A - Chromatin structure & gene expression :

-DNA + histone = nucleosome -> affect ability to transcription.

-Regarding DNA transcription :

Can transcribed	Can not transcribed
Relaxed form of chromatin called <u>euchromatin</u> (active)	Highly condensed form called <u>Heterochromatin</u> (inactive)

■ **Chromatin remodeling** : interconversion of active & inactive forms .

* Two major influences on chromosome structure & activity :

- 1) DNA methylation .
- 2) histone acetylation .

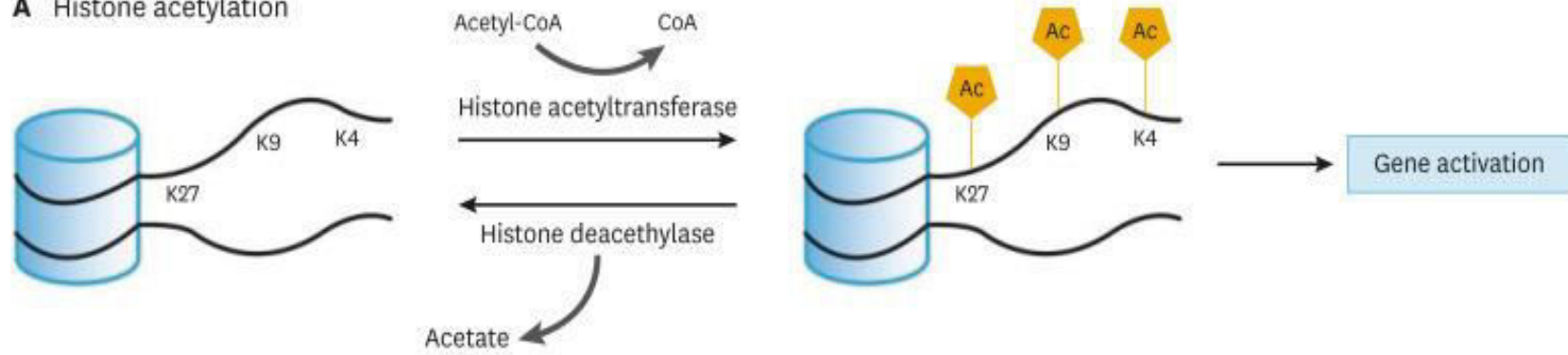
We notice that the genes that are in permanent inactive form ☹️ Have more methylated DNA (5 methylcytosine)
↳ Than the active form .

We take the DNA of one of the X chromosome of a female & we notice that :(see what happen when acetylated or methylated)

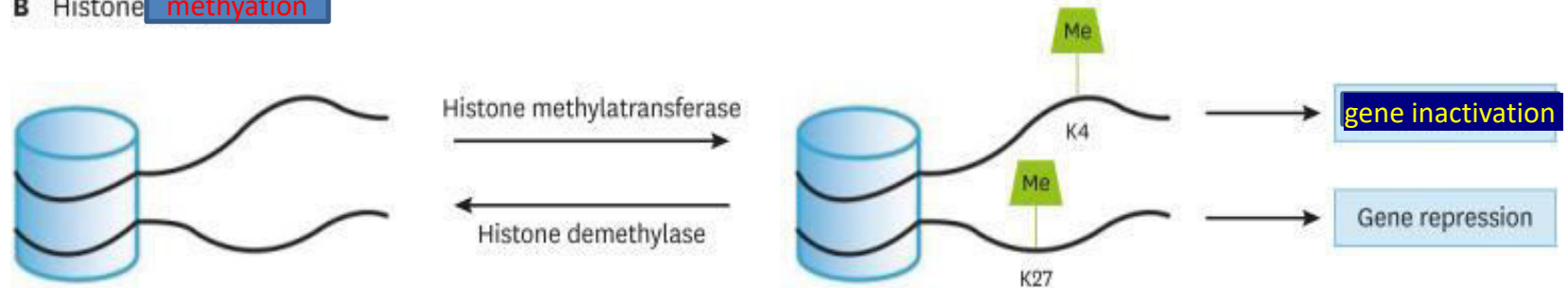
Highly methylated	Histone become Highly acetylated
heterochromatin	Euchromatin
Transcription turned off	Actively transcribed .(chromatin become looser. So, the DNA become more Accessible to transcription)

Histone modification

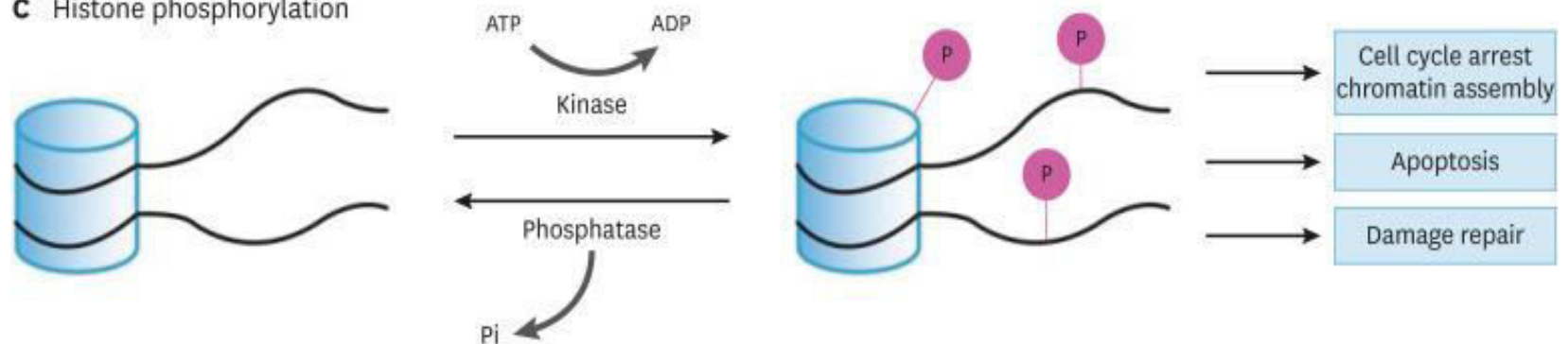
A Histone acetylation



B Histone methylation

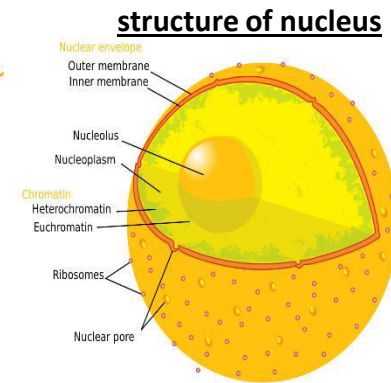


C Histone phosphorylation



B - RNA polymerase in the nucleus of eukaryotes are :

- 1) three classes (each class recognize particular type of genes)
- 2) large enzyme .
- 3) multiple subunit .



RNA polymerase I

synthesize the precursor of the large RNAs (28S, 18S and 5.8 S)

XXXX

in nucleolus (not nucleus)

(note that mRNA & tRNA are synthesized In the nucleoplasm)

RNA polymerase II

synthesize the precursor of the **messenger RNAs** that Translated to produce Protein .

it also synthesize small nuclear **RNA (snRNA)**

in nucleoplasm

(note that it is used to Produce viral DNA by some viruses)

RNA polymerase III

this enzyme produces the small RNAs, including **tRNA**, the **small 5s ribosomal RNA**, & some **snRNAs**.

See there



a) promoters for class II genes:

1. Contain 3 box :

a) Hogness box or TATA box: (ATATAAAA)

- a sequence of DNA nucleotides almost identical to Pribnow box (TATAAT)
- usually found about 25 nucleotides upstream (- 25) of initial base transcription start site of mRNA molecule

b) CAAT box :(GGCCAATCT)

- Found between 70 and 80 nucleotide upstream (-70 or -80)

C) GC box :(GGGCGG)

- Many promoter contain this box

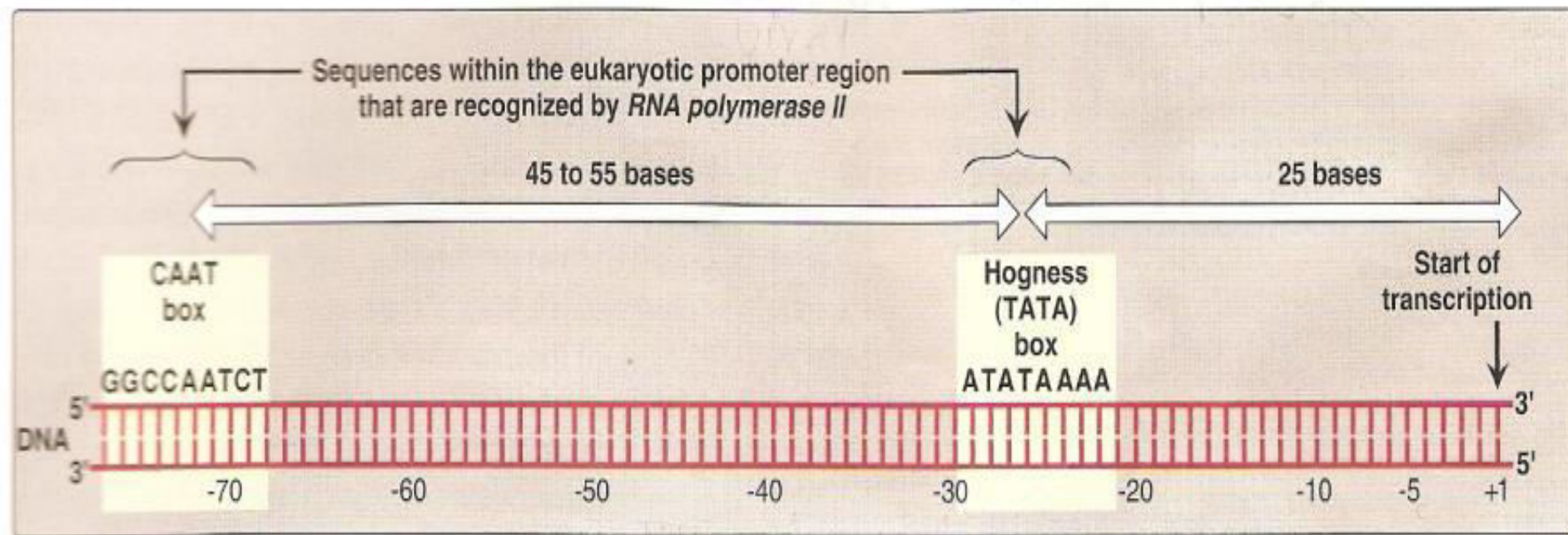


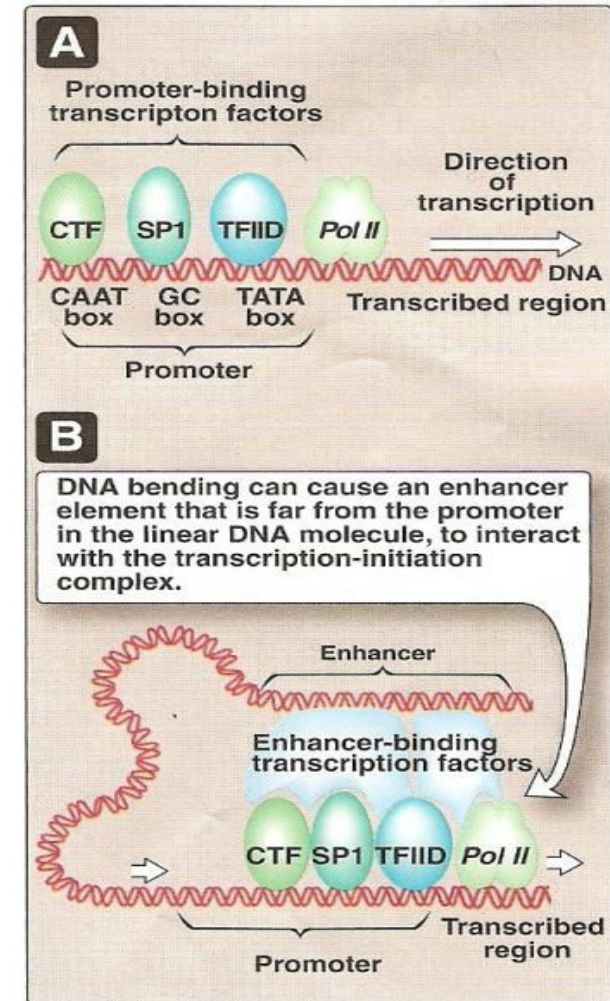
Figure
Eukaryotic gene promoter consensus sequences.

2. Cis-acting genetic elements : (DNA sequence)

- ✓ Because TATA box, CAAT box and GC box are found on the same molecule Of DNA as genes being transcribed they are called cis-acting genetic elements

3. Cis-acting genetic elements serve as binding sites for protein called general transcription factor :

- which in turn interact with each other and with RNA polymerase II
- Transcription factors encode by genes on different chromosomes (not the same gene as cis)
- Because its synthesis in cytosol can diffuse through the cell to their point of action (which may be different chromosome) , they are called trans-acting elements
- (it is protein not DNA sequence note the difference
- they can either stimulate or inhibit transcription of particular genes
- Note : Promoter- binding transcription factors: CTF , SP1, TFIID



Figure

A. Eukaryotic general transcription factors bound to the promoter. CTF, SP1, and TFIID are general transcription factors. B. Enhancer stimulation of RNA polymerase II.

b) Role of enhancer in eukaryotic gene regulation:

■ Note : What is Enhancers ?? ☹

- ✓ special cis- acting DNA sequence (always come in exams)

And note: because it is Cis so it is DNA sequence not protein (IIIIIIIMP):☺.

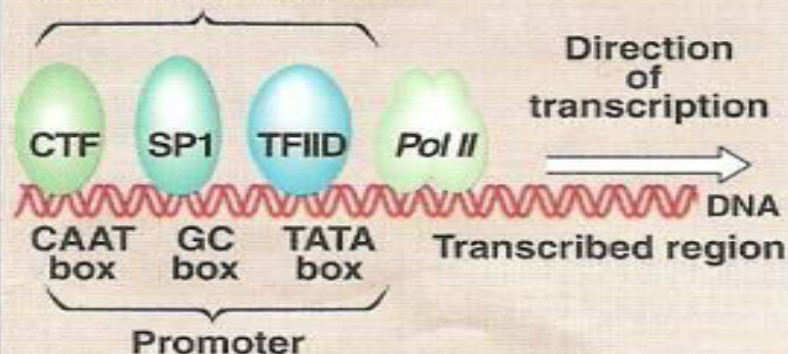
- ✓ increase the rate of initiation of transcription by RNA polymerase *II*
- ✓ must be in the same chromosome as the gene whose transcription they stimulate .
- ✓ they can be located "upstream" (to the 5'-side) or "downstream" (to the 3'-side) of the transcription start site .
- ✓ they can be close to or thousand of base-pair away from the promoter .
- ✓ they can occur on either strand of the DNA .
- ✓ contain DNA sequences called "**response elements**" that bind specific transcription factor called **activator**

So , By bending or looping the DNA, these enhancer- binding factor can interact with transcription factors bound to a promoter & with RNA polymerase *II*, thereby stimulating transcription

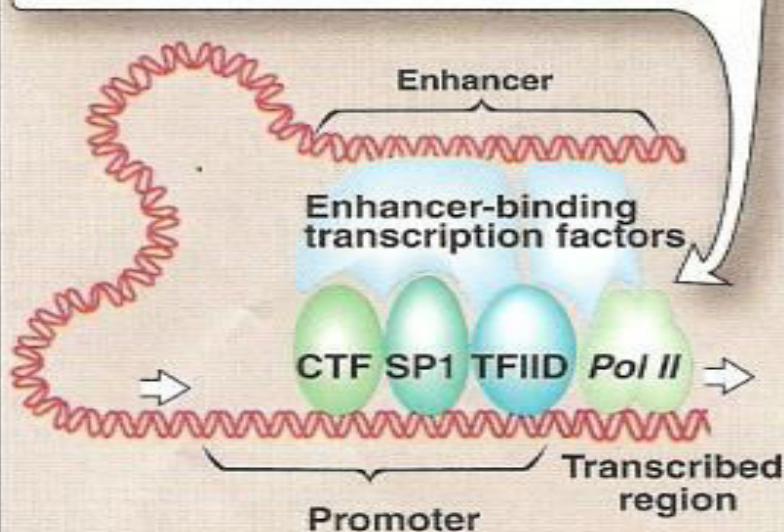
Note : **Silencers** act over long distances to reduce the level of gene expression.

A

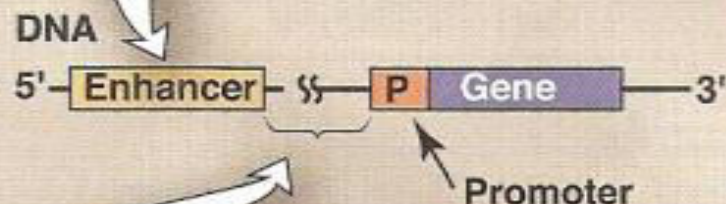
Promoter-binding
transcription factors

**B**

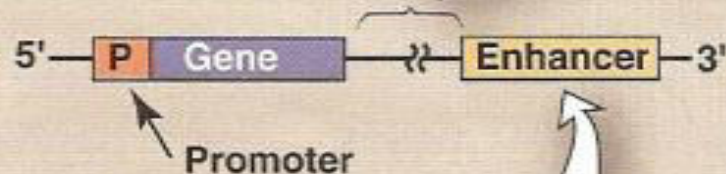
DNA bending can cause an enhancer element that is far from the promoter in the linear DNA molecule, to interact with the transcription-initiation complex.



Enhancer sequence "upstream"
from promoter region.



Up to 2000 base pairs
separate the enhancer sequence
from the gene it regulates.



An enhancer sequence can
be "downstream" from the
promoter region.

Figure

Some possible locations of
enhancer sequences.

c) Inhibitors of RNA polymerase II:

■ α - amanitin :

- inhibit RNA polymerase 11



- a potent toxin produce by the poisonous mushroom Amanita phalloides (sometimes called "death cap" or "destroy angel" –it is said to taste delicious!).

- because it forms a tight complex with the polymerase ,SO):

1) inhibiting mRNA synthesis .

2) inhibit protein sunthesis .

B. Mitochondrial RNA polymerase :

- Mitochondria contain a single RNA polymerase that resembles bacterial RNA polymerase more closely than it does the eukaryotic enzyme.

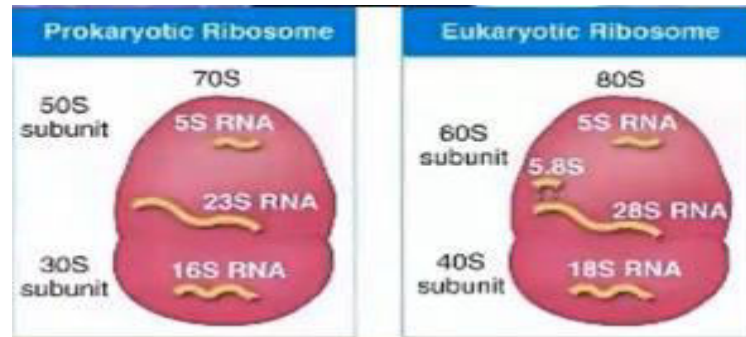
V. POSTTRANSCRIPTIONAL MODIFICATION OF RNA :

A primary transcript :

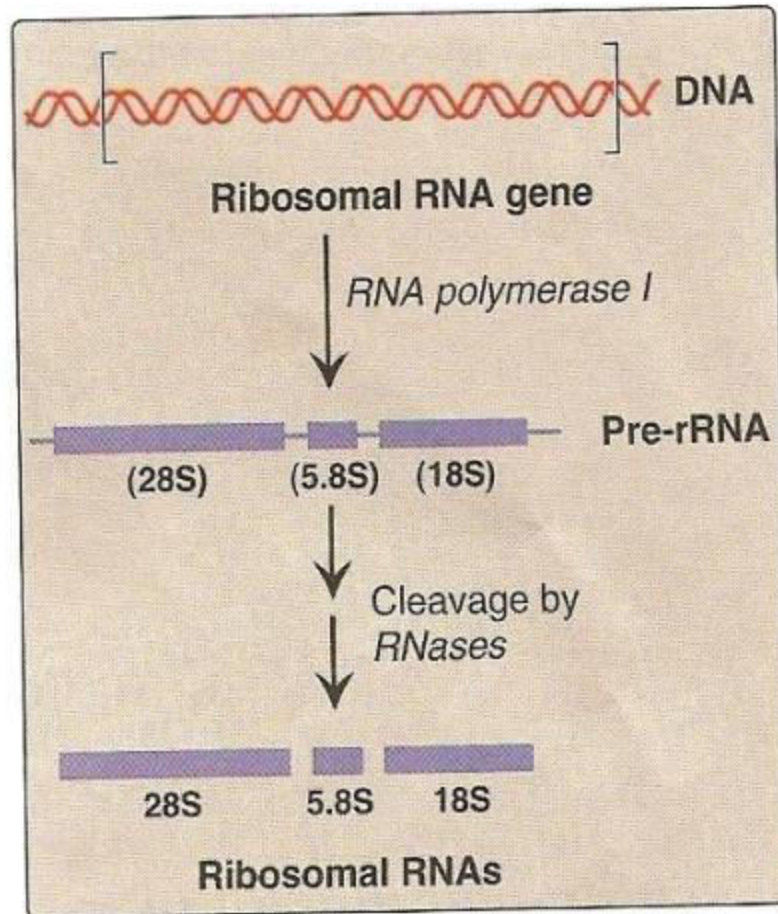
- is a linear copy of a transcriptional unit—the segment of DNA between specific initiation & termination sequences.

	prokaryotes	eukaryotes
rRNA	are post-transcriptionally modified by cleavage of the original transcripts by ribonucleases	are post-transcriptionally modified by cleavage of the original transcripts by ribonucleases
tRNA	are post-transcriptionally modified by cleavage of the original transcripts by ribonucleases . tRNAs are then further modified to help give each species its unique identity	are post-transcriptionally modified by cleavage of the original transcripts by ribonucleases . tRNAs are then further modified to help give each species its unique identity
mRNA	generally identical to its primary transcript(no posttranscription)	extensively modified posttranscriptionally

A. Ribosomal RNA:



prokaryotes	eukaryotes
in the nucleolus	in the nucleolus
are synthesized from long precursor molecules called preribosomal RNAs.	are synthesized from long precursor molecules called preribosomal RNAs.
DNA ----> rRNA gene by (RNA polymerase 1) ----> pre rRNA by (ribonucleases) ----> intermediate – sized piece of rRNA by (trimmed) ----> rRNA species .	DNA ----> rRNA gene by (RNA polymerase 1) ----> pre rRNA by (ribonucleases) ----> intermediate – sized piece of rRNA by (trimmed) ----> rRNA species .
The 23S,16S & 5S ribosomal RNAs of prokaryotes are produced from a <u>single</u> RNA precursor molecule	28S,18S & 5.8S rRNAs of eukaryotes are produced from a <u>single</u> RNA precursor molecule
Other notes : some of the protein destined to become components of the ribosome associate with rRNA precursor prior to & during its posttranscriptional modification in the nucleolus	Other notes : Eukaryotes 5S rRNA is synthesized by RNA polymerase <i>III</i> (not 1 like others) & modified separately



Figure

Posttranscriptional processing of eukaryotic ribosomal RNA by ribonucleases.

B. Transfer RNA : (exactly like what the doctor said)

- Both eukaryotic & prokaryotic transfer RNAs are also made from longer precursor molecules that must be modified .
- **14-nucleotide intron** must be removed from the **anticodon loop** by **splicing** , & sequences at both 5'-&3'- ends of the molecule must be terminated .
- **16 – nucleotide sequence at the 5' –end is cleaved by RNase P .**
- Other posttranscriptional modifications include replacement of uracil by addition of a-CCA sequence by (nucleotidyltransferase) to the 3'-terminal end of tRNA (**found on all mature tRNA**) .
- modification of bases at a specific positions to produce "**unusual bases**".
then
e.g : dihydrouracil .

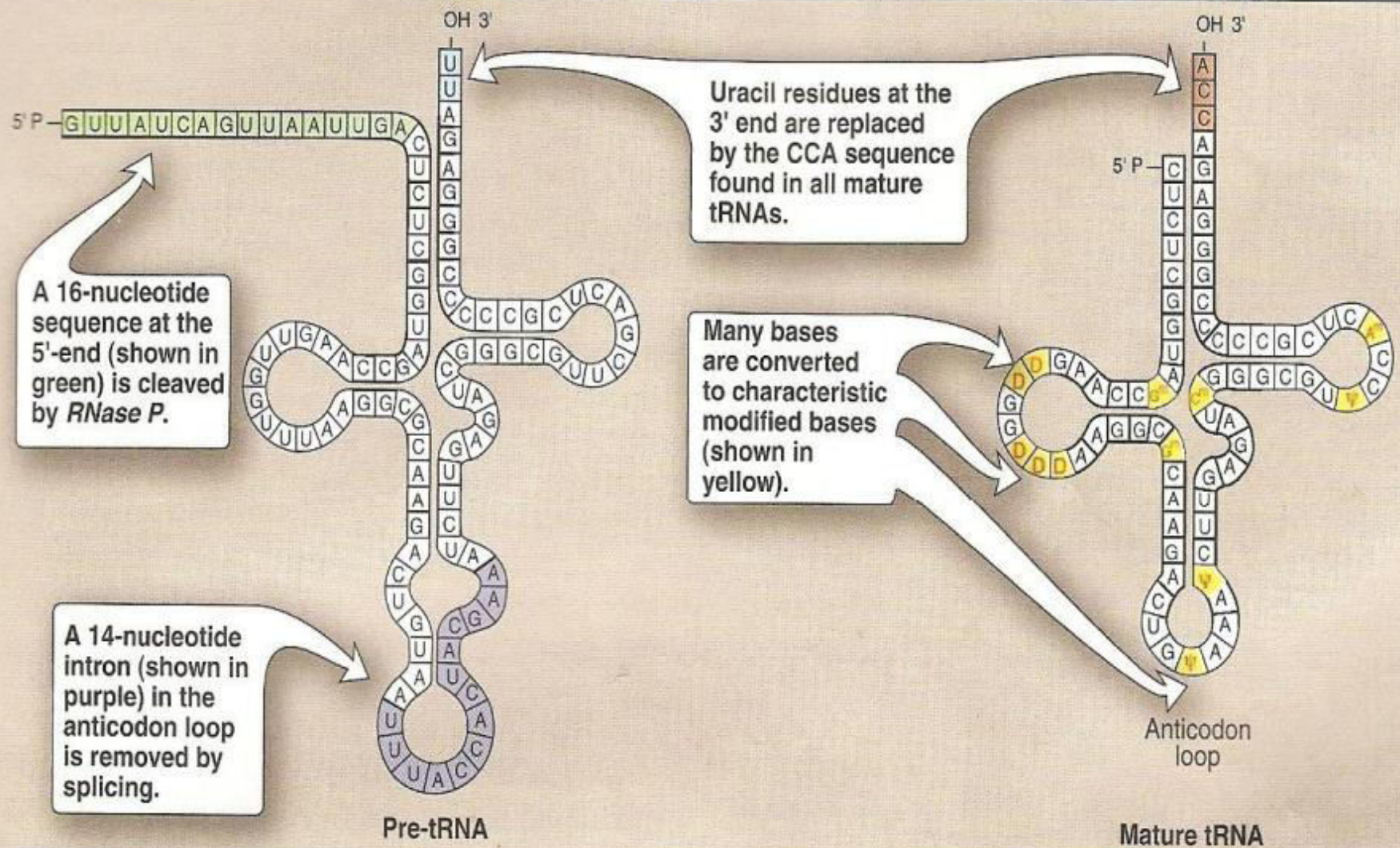


Figure 22-12

A. Primary tRNA transcript. B. Functional tRNA after posttranscriptional modification. Modified bases include D (dihydrouridine), ψ (pseudouridine), and m , which means that the base has been methylated.

C. Eukaryotic messenger RNA :

- The RNA molecule synthesized by RNA polymerase II (**primary transcript**)
(**Not polymerase I always come in exam so RNA polymerase I we find it
In rRNA**)
- contain the sequences that are found in cytosolic mRNA .
- The collection of all the precursor molecules for mRNA is known
heterogeneous nuclear RNA (hnRNA) .
- The primary transcript are extensively modified in the nucleus after
transcription , these modifications include :

1. 5' "capping":

- This is the first processing reactions for hnRNA .
- ✓ The cap is 7-methyl-guanosine attached **backward** to the terminal end of the mRNA forming an unusual 5' → 5' triphosphate linkage .
- ✓ The addition of the guanosine triphosphate part of the cap is catalyzed by the **nuclear** guanylyltransferase (note not cytoplasmis its nuclear) .
- ✓ Methylation of terminal guanine is catalyzed by the cytosolic guanine -7-methyltransferase (note not nuclear its cytoplasmic)
- ✓ , methyl donor is SAM (S-adenosylmethionie) .
 - ✓ Additional methylation steps may occur .
- ✓ Functions of the 7-methyl-guanosine cap are :
 - a. Initiation of translation (eukaryotic mRNAs lacking the cap are not efficiently translated) .
 - b. Stabilizing mRNA .

Are you still confusing turn the page to see how we change the ☹

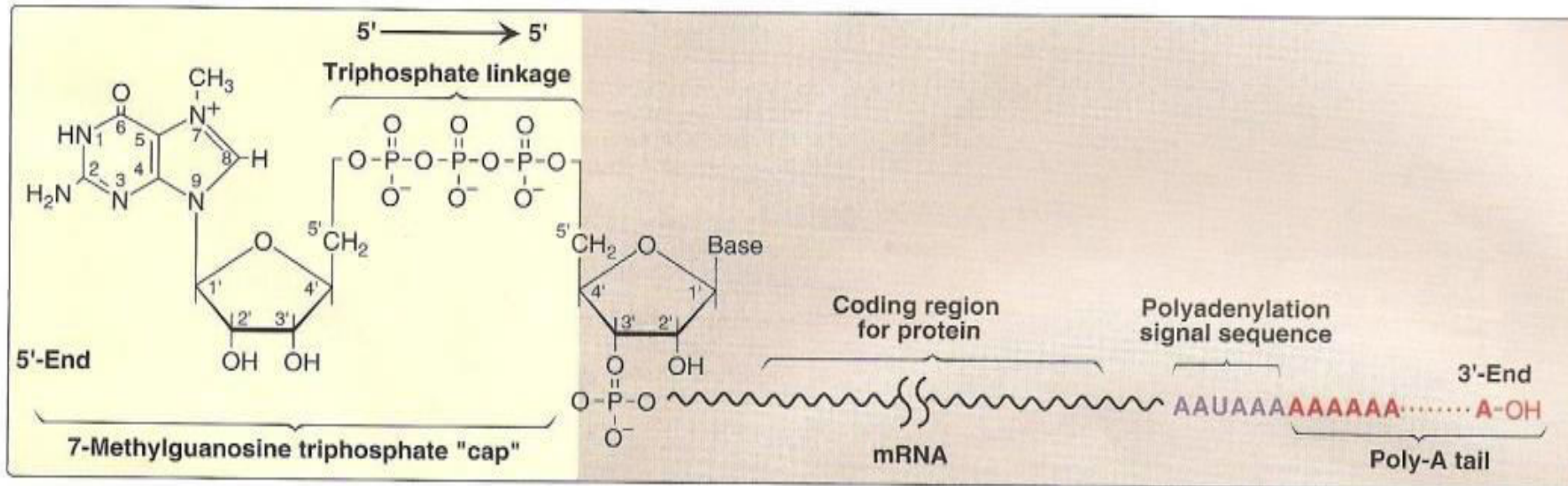


Figure
Posttranscriptional modification of mRNA showing the 7-methylguanosine cap and poly-A tail.

So,

According to 5' “capping”: (☹ ---- > ☺)

Cap : 7-methyl-guanosine .

Bond : unusual 5' → 5' triphosphate linkage .

Site : nucleus & cytosol .

Enzymes :

- 1- nuclear guanylyltransferase .
- 2- cytosolic guanine -7-methyltransferase ,
- 3 - methyl donor is SAM (S-adenosylmethionine) .

Functions :

- a. Initiation of translation (eukaryotic mRNAs lacking the cap are not efficiently translated).
- b. Stabilizing mRNA .

2. Addition of a poly-A tail

- Most eukaryotic mRNAs have a chain of 40-200 adenine nucleotides attached to the 3'-end (poly-A tail) . (MCQ)
 - ✓ Note that mRNAs those coding for the histones & some interferons doesn't have poly-A tail. (T) (always come in final exams)
- ✓ The poly-A tail is not transcribed from DNA , but is added after transcription by nuclear polyadenylate polymerase .
- ✓ A consensus sequence called the polyadenylation signal sequence (AAUAAA) found near the 3'-end of the RNA molecule , signal the adding of poly-A tail to the mRNA .
- ✓ Functions of poly-A tail :
 - a. Stabilizing mRNA .
 - b. Facilitate the exit of mRNA from the nucleus to the cytosol .
- ✓ After mRNA enters the cytosol , the poly-A tail is gradually shortened .

So,

According to **poly-A tail**: (☹️ ----- > 😊)

- **Nature** : a chain of 40-200 adenine nucleotides attached to the 3'-end of mRNA.

Remember :

- Not found in mRNAs of the histones & some interferons 😊 .
- is not transcribed from DNA , but is added after transcription .

- **Site** : nucleus & cytosol .

- **Enzyme of adding** : nuclear polyadenylate polymerase (not cytoplasm) .

- **signal of adding** : polyadenylation signal sequence (AAUAAA) found near the 3'-end of the RNA molecule .

- **Functions** :

- a. Stabilizing mRNA .
- b. Facilitate the exit of mRNA from the nucleus to the cytosol .

Don't forget :After mRNA enters the cytosol , the poly-A tail is gradually shortened 😊 .

3. Removal of introns :

- In the maturation of EUARYOTIC mRNA ... usually RNA sequences are removed (don't code for protein (introns or intervening sequences) from the primary transcripts)
- The remaining coding sequence (exon) are spliced together to form the mature mRNA
- So mature mRNA is smaller than primary transcripts
- This process is accomplished by (spliceosome)

Intron

- Sequence not coding for protein
- No , few or many .
- some contain more than 50% intervening sequences(such as the primary transcripts for the α -chains of collagen) that must be removed before mature mRNA is ready for translation

exon

-Sequence coding protein

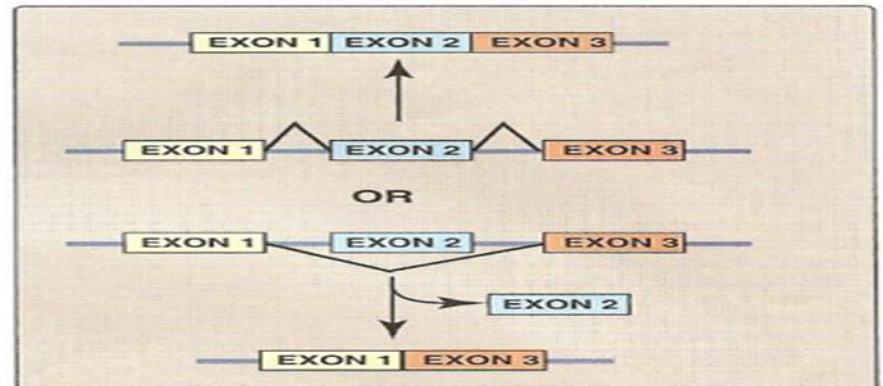


Figure 30.19
Alternative splicing patterns in eukaryotic mRNA.

a. role of small nuclear RNAs (snRNAs):

- (snRNPs or " snurps ") = snRNAs + proteins .
- spliceosome = snRNPs + primary transcripts .
- Function : these facilities the splicing of exon segments by forming base pair with the consensus sequences at each end of the intron .

■ **Note : Removal of intron is an energy dependent mechanism .**

■ **systemic lupus erythematosus :**

- Fatal inflammatory disease
- autoimmune response .
- antibodies against host protein including snRNPs

b. Mechanism of excision of introns :

1- binding of snRNPs .

Function : brings the sequences of the neighboring exons into the correct alignment for splicing .

2- The 2'-OH group of an adenosine (A) residue (known as the branch site)

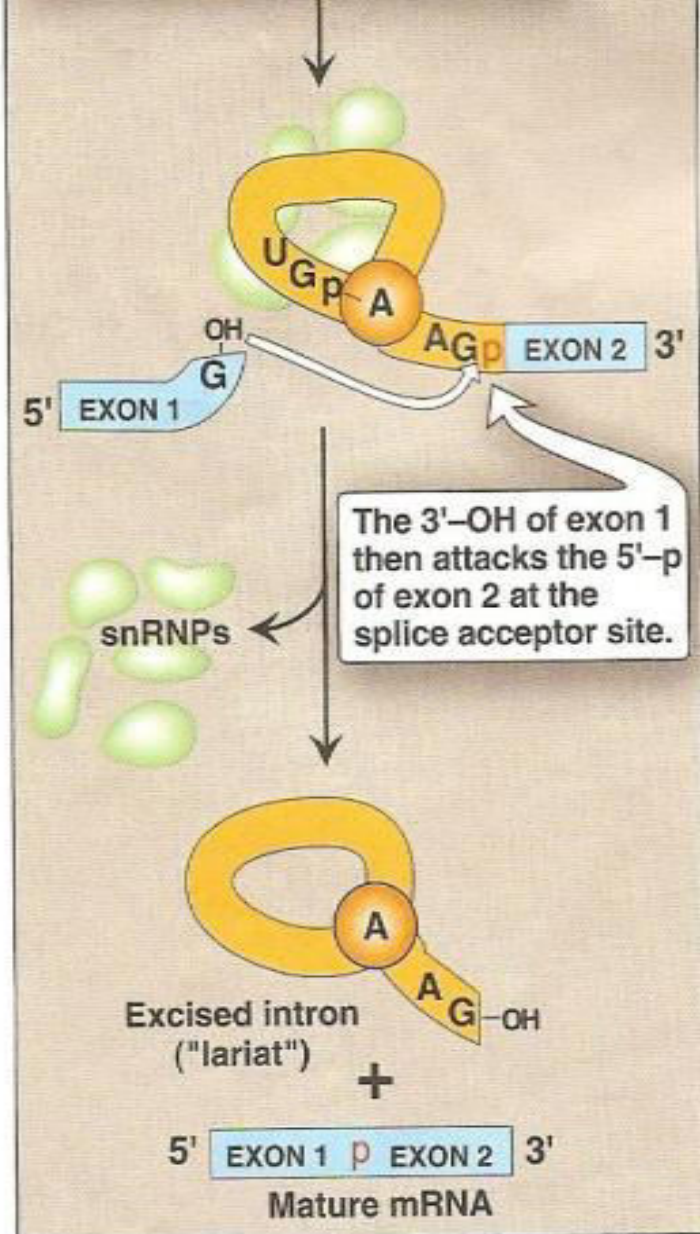
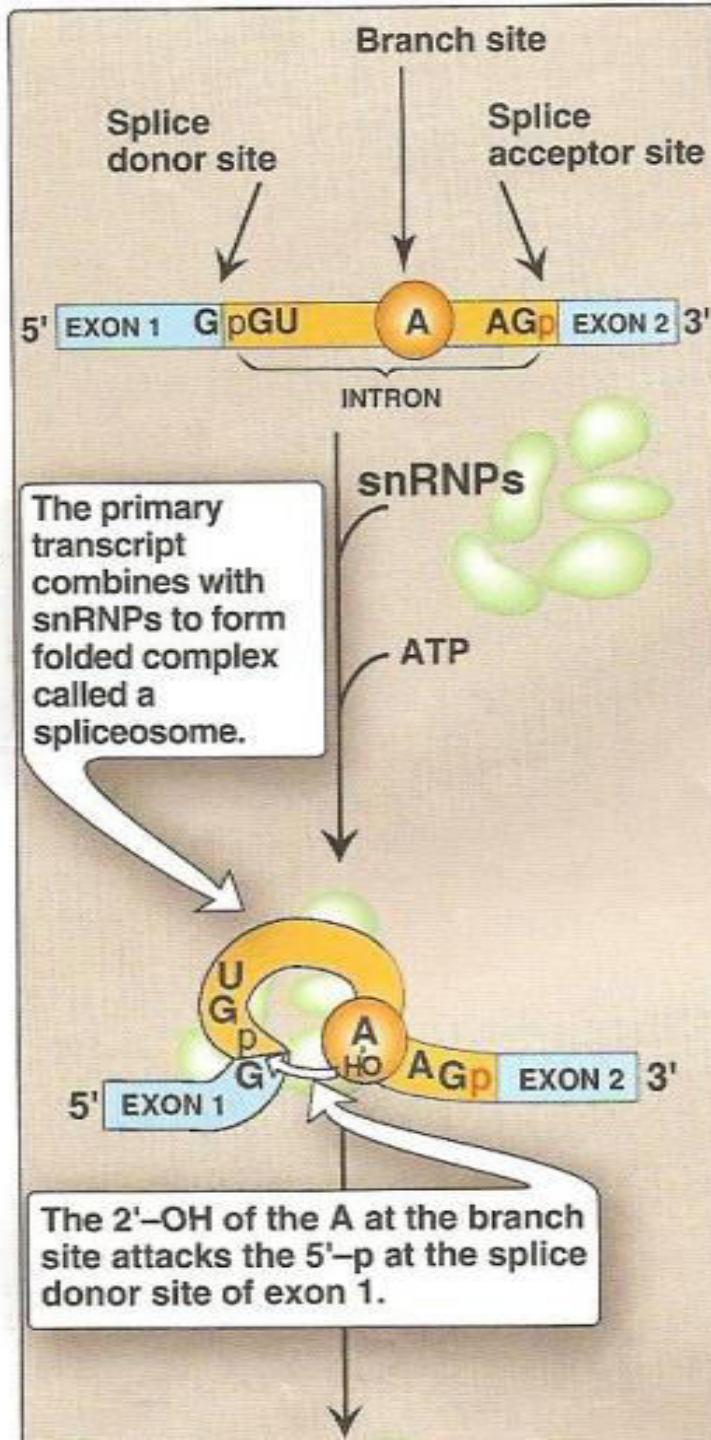
- **Function** : for intron attacks (IMP)(always come in exams) .
- **Type of bond** : forms a phosphodiester bond with the phosphate at the 5'-end of intron 1 .

Note : the GU & AG sequences at the branch site are invariant

3- The newly freed 3'- OH of the upstream exon 1 then forms a phosphodiester bond with the 5'-end of the downstream exon 2 .

4- The excised intron is released as a "lariat" structure , which is degraded.

■ After removal of all introns, the mature mRNA molecules leave the nucleus by passing into the cytosol through pores in the nuclear membrane.



Figure

Removal of introns. snRNP = small nuclear ribonucleoprotein particle.

C . effect of splice site mutation :-

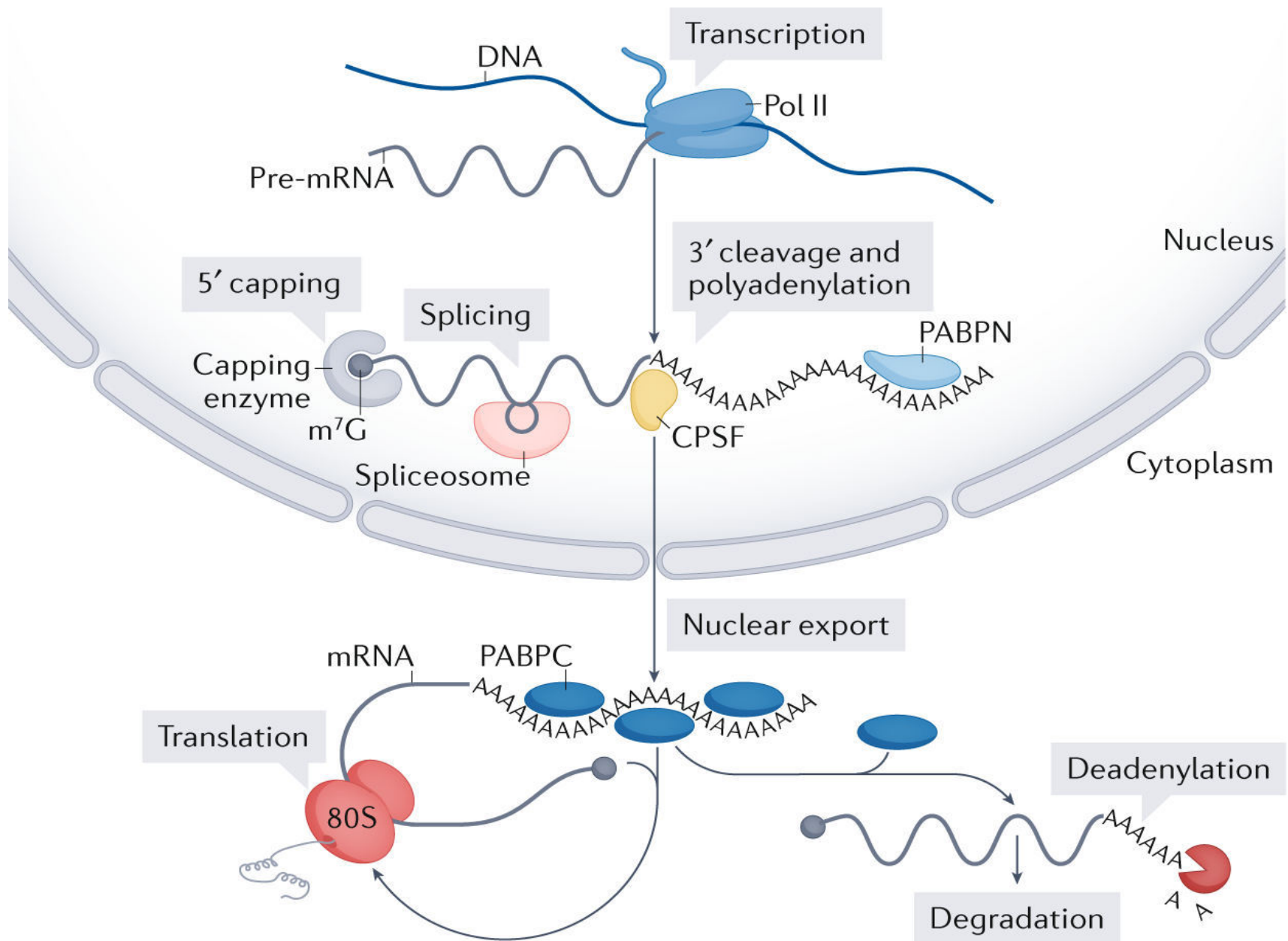
* mutation at splices site can lead to :

- improper splicing
- production of aberrant proteins

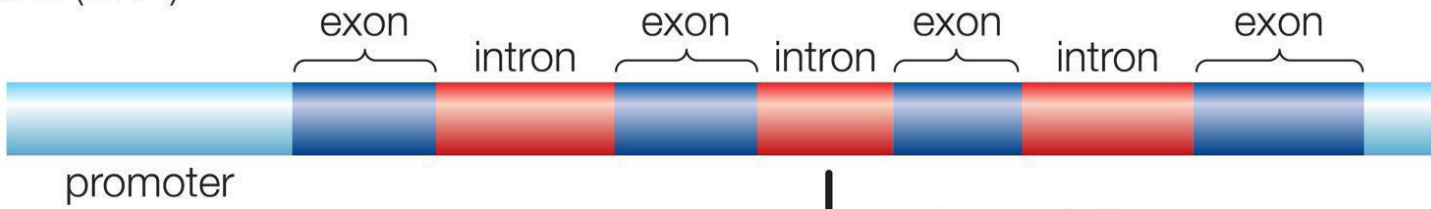
* 15% of genetic disease result from mutation affect RNA splicing

* EX : **β thalassemia** :

- from mutation affect the splicing of RNA
- disease in which the production of the β -globin protien is defective



gene (DNA)



transcription

RNA



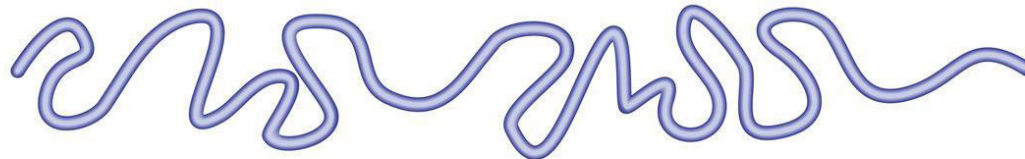
splicing

mRNA



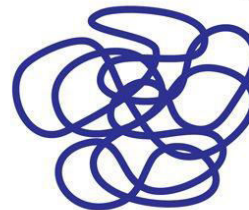
translation

amino acid
chain



posttranslational
modification

protein



Made with **KINEMASTER**

instructions to make proteins are contained in our DNA. DNA contains genes. A gene is a continuous string of

MOLECULAR BIOLOGY OF THE GENE

GENES AND HOW THEY WORK



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DNA Damages

- ⊙ **Due to by physical, chemical & environmental agents.**
- ⊙ **Broadly classified into four categories.**
 1. **Single base alterations** (e.g. depurination, deamination).
 2. **Two-base alterations** (e.g. pyrimidine dimer)
 3. **Chain breaks** (e.g. ionizing radiation)
 4. **Cross-linkages** (e.g. between bases).

- ⊙ Cytosine gets **deaminated** to form uracil while **adenine** forms hypoxanthine.
- ⊙ Spontaneous **depurination**, due to cleavage of glycosyl bonds (that connect purines to the backbone).
- ⊙ The **depurinated** sites are called as **abasic sites**.
- ⊙ They were **detected** in purines & called **apurinic sites (AP sites)** which represent lack of purine.
- ⊙ The term **AP sites** is generally used to represent any base lacking in DNA.

- ① **The production of reactive oxygen species** is often associated with alteration of bases e.g. formation of 8-hydroxy guanine.
- ② **Free radical formation & oxidative damage to DNA** increases with advancement of age.
- ③ **Ultraviolet radiations** result in the formation of covalent links between adjacent pyrimidines along the DNA strand to form pyrimidine dimers.
- ④ **DNA chain breaks** can be caused by ionizing radiations (e.g. x - rays).

Mutations

- ⊙ **Mutation refers to a change in the DNA structure of a gene.**
- ⊙ **The substances (chemicals) which can induce mutations are collectively known as mutagens.**
- ⊙ **The changes that occur in DNA on mutation are reflected in replication, transcription & translation.**

Types of mutations

- ⊙ **Two major types**

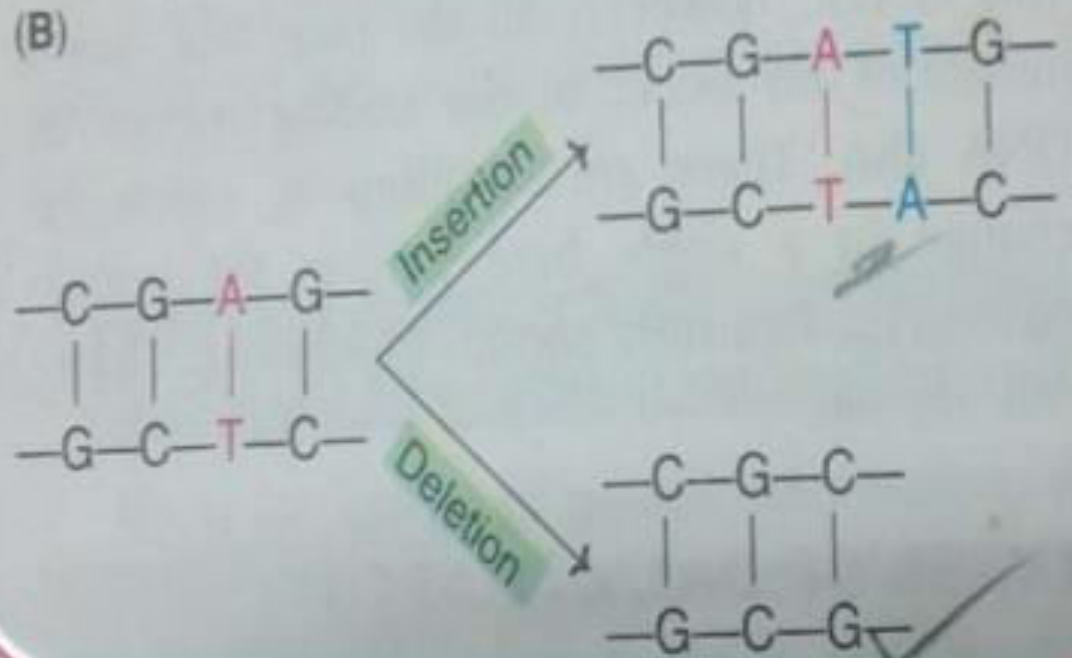
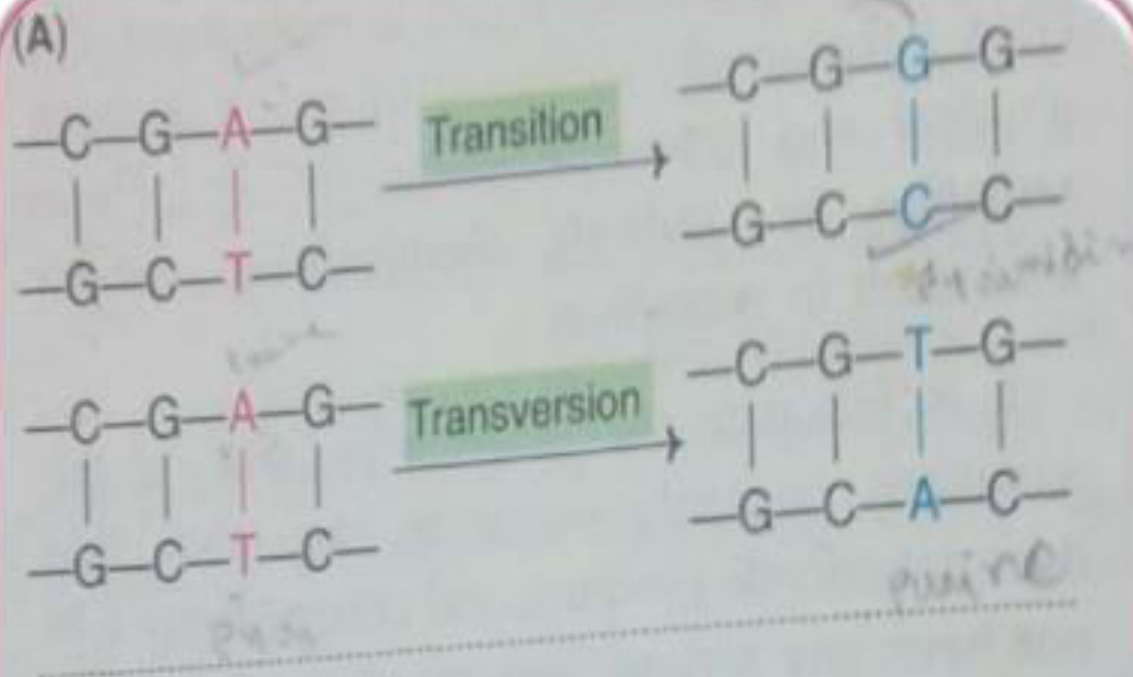
1. **Point mutations**
2. **Frameshift mutations**

- ⊙ **Point mutations:**

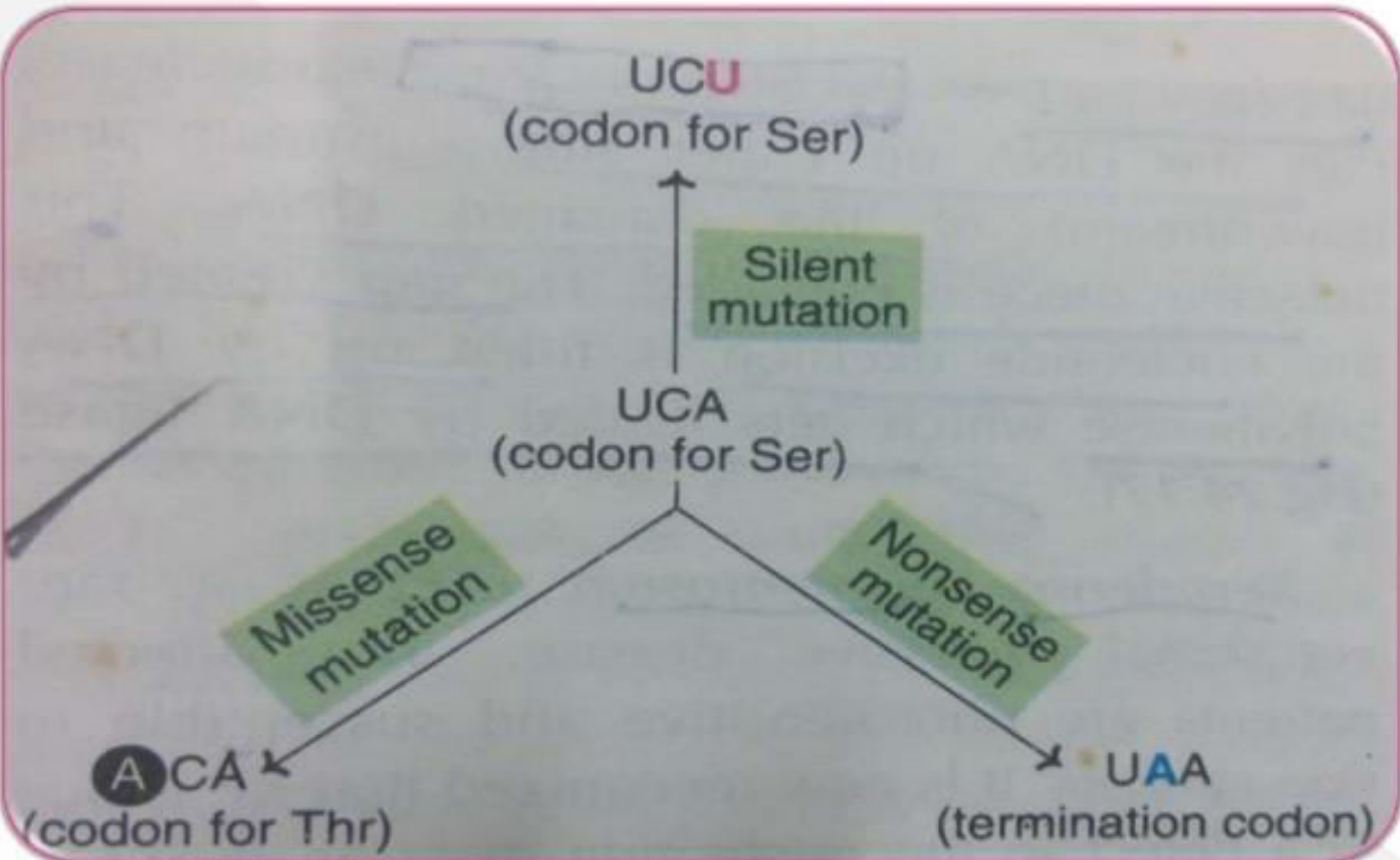
- ⊙ **The replacement of one base pair by another results in point mutation.**

- ⊙ **They are of two sub-types.**
- ⊙ **Transitions:**
- ⊙ **In this case, a purine (or a pyrimidine) is replaced by another.**
- ⊙ **Transversions:**
- ⊙ **These are characterized by replacement of a purine by a pyrimidine or vice versa.**

Point & frameshift mutations



Point mutations



Altering the nucleotide sequence in a codon can cause

silent mutations

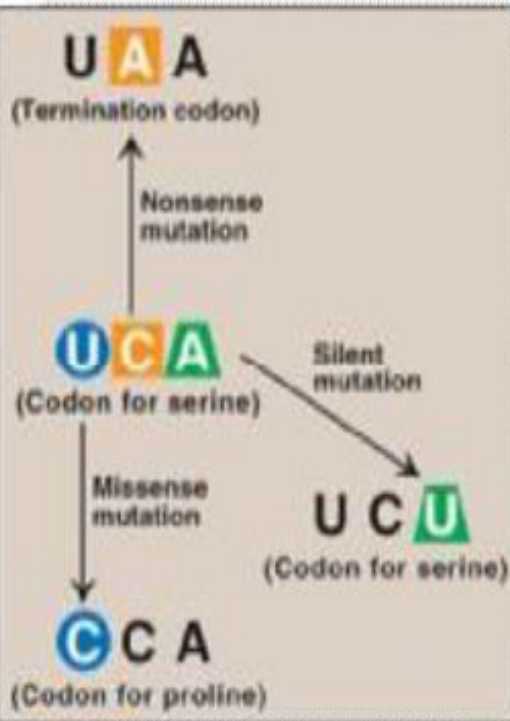
the altered codon codes for the original amino acid

missense mutations

the altered codon codes for a different amino acid

nonsense mutations

the altered codon is a termination codon



Consequences of point mutations

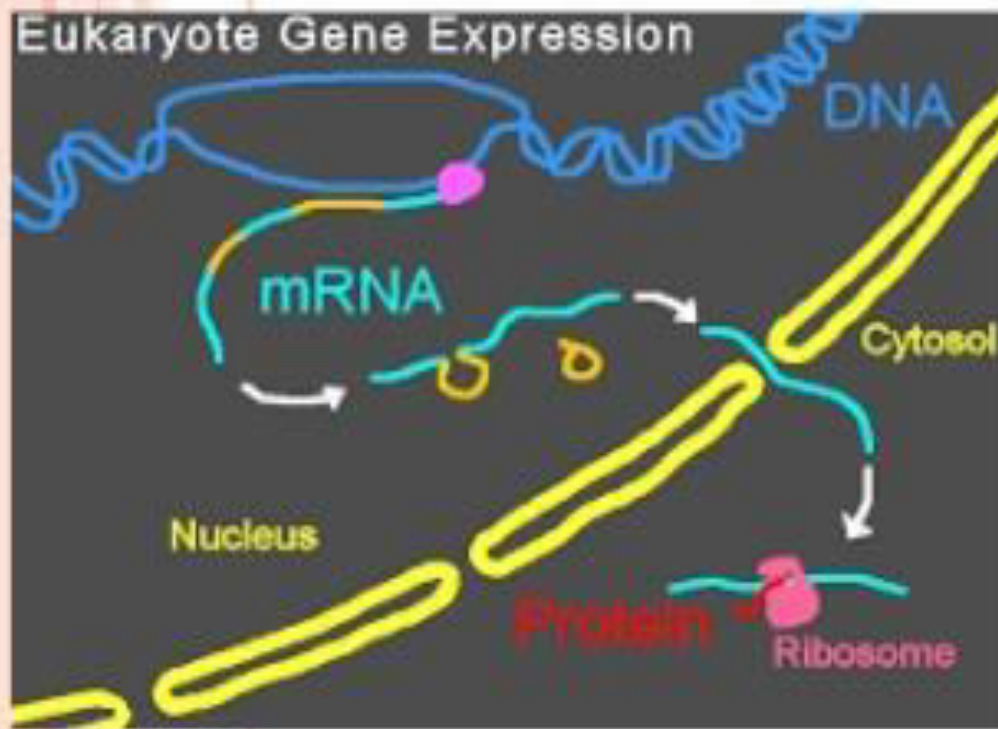
- ⊙ **Silent mutation:**
- ⊙ **The codon (of mRNA) containing the changed base may code for the same amino acid.**
- ⊙ **UCA codes for serine & change in the third base (UCU) still codes for serine.**
- ⊙ **This is due to degeneracy of the genetic code.**
- ⊙ **There are no detectable effects.**

- ⊙ **Missense mutation:**
- ⊙ **In this case, the changed base may code for a different amino acid.**
- ⊙ **UCA codes for serine while ACA codes for threonine.**
- ⊙ **The mistaken (or missense) amino acid may be acceptable, partially acceptable or unacceptable with regard to the function of protein molecule.**
- ⊙ **E.g. Sickle-cell anemia.**

- ⊙ **Nonsense mutation:**
- ⊙ **The codon with the altered base may become a termination (or nonsense) codon.**
- ⊙ **Change in the second base of serine codon (UCA) may result in UAA.**
- ⊙ **The altered codon acts as a stop signal & causes termination of protein synthesis.**
- ⊙ **Frameshift mutations:**
- ⊙ **These occur when one or more base pairs are inserted in or deleted from the DNA, respectively causing insertion or deletion mutations.**

Consequences of frameshift mutations

- ⊙ **The insertion or deletion of a base in a gene results in an altered reading frame of the mRNA.**
- ⊙ **The machinery of mRNA (containing codons) does not recognize that a base was missing or a new base was added.**
- ⊙ **No punctuation in the reading of codons, translation continues.**
- ⊙ **The result is that the protein synthesized will have several altered amino acids and/or prematurely terminated protein.**



REGULATION OF GENE EXPRESSION

WHAT THE I MBBS STUDENTS SHOULD KNOW ?

- What do you mean by term gene expression and its regulation ?
- Why gene regulation is important for life ?
- Explain gene regulation in prokaryotes : Jacob Monad experiment (Lac operon model)
- Explain the mechanisms or sites of gene regulation in Eukaryotes



GLOSSARY

- What is gene ?
- What is genome ?
- Total no of genes in our body ?
- Gene expression ?
- Gene product ?
- Positive regulation ? Negative regulation ?

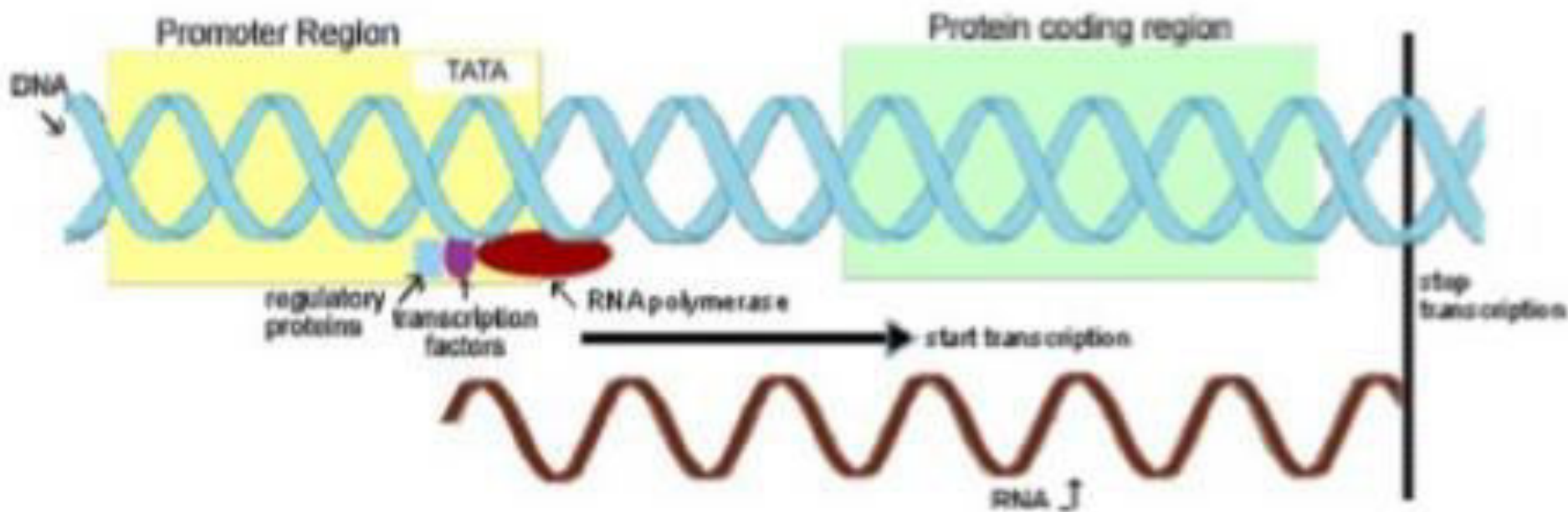
Inducer

repressor

derepressor

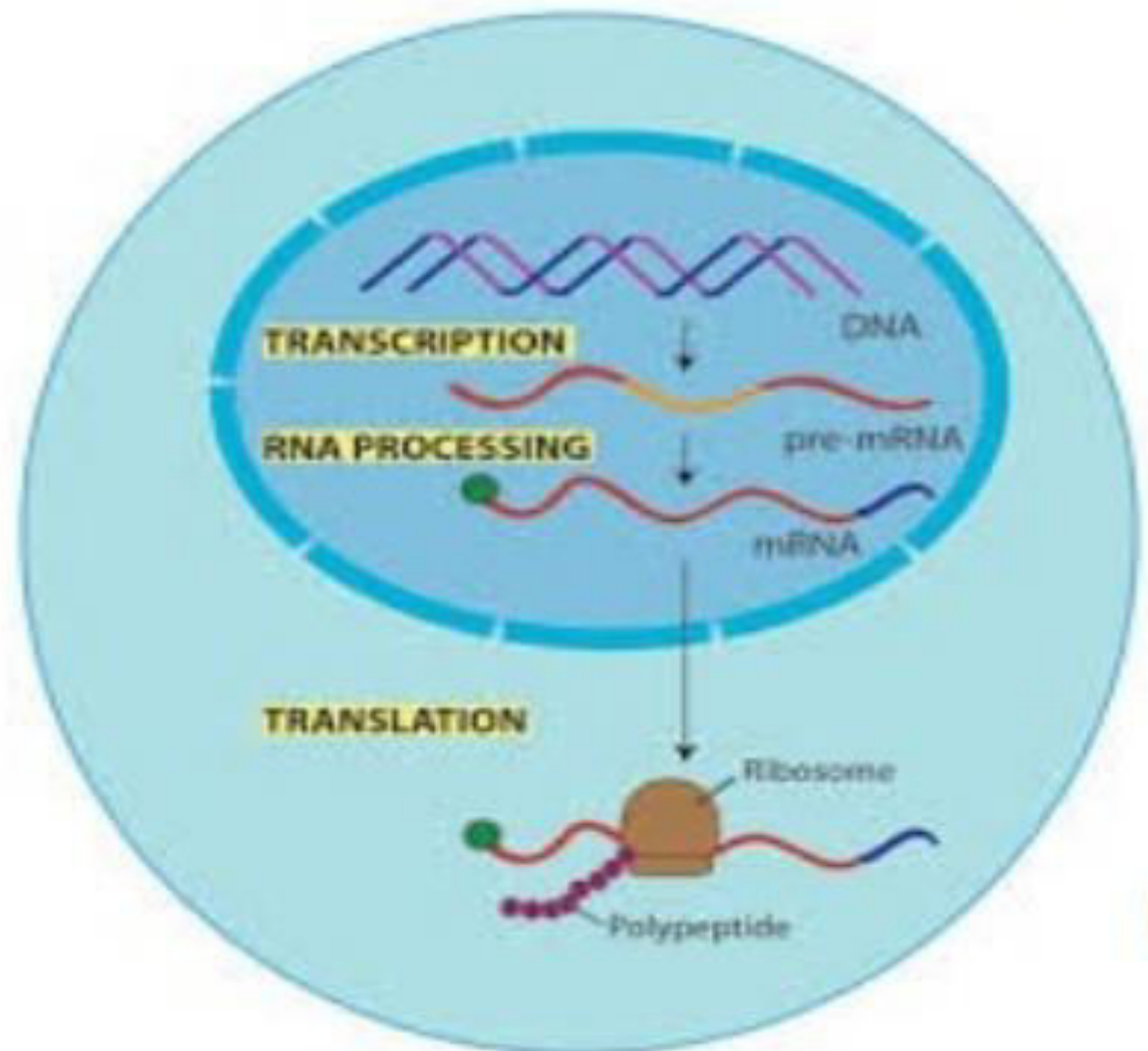


Typical gene organization



REGULATION OF GENE EXPRESSION

FLOW OF GENETIC INFORMATION



TYPES OF GENES

- **Constitutive genes** (House keeping genes):
Expressed at a reasonably constant rate
Not subjected to regulation
eg: enzymes of glycolysis
- **Regulated genes**
expressed only under certain conditions
expressed in all cells / subset of cells
eg: expression of insulin gene in pancreas



TYPES OF GENES

- **Inducible gene:**

expression increases in response to inducer or activator

eg: bile acid induces ALP

Gratuitous inducers

Compounds structurally similar to substrates may act as inducers

eg: IPTG (isopropylthiogalactoside) – lactose analog



- **Repression**

inhibition of gene expression by repressor
eg: inhibition of ALA synthase by heme

Catabolite repression:

Catabolite of a molecule inhibit gene expression



TYPES OF GENE EXPRESSION REGULATION

- **Positive regulation**

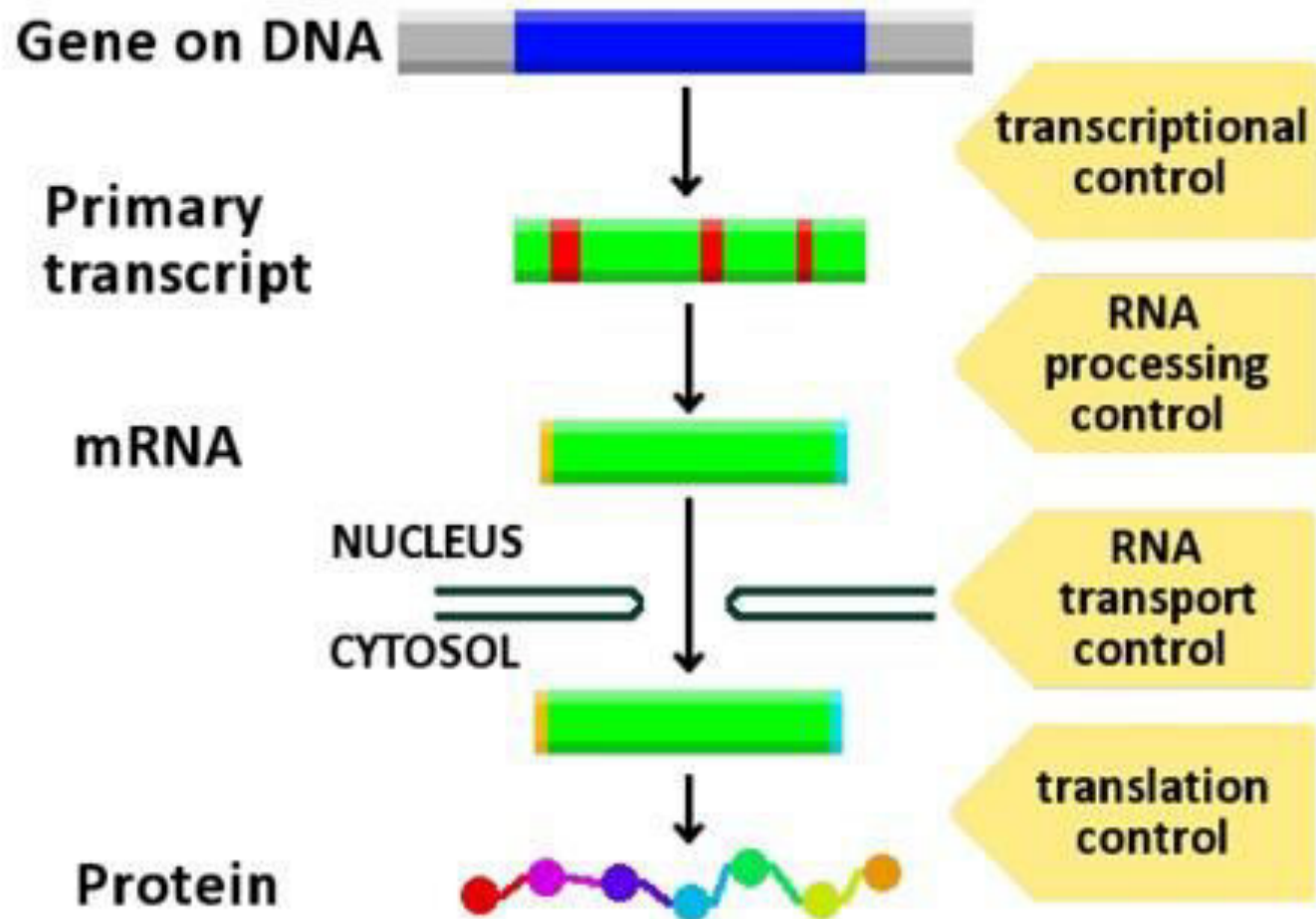
increased gene expression mediated by
positive
regulator / enhancer / activator

- **Negative regulation**

reduced gene expression mediated by
negative
regulator / silencer / repressor



REGULATION OF GENE EXPRESSION



REGULATION OF GENE EXPRESSION

- Gene expression can be modulated by
 - Control of transcription
 - Post transcriptional modifications
 - Gene amplification
 - Gene rearrangements
 - Control of translation
 - Protein modification / stabilization
- Gene regulation is influenced by hormones, heavy metals and chemicals



REGULATION OF GENE EXPRESSION IN PROKARYOTES

- Regulation occurs at transcription
- Operon

Genes involved in a metabolic pathway
when present in a linear
array
- Cistron

Smallest unit of gene expression
- Polycistronic mRNA: A single mRNA that
encodes more than one separately
translated protein.

eg - lac operon mRNA

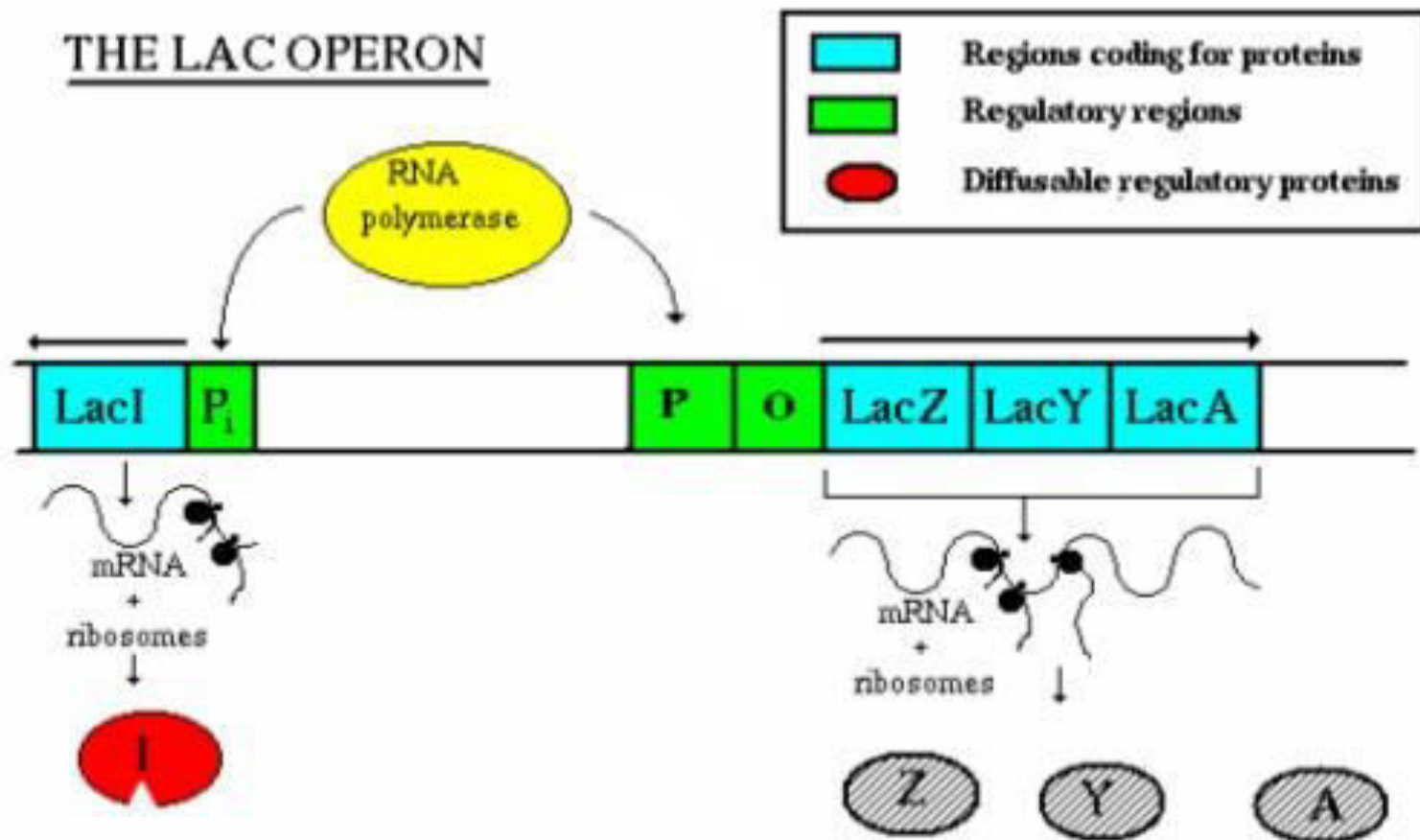


OPERON CONCEPT OF GENE REGULATION

- Francois Jacob and Jacques Monod, 1961
- Lactose metabolism in E.Coli (intestinal bacteria)
- Lac Operon
 - structural gene
 - inhibitor gene
 - promoter / operator areas
- Repression – derepression mechanism



THE LAC OPERON



- Structural gene
 - lacZ* : β - galactosidase (lactase)
 - lacY*: galactoside permease
 - lacA*: thiogalactoside transacetylase
- Inhibitor gene
 - lacI*: LacI (lac operon repressor protein)
- Promoter site
 - transcription of structural genes
- Operator site
 - binding of lac repressor



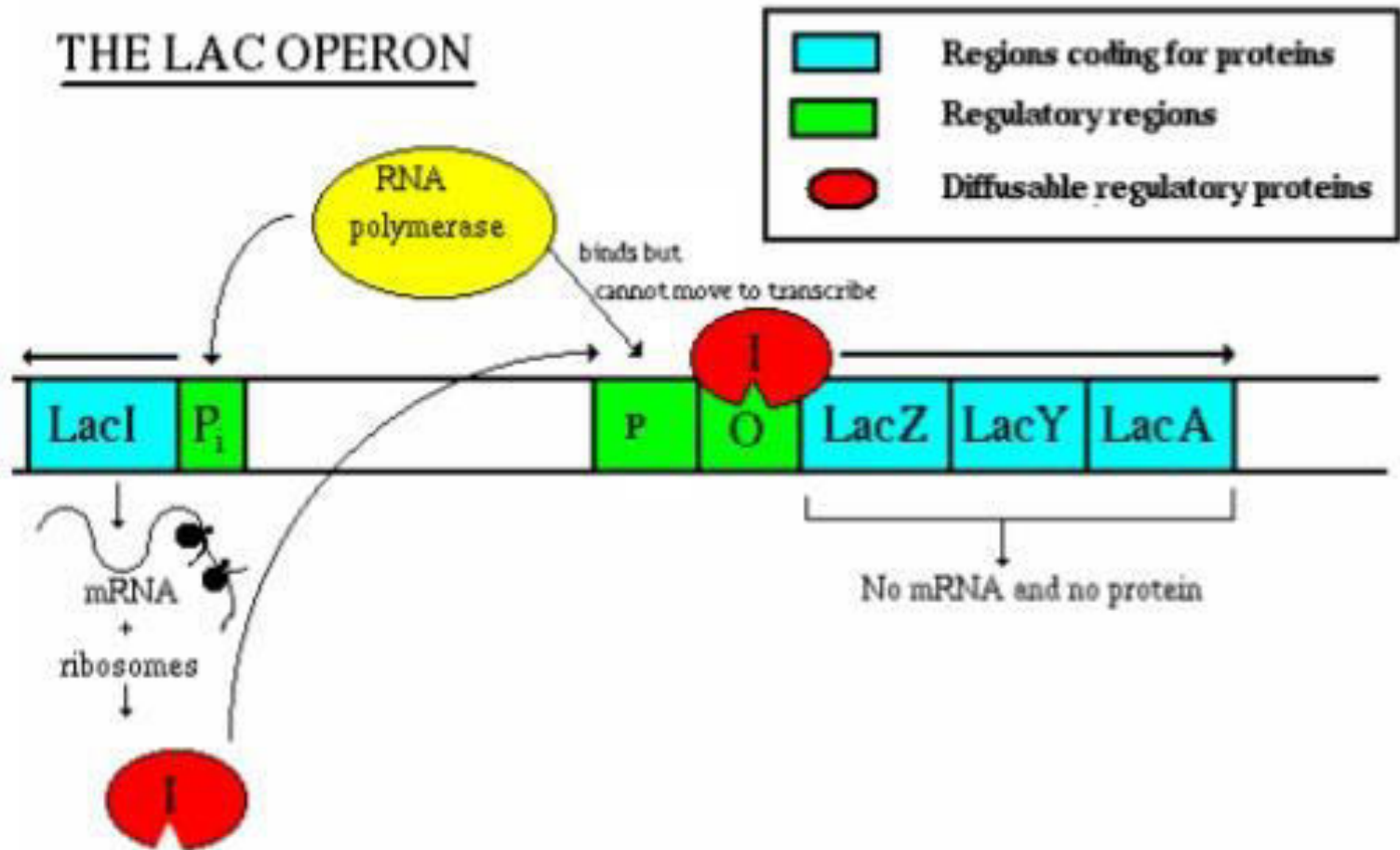
REGULATION LAC OPERON

- Absence of lactose – reduced synthesis of lactose metabolizing enzymes
- presence of lactose – increased enzyme synthesis
- Presence of lactose and glucose – reduced synthesis of enzymes



REPRESSION OF LAC OPERON (ABSENCE OF LACTOSE)

THE LAC OPERON



REPRESSION OF LAC OPERON (ABSENCE OF LACTOSE)

Regulatory gene



lac repressor



binds to operator site



RNAP cannot act on structural genes

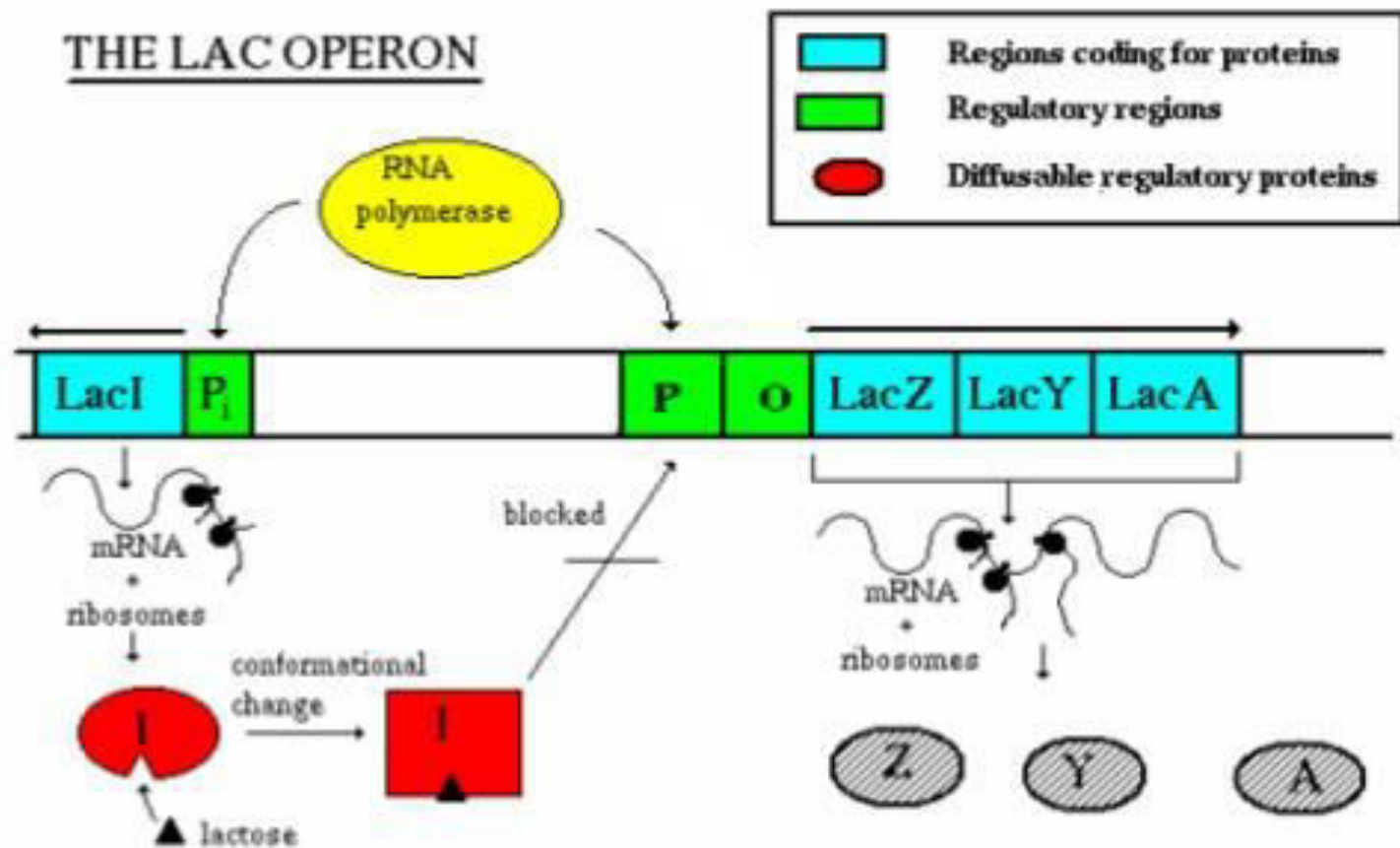


transcription not → lactose utilizing enzymes not synthesized

synthesized



DEREPRESSION OF LAC OPERON (PRESENCE OF LACTOSE)



DEREPRESSION OF LAC OPERON (PRESENCE OF LACTOSE)

Lactose + repressor



conformational changes in repressor



repressor will not bind to operator site



RNAP transcribe structural genes

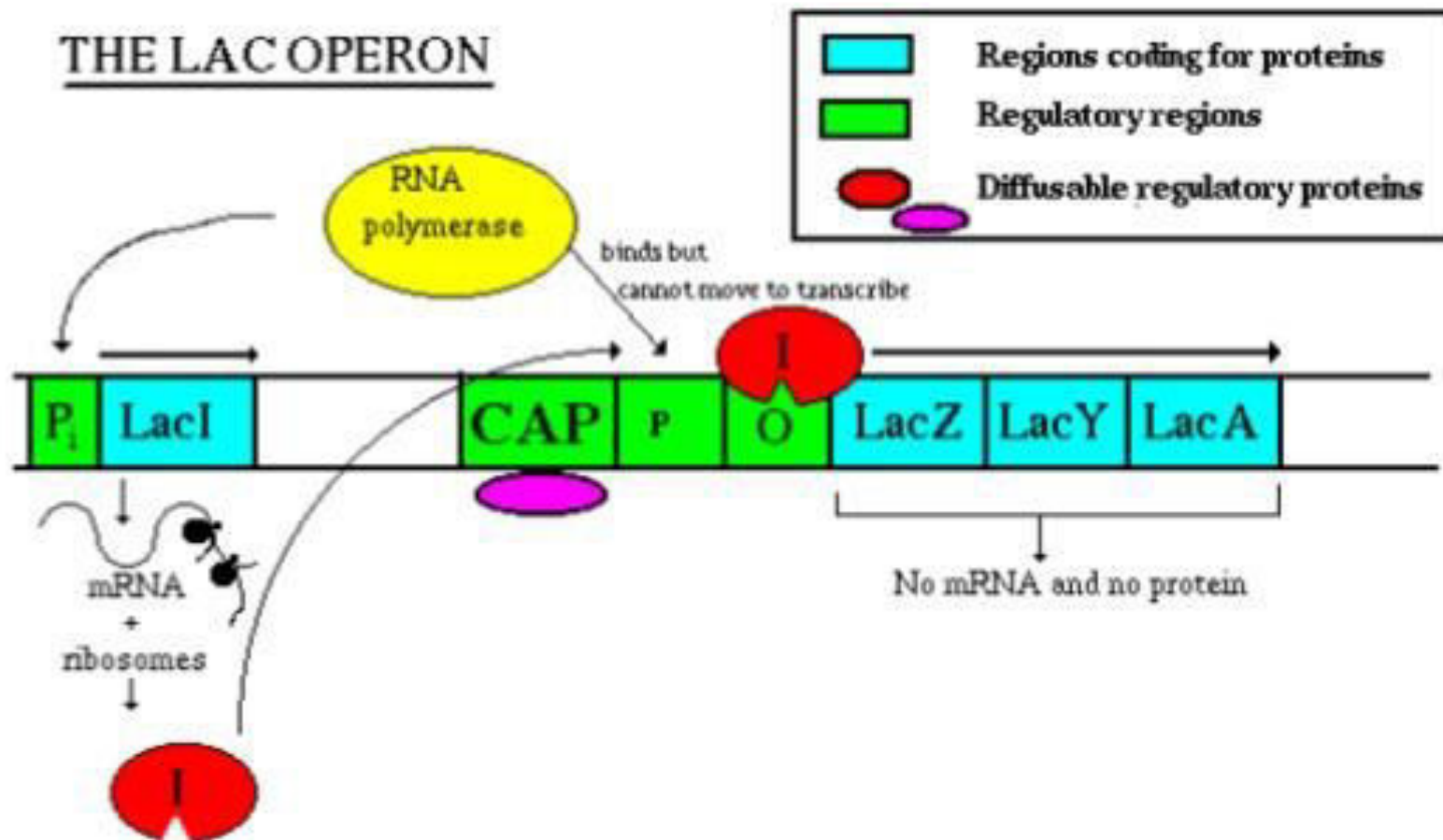


synthesis of lactose utilizing enzymes



CATABOLITE REPRESSION (BOTH GLUCOSE AND LACTOSE PRESENT)

THE LAC OPERON



CATABOLITE REPRESSION

- Repression of lac operon by catabolite of glucose (catabolite activator protein)

↓
CAP + c AMP

↓
facilitate binding of RNAP to the promoter

↓
transcription of structural gene



EUKARYOTIC GENE EXPRESSION

- Eukaryote genome is complex
DNA is extensively folded and packed into chromatin (protein – DNA complex)

Differs from prokaryotic gene regulation

- Genes are not organized into operon
- Separation of transcription and translation
- RNA processing : capping at 5' ends,
addition of poly A tail at 3' end
splicing



DNA ORGANIZATION

RECOLLECT HOW DNA IS ORGANISED

BUZZ SESSION!!!



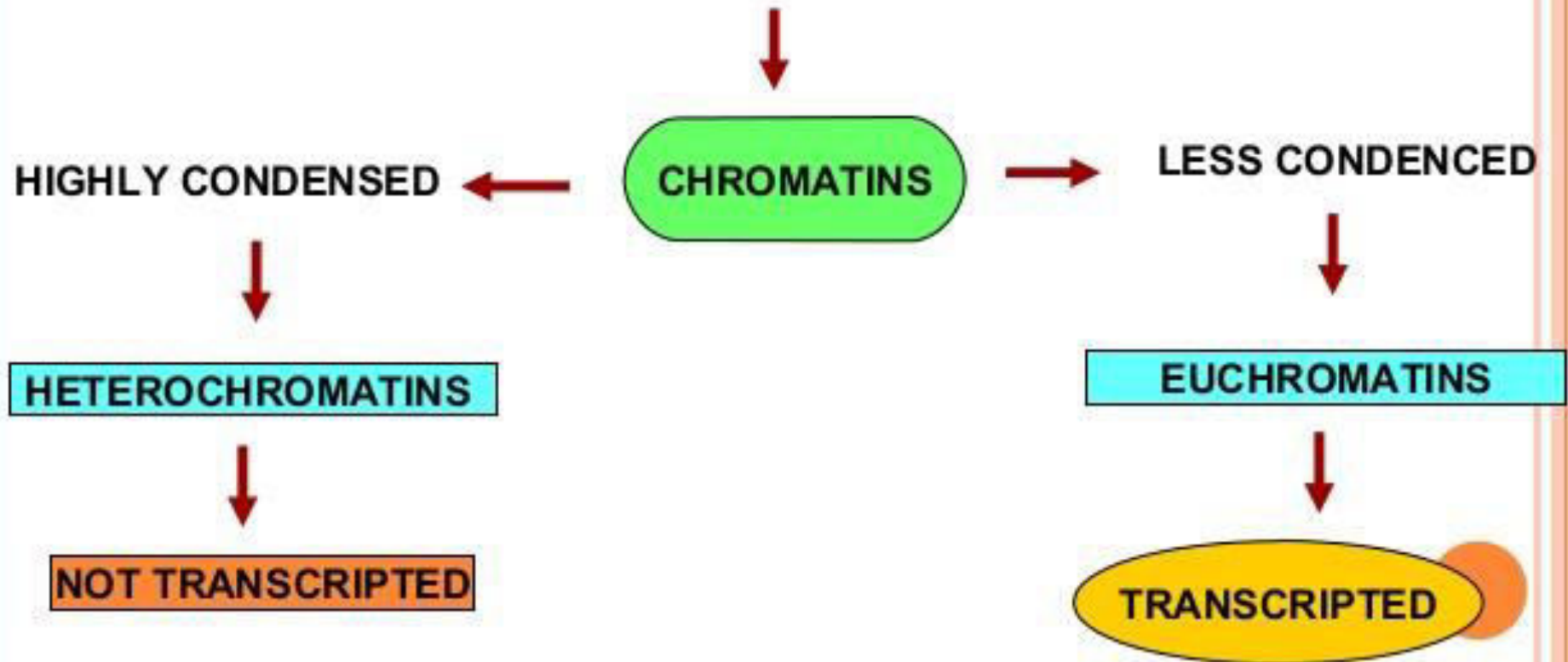
REGULATION OF EUKARYOTIC GENE EXPRESSION

- Regulation at transcription level
 - Chromatin remodeling
 - Enhancers
 - Trans acting molecules – DNA binding proteins
 - function as transcriptional factors
 - Cis-acting regulatory elements
 - eg: Hormone response elements
- Regulation by post transcriptional process
 - alternative mRNA splicing
 - mRNA editing
- Gene amplification
- Gene rearrangement

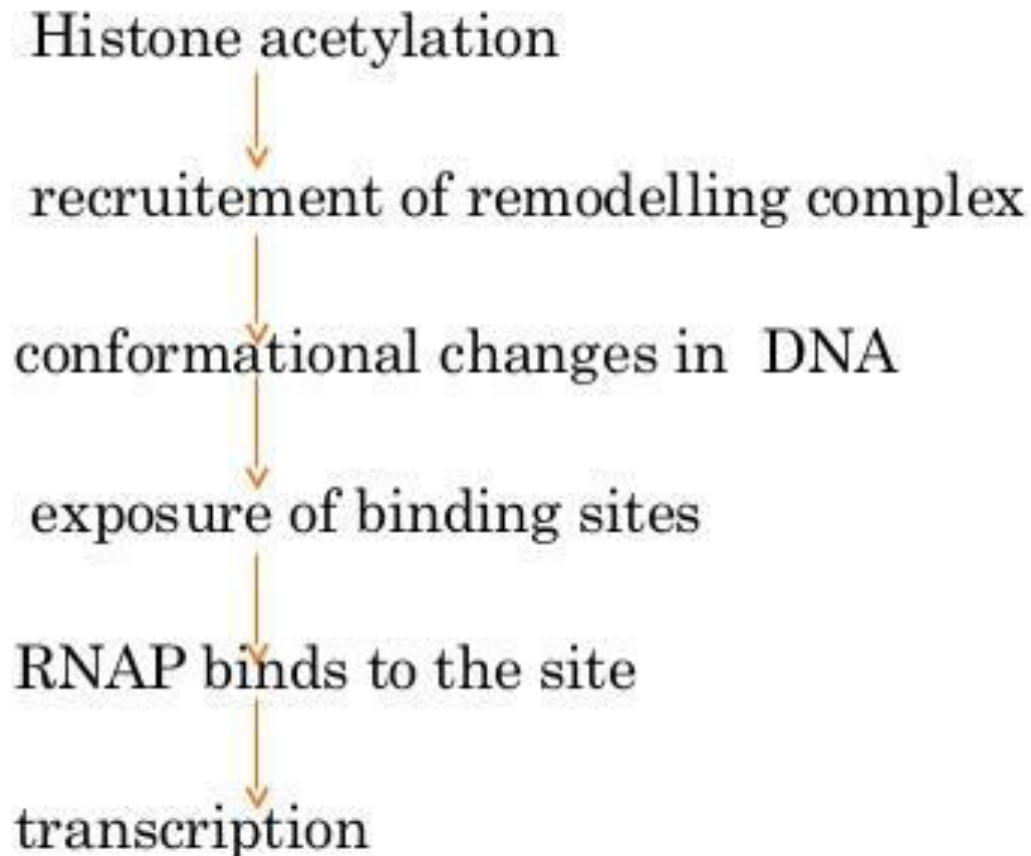


Eukaryotic DNA:

DNA + Histones



MECHANISM



- Chromatin remodeling

Heterochromatin / Euchromatin.

eg, B-globin gene in reticulocytes and muscle

Importance:

1) increases the accessibility of promoters and regulatory regions

2) increases binding of RNA polymerase

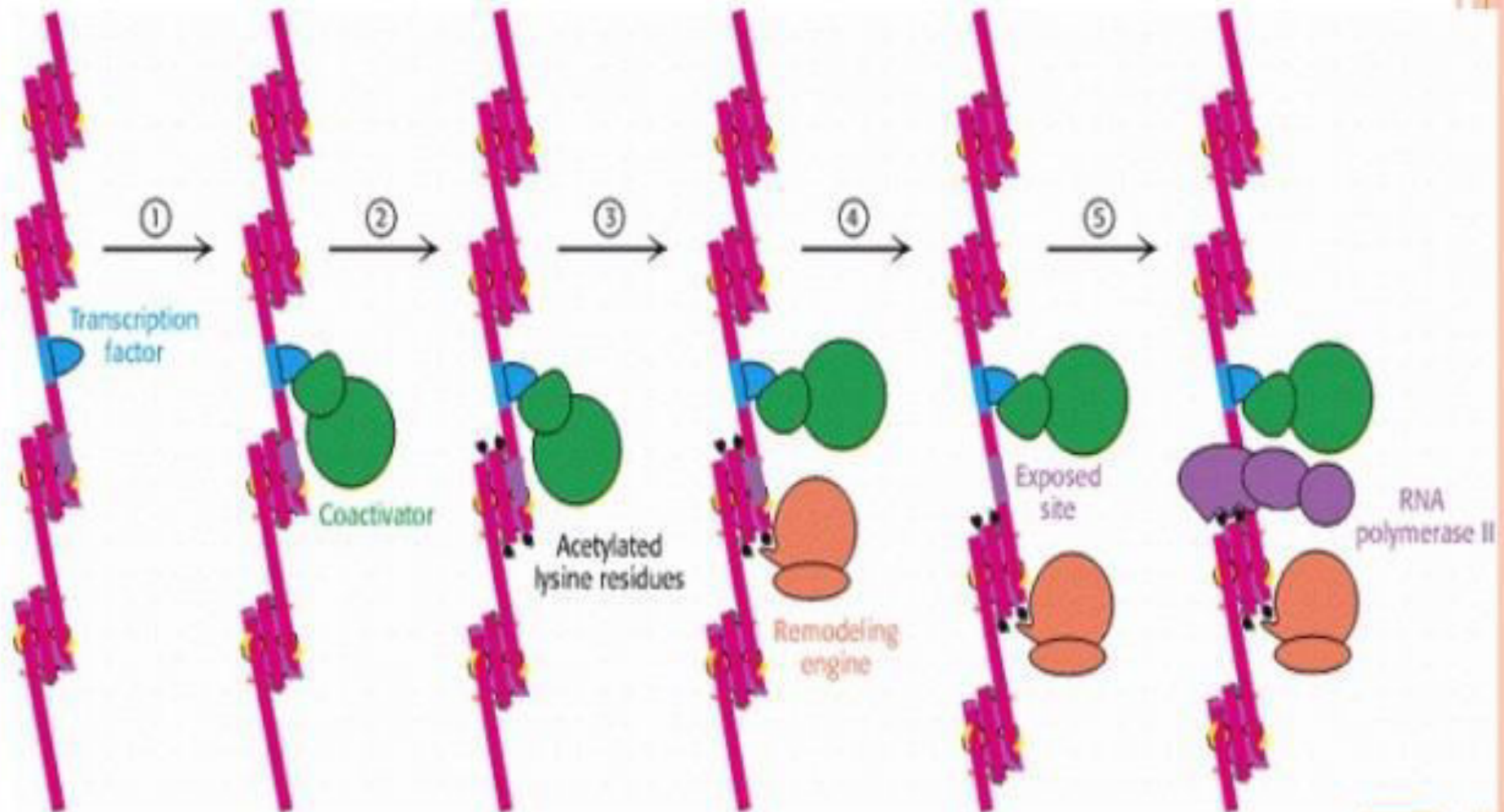
- Chromatin remodeling is facilitated by histone acetylation / deacetylation

Histone acetylase

○ **Histone**  **Acetylated**
Histone

Histone deacetylase

CHROMATIN REMODELING



ENHANCERS

- DNA elements which facilitate / enhance the initiation of transcription at the promoter

Features

- Is active only when it exists within the same DNA molecule (cis to the promoter)
- Work when located long distances from the promoter
- Work when upstream or downstream from the promoter
- Work when oriented in either direction



MECHANISM OF ACTION

Enhancer work by :

- binding one or more proteins
- facilitating binding of the basal transcription complex to the promoter
- recruiting chromatin-modifying co regulatory complexes
- Insulators: prevents the enhancer function



DNA BINDING MOTIFS

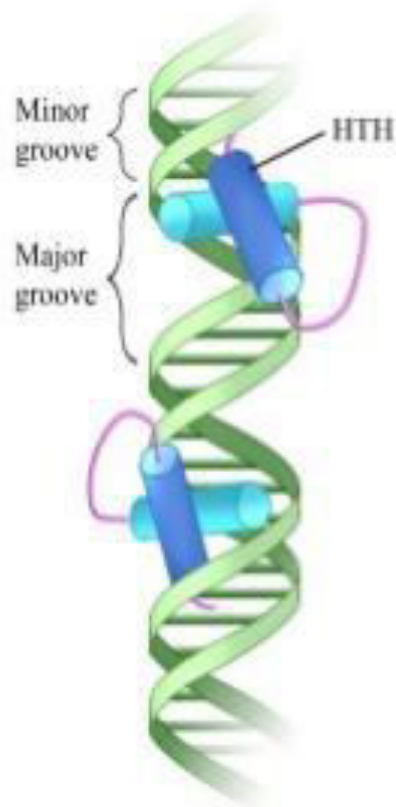
- Mediate DNA-protein interactions

Features

- Binds with high affinity to the specific site
- Maintained by hydrogen bonds, ionic interactions and van der Waals forces
- Involved in providing:
trans -activation domains / ligand binding sites
surfaces for interaction with co-
activators/repressors



TYPES OF MOTIFS



Helix turn Helix

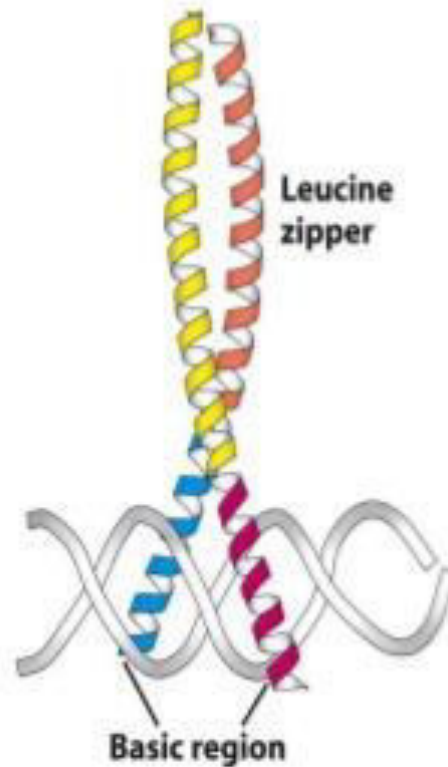
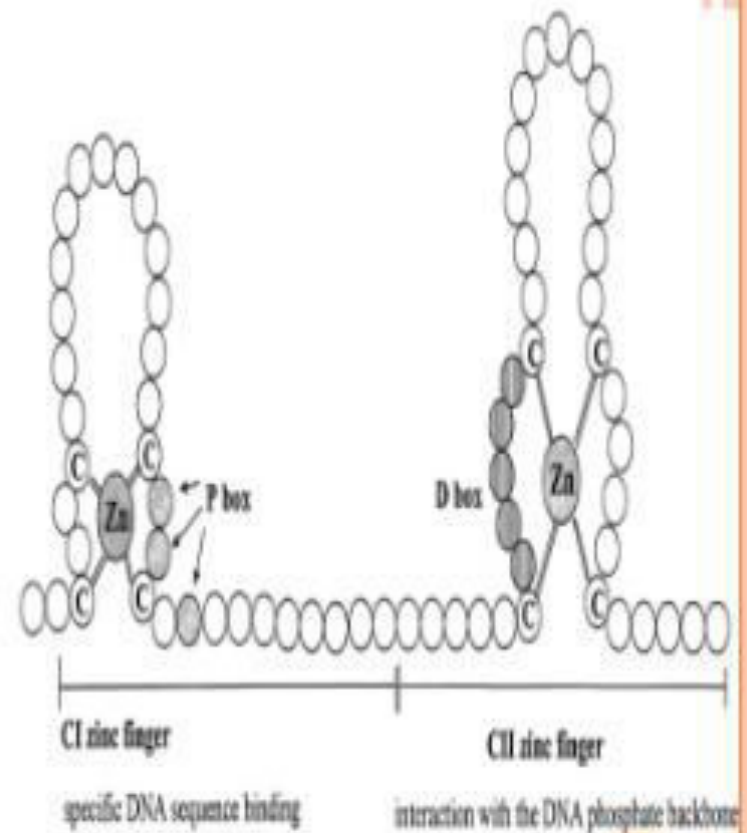


Figure 32.7
Biochemistry, Seventh Edition
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Leucine zipper



Zinc finger

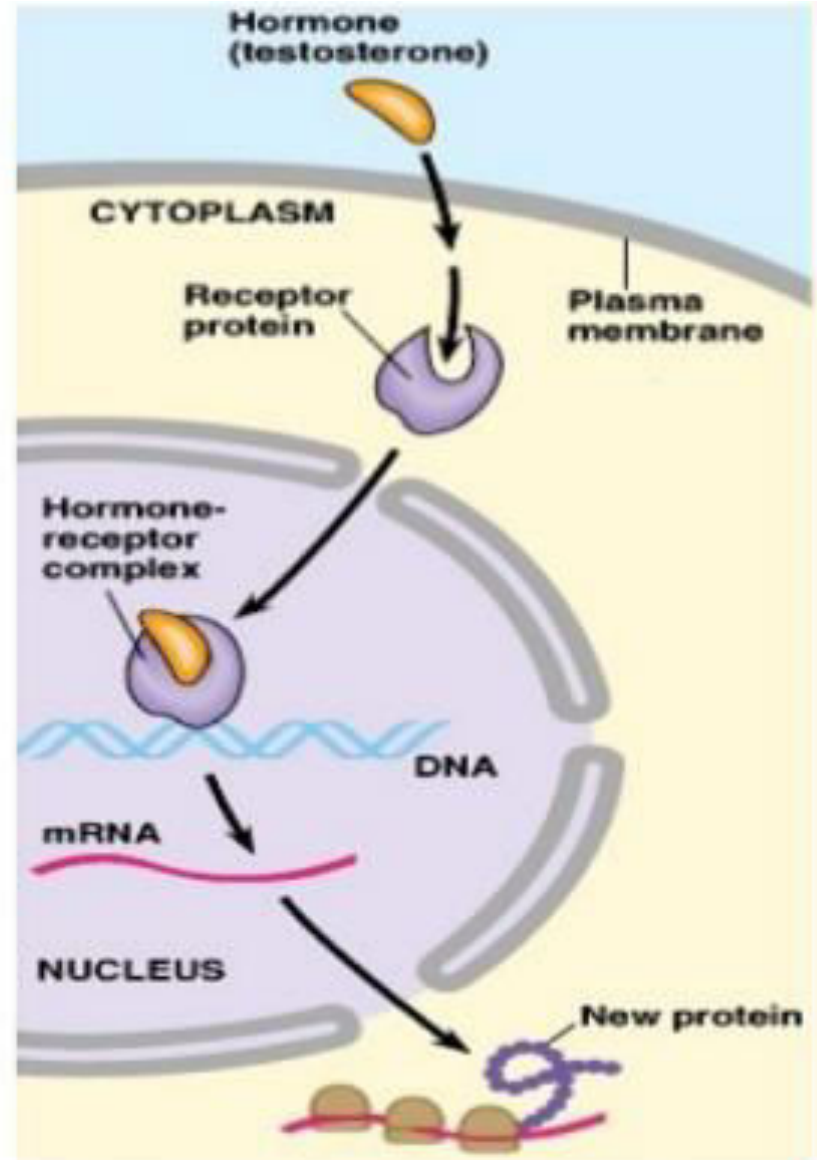
TYPES OF MOTIFS

Binding motif	Organism	Regulatory protein
Helix-turn-Helix	E.Coli	Lac repressor CAP
	mammals	Pit 1, Oct 1
Zinc finger	E.Coli	Gene 32 protein
	mammals	Steroid receptor family
Leucine Zipper	Yeast	GCN4
	mammals	Fos, jun, c-myc



Transcription Factors that are Steroid/Thyroid Hormone Receptor

- Nuclear receptor binds to **HRE** and induce or repress the transcription of target gene.
- Receptor contains separate:
 - Ligand binding domain
 - DNA binding domain



GENE AMPLIFICATION

- Increases the number of genes available for transcription
- Eg: resistance to methotrexate by cancer cells

cancer cells



increase the number of genes for DHFR



drug resistance



GENE REARRANGEMENT

- Occurs during immunoglobulin synthesis
- IgG light chain mRNAs are coded by different segments/domains
 - variable (VL) = 300 coding sequences
 - joining (JL) = 5
 - constant (CL) = 10
- Difficult for immune cells to achieve immunological specificity



IMMUNOGLOBULIN GENE REARRANGEMENT

Figure 5: V(D)J Recombination

Germline configuration



D to J recombination



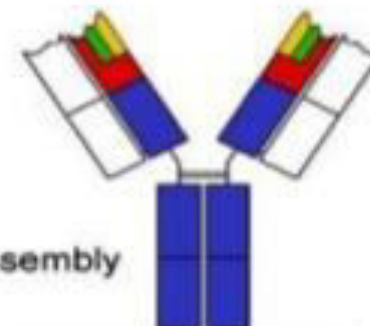
V to DJ recombination



transcription, splicing



translation, assembly



(adapted from Janeway 2001)

IMMUNOGLOBULIN GENE REARRANGEMENT

- Single V, J, C coding sequences recombine
- ↓
- Generate single contiguous transcription unit
- ↓
- RNAP transcribe into single mRNA



ALTERNATIVE RNA PROCESSING

- Primary transcript is processed differently in different tissues

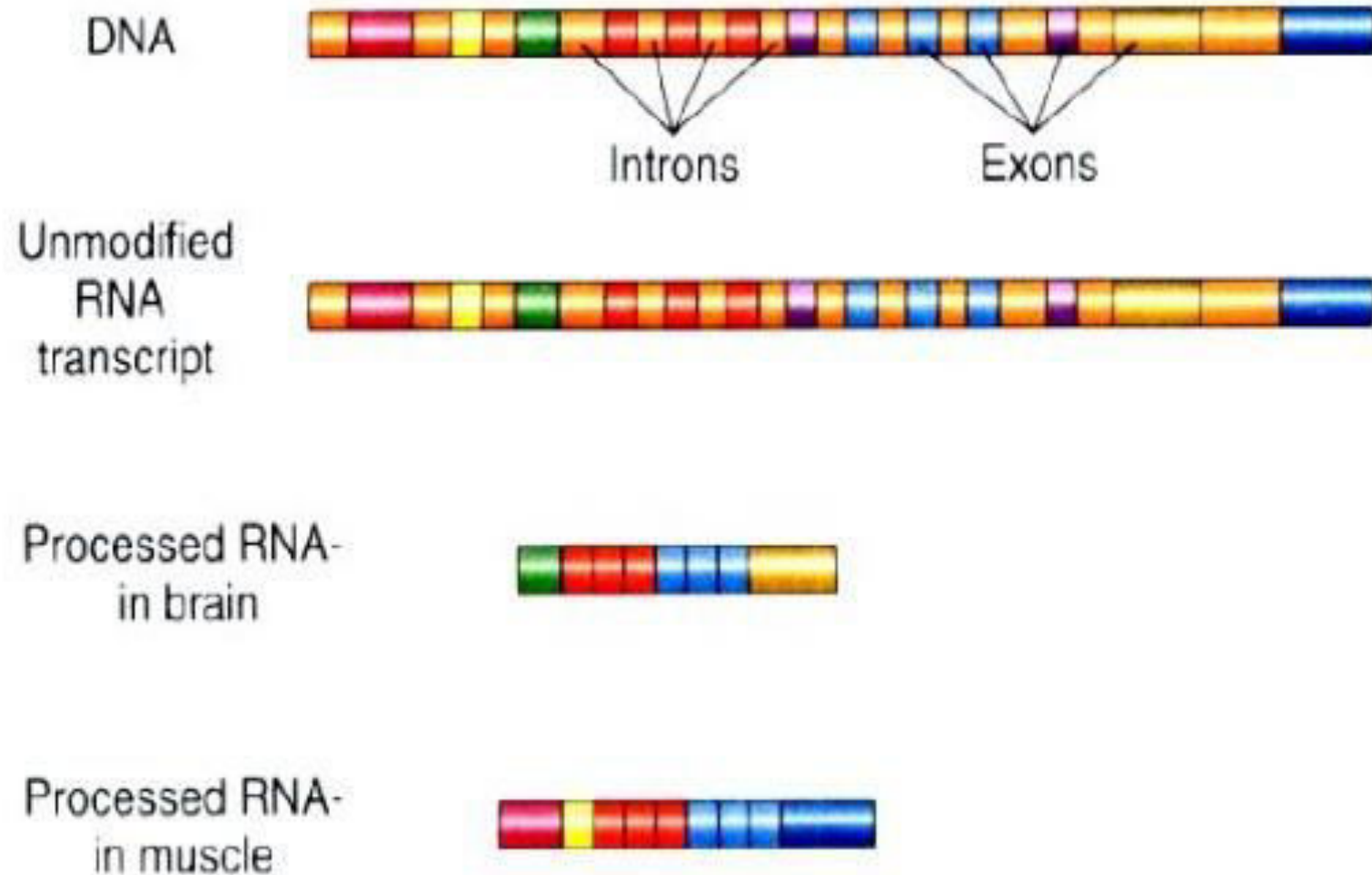
Achieved by

- Alternative transcription start sites
- Alternative polyadenylation site
- Alternative splicing and processing

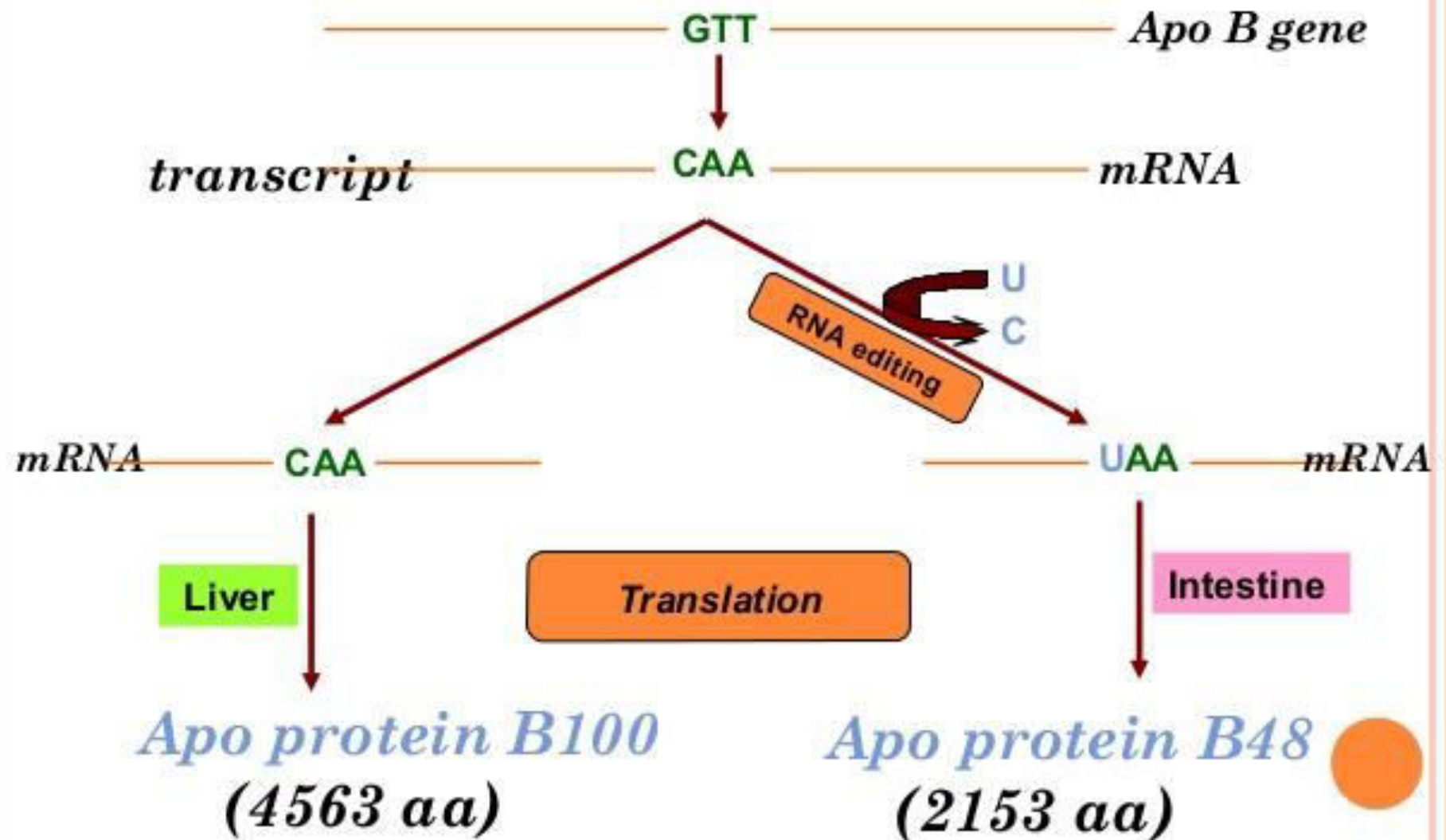
REGULATION OF MRNA STABILITY

- Ends of mRNA
 - 5' cap structure: prevents attack by 5' exonuclease
 - 3' poly (A) tail: prevents attack by 3' exonuclease
- Ribonucleoprotein particles

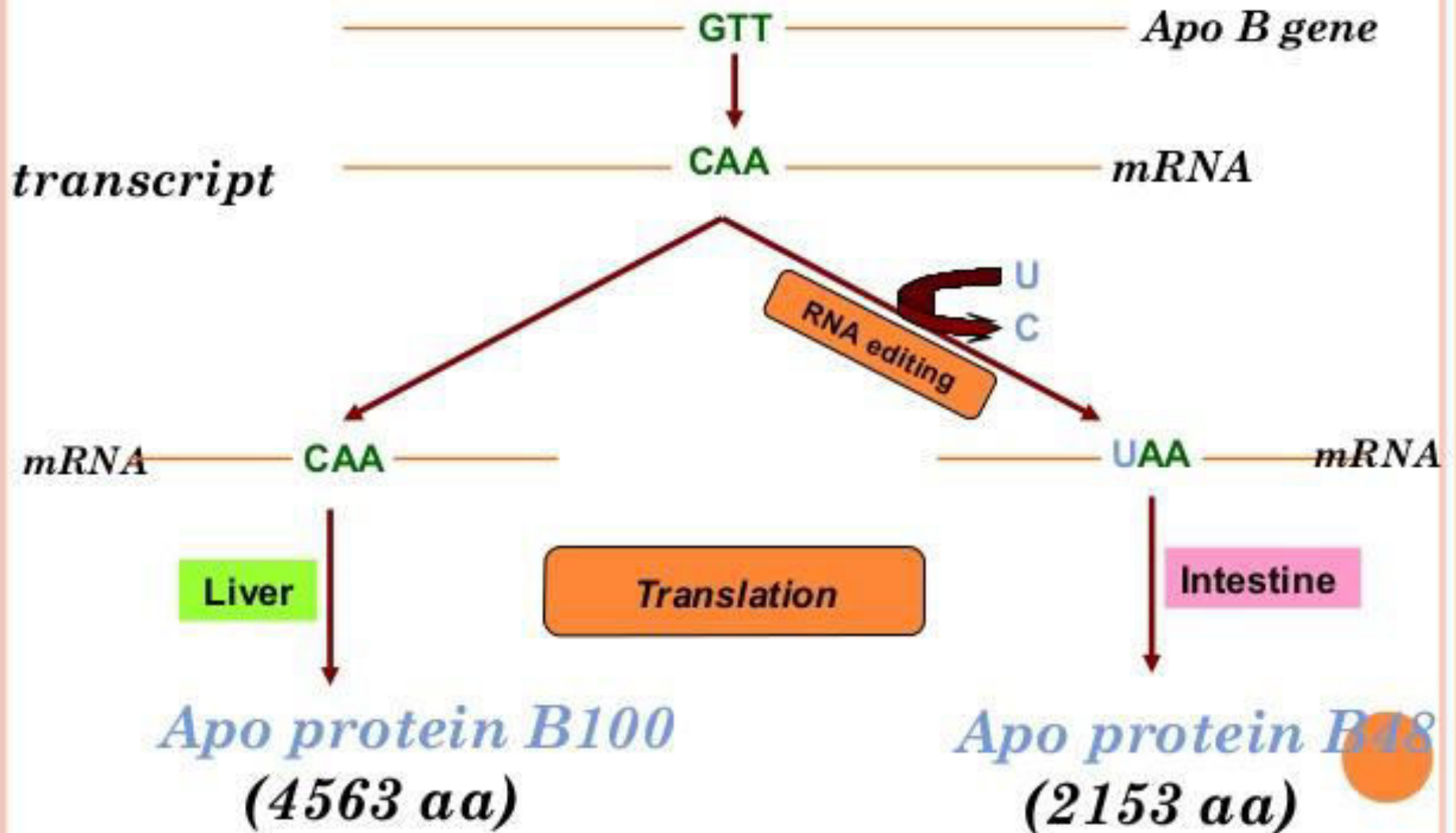
DIFFERENT FORMS OF TROPOMYOSIN BY ALTERNATIVE GENE SPLICING.



Example of RNA Editing



Example of RNA Editing



QUESTIONS ?

- Must be able to explain the mechanisms by which Eukaryotic gene expression are regulated and explain with suitable illustration.
- Must be able to explain the lac operon concept in prokaryotes.
- Must be able to describe schematically the different mechanisms of eukaryotic gene regulation.
- Must be able to effectively answer Applications of lac operon in medicine?



A ALWAYS

S SEEK

K KNOWLEDGE



